

Differential Climate Impacts on Fishing Effort in Chilean Small Pelagic Fisheries

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Abstract

Climate-driven shifts in stock productivity reach fishing fleets unevenly when species portfolios differ across fleet segments and the sector-level allocation of catch rights cannot adjust to changes in relative productivity. We quantify this distributional channel for the multi-species small pelagic fishery of Central-South Chile (anchoveta, sardine, jack mackerel), managed under sector-specific quota regimes with a fixed industrial–artisanal allocation. Our framework couples a Bayesian state-space model of stock dynamics, fit to 2000–2024 official assessment data, with a vessel-level trip equation; climate enters through an indirect biomass channel and a direct weather channel. Under a six-model CMIP6 ensemble spanning two emissions scenarios and two time windows, projected fleet-level effort falls by 13–15 percent for the artisanal fleet but only 0.8–1.0 percent for the industrial fleet — an asymmetry of roughly fifteen to one driven by differential portfolio exposure and weather sensitivity, with portfolio exposure decomposed between fleets and within the artisanal fleet by vessel type. The findings imply that climate-adaptation policy in Chile should revisit the sector-level allocation of catch rights, allowing it to respond to relative productivity shifts rather than remain fixed independent of the climate state.

Keywords: Bayesian state-space model; Bioeconomic projection; Chilean small pelagic fishery; Climate change; Fishing effort; Fleet heterogeneity; Quota allocation

JEL Codes: Q22, Q54, Q57, Q58

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1 Introduction

Climate change is reshaping the productivity of marine fisheries, but the incidence of this reshaping across fleet segments is rarely uniform. Quota-allocation regimes that codify the relative shares of industrial and artisanal (small-scale) fleets were typically designed under stationary-climate assumptions, so a fleet-asymmetric productivity shock acts as an unscripted reallocation of harvest rights. Whether such regimes should anticipate this reallocation depends on the projected magnitude and asymmetry of fleet-level effort responses under credible climate scenarios—a quantity the empirical literature has yet to deliver for most major multi-species fisheries.

We provide that quantity for the small pelagic fishery (SPF) of Central-South Chile (Regions V–X and XIV), part of the country’s largest fishery—approximately 86% of national fish landings by industrial and artisanal sectors in 2019 (SUBPESCA, 2020)—and the segment in which all three target species, anchoveta (*Engraulis ringens*), sardina común (*Strangomera bentincki*), and jack mackerel (*Trachurus murphyi*), coexist in non-negligible shares of both fleet portfolios. These three species share habitat and are coupled through trophic and market linkages (Alheit & Niquen, 2004; Arancibia et al., 2019; Dresdner et al., 2013). The fishery is managed through a sector-specific quota architecture: a fixed industrial-versus-artisanal allocation splits the Total Allowable Catch between the two segments, while within-sector distribution operates through Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP) for the industrial fleet and the *Régimen Artesanal de Extracción* (RAE) for the artisanal fleet, with limited cross-sector transferability. Because the two segments target different species portfolios, a stock-asymmetric climate shock translates directly into a fleet-asymmetric effort response. Cross-country evidence indicates that Chile is projected to lose part of its maximum catch potential under climate change: Cheung et al. (2010) classify Chile among a group of countries with moderate projected declines (6–13% by 2055 under SRES A1B), including the contiguous United States, China and Brazil; the available climate-fisheries economics literature is patchy globally and concentrated in a few well-studied regions (Sumaila et al., 2011). To our knowledge, no existing study for Chile combines multi-species stock dynamics with vessel-level effort responses under projected climate scenarios.¹

Our bioeconomic framework combines two empirical components. First, a Bayesian state-space model of multi-species stock dynamics — fit to 2000–2024 official IFOP and SPRFMO assessments — estimates, for each species, two semi-elasticities of productivity with respect to sea-surface temperature and log-chlorophyll-a anomalies. We call these parameters *climate shifters*. Because they enter the biological law of motion as coefficients on exogenous climate forcings, they remain interpretable when applied to climate states outside the 2000–2024 sample. Second, a vessel-level negative binomial trip equation — estimated separately for the industrial and artisanal fleets, with year fixed effects that absorb aggregate non-climate shocks (notably the 2019–2020 social and pandemic disruptions) — maps prices, allocated harvest, weather, and regulatory closures

¹The only empirical study of Chilean fisher behavior that we are aware of is Peña-Torres et al. (2017), which analyses how the El Niño–Southern Oscillation (ENSO) affects location choices in the jack mackerel fishery.

into annual fishing effort. Climate reaches fleet-level effort through two channels: an *indirect biomass channel* via the climate shifters, and a *direct weather channel* via vessel-specific exposure to severe winds. We evaluate the framework against a six-model CMIP6 ensemble under SSP2-4.5 and SSP5-8.5, at mid-century (2041–2060) and end-of-century (2081–2100). Throughout, we report two sources of uncertainty separately: dispersion across climate models (which future climate trajectory we project against) and dispersion in the Bayesian posterior of the structural parameters (uncertainty about the elasticities themselves)

We obtain three sets of results. First, the distributional incidence of climate-driven productivity shocks across fleets is sharply asymmetric: the artisanal fleet contracts by 13.2–15.0% across CMIP6 scenarios, while the industrial fleet contracts by only 0.8–1.0% — a ratio of roughly fifteen to one. Two economic mechanisms compound to produce this asymmetry. Each fleet is exposed to climate-driven productivity shifts through a distinct species portfolio, and the fixed industrial–artisanal allocation binds, preventing the cross-sector reassignment of harvest rights that would otherwise equalize marginal returns under the new relative productivities. The result instantiates, in a climate-shock setting, the portfolio-and-access framework of Kasperski & Holland (2013) and Cline et al. (2017): segments with narrower species portfolios bear sharper variance, and the welfare cost of that exposure rises when institutional access constraints prevent rebalancing (Oken et al., 2021). Because adaptation responses at the individual fisher level are themselves heterogeneous and context-dependent (Jardine et al., 2020; Zhang & Smith, 2011), the policy reading is direct: cross-sector transferability is the first-order rent-reallocation margin for attenuating disproportionate fleet-level losses, and any reform—the size, direction, and shadow price of admissible transfers—is quantifiable from the posterior we report. Second, the long-run change in intrinsic productivity is large and negative for anchoveta and sardine—the two coastal-upwelling stocks that dominate the artisanal portfolio—with floor effects for sardine under SSP5-8.5 end-of-century. Third, the climate shifters are sharply identified for these two stocks but not for jack mackerel, the species that anchors the industrial portfolio. We document this null with four converging tests (spatial-domain robustness across nested coastal aggregations, dual-source biomass with the Northern Chilean acoustic series, an OROP-PS coherence diagnostic, and basin-scale ENSO Niño 3.4 as an alternative shifter) and treat the jack mackerel productivity factor as fixed at its prior mean throughout the projections.

The remainder of the paper is organized as follows. Section 2 describes the economic and institutional setting of the Chilean SPF. Section 3 presents the data, the stock-dynamics model, the trip equation, and the CMIP6 projection design (Section 3.4). Section 4 reports the identified climate shifters, the comparative-statics projections of stock productivity, and the projections of fleet-level effort (Section 4.3). Section 5 discusses policy implications and limitations; Section 6 concludes.

2 The small pelagic fishery in Chile

The Chilean small pelagic fishery (SPF) is the largest fishery in the country, accounting for approximately 86% of national fish landings by the industrial and artisanal sectors in 2019 (SUBPESCA, 2020). The fishery is divided into two latitudinal zones with distinct species compositions. The northern zone (Regions I–IV) is dominated by anchoveta (*Engraulis ringens*) and jack mackerel (*Trachurus murphyi*), while sardine (*Strangomera bentincki*) is effectively absent. In contrast, the Central-South zone (Regions V–X and XIV) includes all three species in non-negligible shares within both artisanal and industrial landings. Because this region supports a genuinely multi-species fishery operating under a fixed sector-level allocation regime, it provides the relevant setting for evaluating the distributional consequences of climate-induced shifts in stock productivity, and is therefore the focus of this paper. The principal ports servicing this zone include San Antonio, Tomé, Talcahuano, San Vicente, Coronel, Lota, and Corral. The three species are harvested primarily by purse-seine fleets that differ in vessel size, operating range, and species composition.

The historical evolution of the fishery illustrates the portfolio mechanism developed in the remainder of the paper. The jack mackerel fishery was initially concentrated in northern Chile but shifted toward the Central-South zone during the mid-1980s (Peña-Torres et al., 2017), where fishing grounds have traditionally operated within 50 nautical miles of the coast. Between 1987 and 2004, approximately 85% of jack mackerel landings were directed to the reduction industry (Peña-Torres et al., 2017), reflecting the importance of fishmeal and fish-oil export markets in determining the value of the industrial portfolio. Sardine and anchoveta, by contrast, are jointly harvested using the same gear and fishing grounds. The two species share habitat and fishing technology, making species differentiation at the point of capture effectively impossible (Dresdner et al., 2013). As a result, substitution between them is constrained primarily by quota regulations rather than by biological or technological limitations.

The status of the three stocks has evolved substantially over time. Central-South anchoveta was classified as collapsed through 2018, reclassified as overexploited in 2019, and has been harvested within maximum sustainable yield (MSY) limits since 2020. Sardine has generally remained within MSY reference levels, except in 2021 and 2023 when it was classified as overexploited. Jack mackerel remained overexploited until 2018 and has since been harvested within MSY limits.

The Chilean SPF is managed through an annual Total Allowable Catch (TAC; *Cuota Global*) that is divided between the industrial and artisanal sectors, with small fractions reserved for research and contingency purposes. TACs are further subdivided across regions and seasons, and unused quotas may be reassigned within the fishing year. Anchoveta and sardine are managed under a mixed-species quota system: although each species has an independent quota, substitution between them is permitted within the regulatory framework.

Since 2013, the industrial sector has operated under an individual transferable quota (ITQ) system known as Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP). Class A

licenses were allocated according to historical catch shares, whereas Class B licenses—representing up to 15% of the industrial allocation—have been distributed through sealed-bid first-price auctions since 2015. Peña-Torres et al. (2022) document that these auctions have exhibited low participation, limited capacity to reflect economies of scale, and indications of coordinated bidding behavior. The artisanal sector operates under a regulated open-access regime, with regional TACs distributed through the *Régimen Artesanal de Extracción* (RAE) according to area, vessel size, landing site, or fisher organization.

Biological closures (*vedas*) constitute a central management instrument for sardine and anchoveta. In the Central-South zone, reproductive closures extend from July to October in Regions IX–XIV and from August to October in Regions V–VIII, while recruitment closures occur from January through February across all regions. In total, these regulations generate 151 closed fishing days per year in Regions V–VIII and 182 days in Regions IX–XIV. Jack mackerel is not subject to seasonal biological closures.

Beyond the within-sector quota architecture described above, the partition of each Total Allowable Catch between the artisanal and industrial sectors is fixed by statute at the legislative level. Article~47 of the General Law of Fisheries and Aquaculture (Ley General de Pesca y Acuicultura, LGPA) governed this de jure partition during the estimation window of the present paper (2018–2024); a 2025 reform, Ley~21.752, fixes a new partition in force until 2040. Table~1 summarises the two regimes for the three Central-South small pelagic stocks of interest. Under the pre-2025 regime, the artisanal sector held 78% of the Central-South anchoveta and sardina com'un TACs and 10% of the Central-South jurel TAC across the V–Los~Lagos macrozone; Ley~21.752 raises the artisanal share to 90% in both small pelagic stocks and partitions jurel into a 30% artisanal share in V–Los~R'ios and a 15% share in Los~Lagos, for a quota-weighted aggregate of approximately 28% across the V–Los~Lagos macrozone. The de jure partition admits a single inter-sectoral re-allocation channel, governed by Article~55T of the LGPA: industrial titulares and artisanal armadores may cede assigned tonnage across the sectoral boundary subject to ex-ante SUBPESCA authorisation, with the transferred volume drawn from the cedent's residual quota and added to the cessionary's effective quota for the calendar year. The SERNAPESCA quota control reports for 2013–2024 document that this channel is heavily used in the Central-South small pelagic fishery (Appendix~H, Section~H.3): the industrial sector cedes a stable 85–99% of its assigned anchoveta and sardina com'un TACs to artisanal cessionaries every year, in volumes ranging from 25 to 77~kt per species-year, while jurel cessions are an order of magnitude smaller and shift direction across years. Two consequences for the present paper follow directly. First, the empirical specifications in Sections~3.3.1 and~4.2 are identified on realised landings by sector, which are post-cession; the asymmetric climate sensitivity we report therefore reflects the operational allocation produced by the joint Article~47–Article~55T regime, not the de jure Article~47 fractioning alone. Second, Article~6 of Ley~21.752 imposes a specific patent of 1.3 monthly tax units per transferred ton on industrial-to-artisanal cessions, with artisanal armadores exempt from the patent in either direction; this is a transactional cost on the pre-existing Article~55T channel rather than the introduction

of a new wedge. Within sectors, transferability also differs sharply: industrial coefficients trade under the LTP regime introduced in 2013, while artisanal RAE assignments remain locked to the regional registry. We return to the welfare implications of the 2025 redistribution under climate stress in Section~5 and Appendix~H.

Table 1: Statutory sectoral partition of the Central-South small pelagic TACs, pre- and post-Ley 21.752 (2025).

Stock	TAC 2024 (kt)	Artisanal share of TAC (%)	
		Art. 47 LGPA	Ley 21.752
Anchoveta (V–Los Lagos)	211.3	78	90
Sardina común (V–Los Lagos)	290.6	78	90
Jurel (V–Los Ríos)	578.6	10	30
Jurel (Los Ríos–Los Lagos)	80.6	10	15

Notes. TAC 2024 is the Cuota Global de Captura fixed by SUBPESCA for the calendar year, in thousand tonnes. The pre-2025 fractioning regime follows the Article 47 schedule of the LGPA (DS 430/1991 of the Ministerio de Economía, Fomento y Reconstrucción, consolidated text of Law 18.892/1989) [SERNAPESCA2024-cuota]. The 2025 regime corresponds to Article 1 of Ley N° 21.752 [Ley21752-2025; Boletín17096-21], in force until 31 December 2040. Items 6 and 7 of Article 1 assign 90% of the anchoveta and sardina común TACs to the artisanal sector throughout the V–Los Lagos macrozone (single block). Items 4 and 5 partition jurel into two sub-macrozones: 30% artisanal in V–Los Ríos and 15% artisanal in Los Lagos. The SERNAPESCA quota report aggregates Los Ríos and Los Lagos into a single jurel quota of 80.6 kt; the row in the table applies the Los Lagos share (15%) as a conservative approximation, with the understanding that the true sub-block share lies between 15% and 30% depending on the Los Ríos proportion of the aggregate quota. A use-it-or-lose-it adjustment rule applies in the northern macrozone and is omitted from the table because the present paper focuses on the Central-South. Industrial-to-artisanal inter-sectoral transfers are subject under Article 6 to a specific patent of 1.3 monthly tax units per transferred ton, with a minimum of 250 monthly tax units; artisanal armadores are exempt from the patent on transfers in either direction.

3 Methods

Our empirical strategy targets two structural objects—climate shifters of stock productivity and fleet-specific elasticities of fishing effort—and combines them to compute counterfactual fleet responses under future climate paths. First, we identify the climate shifters: species-specific semi-elasticities of intrinsic stock productivity with respect to sea-surface temperature and chlorophyll-a anomalies. We estimate them within a Bayesian state-space model of multi-species stock dynamics, with a Schaefer law of motion and priors taken from official IFOP and SPRFMO assessments. Second, we estimate fleet-specific elasticities of annual fishing effort with respect to allocated harvest, ex-vessel prices, weather exposure, and regulatory closures, using a negative binomial count model fitted separately for the artisanal and industrial fleets, with year fixed effects that absorb aggregate

non-climate shocks. Third, we compute long-run comparative statics under a six-model CMIP6 ensemble (SSP2-4.5 and SSP5-8.5, mid- and end-of-century), propagating the climate shifters through the stock-dynamics steady state to long-run productivity and through the trip equation to fleet-level effort.

3.1 Historical data

We use data requested from the Chilean Fisheries Development Institute (Instituto de Fomento Pesquero, IFOP) covering the 2013–2024 period. The dataset includes trip-level microdata with detailed records on vessel identifiers, departure and arrival times, vessel capacity, fleet and gear type, ports of departure and landing, fishery codes, haul timing and location, species composition, retained catch, and trip activity. We also requested annual information on stock abundance and vessel landings by port, county, region, country, and species. Finally, we use ex-vessel prices reported monthly or annually by port, county, region, country, and species. These prices reflect payments from processing plants to fishers at the point of first sale and are obtained through IFOP’s landing surveys, which do not necessarily cover all market transactions.

Environmental covariates are obtained from the E.U. Copernicus Marine Service Information, accessed through the Copernicus Marine Toolbox API. Salinity, sea surface temperature, and current speed and direction come from the Global Ocean Physics Reanalysis (GLORYS12V1), which provides data at $1/12^\circ$ horizontal resolution with 50 vertical levels ([E.U. Copernicus Marine Service Information, 2025c](#)). Surface wind speed and direction are taken from the Global Ocean Hourly Reprocessed Sea Surface Wind and Stress from Scatterometer and Model dataset, at 0.125° horizontal resolution and hourly frequency ([E.U. Copernicus Marine Service Information, 2025b](#)). Chlorophyll-a concentrations come from the Global Ocean Colour dataset at 4 km horizontal resolution ([E.U. Copernicus Marine Service Information, 2025a](#)). All environmental data were retrieved daily (hourly for winds) for the 2013–2024 period, covering the Chilean Exclusive Economic Zone (EEZ) between 32°S and 41°S (Figure 1).

Sample structure for jack mackerel biomass identification The jack mackerel biomass sample over the 2000–2024 estimation window is informative for identification in 18 of 25 years. Sixteen years contribute a point estimate from the official IFOP hydroacoustic assessment, and two additional years (2012, 2015) enter the likelihood as left-censored observations at the assessment’s lower detection limit. The remaining seven years carry no biomass observation; their latent state is identified dynamically by the law of motion of Section 3.3.1, together with the priors on $(r^0, K, \sigma_{\text{proc}}, \sigma_{\text{obs}})$ from official IFOP and SPRFMO assessments and on (ρ^{SST}, ρ^{CHL}) from the reduced-form stress tests of Appendix A. This reduced informational content is one of the load-bearing facts behind the non-identification result for the jack mackerel climate shifter (Section 4.1).

Two external series—jack mackerel biomass from the Northern acoustic zone (Arica–Iquique–Antofagasta, available since 2010) and the SPRFMO transzonal spawning biomass index for the Southeast Pacific—were considered as auxiliary information. The Northern series correlates with the Centro-Sur (CS) series at $r = 0.82$ over $N = 7$ overlapping years, while the SPRFMO index correlates with CS at $r = 0.11$ over $N = 17$ overlapping years. We treat these numbers as suggestive rather than conclusive: a seven-point Pearson correlation is at best descriptive, but the contrast in sign and magnitude is consistent with a Centro-Sur stock unit that shares short-run dynamics with the Northern Chilean assessment and is only weakly coupled to the SPRFMO-scale aggregate. Neither external series enters the structural model’s likelihood—as a prior or as an observation—to avoid introducing information about coastal climate forcing through the indirect route of Northern or transzonal biomass.

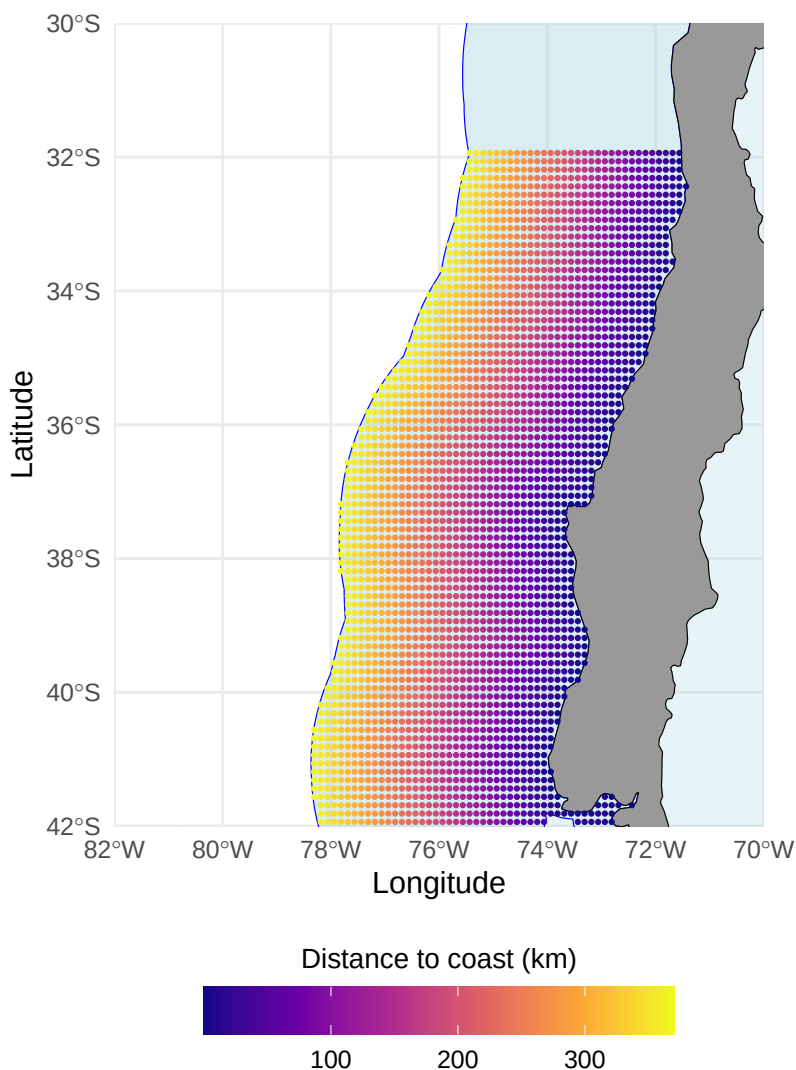


Figure 1: Geographical extent of the study area used for environmental covariates limited to the Chilean Exclusive Economic Zone

3.2 Future climate data

Climate projections for SST, chlorophyll-a, and near-surface wind speed are obtained from the CMIP6 archive using a six-model ensemble (IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, MPI-ESM1-2-HR), selected to span the CMIP6 equilibrium climate sensitivity distribution. We use monthly outputs for two future scenarios: SSP2-4.5 (moderate) and SSP5-8.5 (high emissions). Section 3.4 describes the projection horizons, baseline period, and delta-method procedure used to translate CMIP6 outputs into our empirical models. The per-model decomposition of projection uncertainty appears in Appendix F for stock-level intrinsic productivity and in Appendix G for fleet-level trips.

3.3 Econometric models

3.3.1 Stock dynamics

Our identification target in this subsection is the vector of climate semi-elasticities $(\rho_i^{SST}, \rho_i^{CHL})_{i=1}^3$ —the structural parameters that translate sea-surface temperature and chlorophyll-a anomalies into shifts in long-run stock productivity. We recover them by inverting a Bayesian state-space model of multi-species stock dynamics in which a Schaefer transition with a log-linear climate shifter on the intrinsic growth rate governs the latent biomass $B_{i,t}$ of stock $i \in \{\text{anchoveta, sardina común, jack mackerel}\}$. The Schaefer form (the $\theta = 1$ case of the Pella–Tomlinson family) is adopted on identifiability grounds: with $N = 25$ annual observations per stock, the shape parameter θ is notoriously weakly identified, and imposing $\theta = 1$ is a standard identification device in the small-pelagic literature (Hilborn & Walters, 1992; Maunder & Punt, 2013). The law of motion is

$$B_{i,t+1} = B_{i,t} + r_{i,t} B_{i,t} \left(1 - \frac{B_{i,t}}{K_i}\right) - C_{i,t} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim \mathcal{N}(0, \sigma_{\text{proc},i}^2), \quad (1)$$

where $C_{i,t}$ is the observed annual catch, K_i is the stock-specific carrying capacity, and $\sigma_{\text{proc},i}$ is the standard deviation of the stochastic recruitment innovation. Climate enters the dynamics through a log-linear shifter on the intrinsic growth rate,

$$r_{i,t} = r_i^0 \exp\left(\rho_i^{SST} (SST_{t-1} - \overline{SST}) + \rho_i^{CHL} (\log CHL_{t-1} - \overline{\log CHL})\right), \quad (2)$$

so ρ_i^{SST} and ρ_i^{CHL} are the semi-elasticities of intrinsic stock productivity to a one-degree Celsius SST anomaly and to a one-log-point chlorophyll-a anomaly, respectively. Because these parameters enter the law of motion as coefficients on exogenous climate forcings, they are invariant to the joint historical distribution of (SST, B, C) —transportable, in the structural sense, to climate regimes outside the 2000–2024 estimation sample. This transportability matters because the comparative-

statics exercise of Section 4.3 evaluates the climate shifter at projected SST anomalies that lie several historical standard deviations beyond the estimation envelope.

Latent biomass is linked to the official survey-based estimates of spawning stock biomass $B_{i,t}^{\text{obs}}$ —published by IFOP for anchoveta and sardina común in Centro-Sur Chile and by SPRFMO for jack mackerel—through a log-normal measurement equation,

$$\log B_{i,t}^{\text{obs}} = \log B_{i,t} + u_{i,t}, \quad u_{i,t} \sim \mathcal{N}(0, \sigma_{\text{obs},i}^2). \quad (3)$$

Environmental conditions are summarized annually by sea surface temperature (SST) and chlorophyll-a concentration (CHL), averaged over the Centro-Sur region within the Chilean EEZ and lagged one year to respect the pre-determined ordering of environmental forcing relative to year- t biomass. The choice of these two covariates follows a large biological literature that identifies thermal regime and primary productivity as the dominant local drivers of small-pelagic dynamics in the Humboldt Current System (Axbard, 2016; Cahuin et al., 2009; Yáñez et al., 2014; see Cheung et al., 2008; Jennings et al., 2008).

We estimate the state-space specification by Bayesian inference, for two reasons standard in the small-sample identification literature. First, with $N = 25$ annual observations per stock, joint identification of the stock-dynamics parameters (r_i^0, K_i) and the climate shifters $(\rho_i^{SST}, \rho_i^{CHL})$ requires the prior information embedded in the IFOP and SPRFMO single-species assessments to avoid degenerate likelihood maxima at the boundary of the parameter space; Appendix A demonstrates this boundary problem explicitly. Second, posterior credible intervals provide a natural framework for propagating uncertainty in the shifters through to the comparative-statics exercise of Section 4.3. Estimation is implemented in Stan via Hamiltonian Monte Carlo. Priors on the biological parameters $(r_i^0, K_i, \sigma_{\text{proc},i}, \sigma_{\text{obs},i})$ are adopted from the IFOP and SPRFMO single-species assessments. Priors on the climate shifters $(\rho_i^{SST}, \rho_i^{CHL})$ are weakly informative normal densities centred at the point estimates of an independent stress-test regression of biomass growth increments on SST and log-CHL anomalies, with unit standard deviation (see Appendix A for the full prior elicitation protocol). We estimate three nested specifications of increasing flexibility: (i) species-independent dynamics without climate shifters (`ind''`), (ii) the same with a cross-stock residual covariance Ω (`omega''`), and (iii) the full model augmented with the climate shifter of Eq. (2) (`full''`). The posterior of the `full''` specification is our primary object of interpretation; the nested predictive comparison is relegated to Appendix B. All reported posteriors are based on four independent Hamiltonian Monte Carlo chains of 2,000 post-warmup iterations each, evaluated against standard convergence diagnostics. The Gelman–Rubin statistic $\hat{R}$ (which compares within-chain to between-chain variance and should be close to one if chains have mixed) is below 1.01 for all top-level parameters, and the effective sample size—in both the bulk and the tails of the posterior—exceeds 400 for all top-level parameters, indicating that posterior summaries are not contaminated by autocorrelation between adjacent draws. These thresholds are conventional in the applied Bayesian literature; see Appendix D for the convergence diagnostics by parameter

family. The seven jack mackerel Centro-Sur years without a survey estimate enter the likelihood as unobserved latent biomass states, identified only through the Schaefer transition equation; two further years (2012 and 2015) enter through a log-normal censored-observation term at the stock-assessment detection limit. All associated uncertainty propagates directly into the posterior of $(\rho_{\text{jurel}}^{SST}, \rho_{\text{jurel}}^{CHL})$, so no imputation step is embedded in the Stan likelihood.

Three features of the state-space formulation in Eq. (1)–(3) are material for this paper. First, it separates process noise σ_{proc} from observation noise σ_{obs} , which is essential given the documented coefficient-of-variation of acoustic and egg-production surveys in this fishery. Second, it enforces a pre-determined ordering between year- $(t - 1)$ environmental forcing and year- t latent biomass, ruling out contemporaneous feedback into the climate covariates. Third, it yields structural climate semi-elasticities $(\rho_i^{SST}, \rho_i^{CHL})$ that are invariant to the joint historical distribution of (SST, B, C) and therefore transportable to climate regimes outside the 2000–2024 estimation window. This transportability is essential because the comparative-statics exercise of Section 4.3 evaluates the climate shifter at projected SST anomalies several historical standard deviations beyond the estimation envelope.

Figure 2 shows that anchoveta, sardina común, and jack mackerel exhibit distinct biomass trajectories over the 2000–2024 window, with no evidence of strong interannual co-movement across species once scale differences are accounted for. Harvest pressure is visibly associated with these fluctuations; jack mackerel in particular experienced an abrupt decline during the late 2000s, consistent with the combined effects of intense fishing and unfavorable environmental conditions.

3.3.2 Total annual trips

We model the annual number of fishing trips taken by vessel v in year y as a count process following Kasperski (2015). Because the available logbook data record only purse-seine operations, the unit of observation is a vessel–year. We estimate separate effort models for the industrial and artisanal fleets, allowing fishing activity to respond differently to economic conditions and regulatory constraints across sectors. This specification explicitly accommodates technological heterogeneity across fleet segments, reflecting differences in vessel capacity, operating scale, and production technology. A complementary approach, used in Quezada-Escalona et al. (2026) for the U.S. West Coast Coastal Pelagic Species fishery, models daily participation and target-species choice through a discrete choice framework with species distribution models as proxies for daily availability. The annual NB specification we adopt here is the natural counterpart in a setting where vessel-year is the operational unit at which TACs are allocated under the LTP, and where daily logbook coverage is too sparse to support a daily DCM for the Centro-Sur Chilean SPF over the 2013–2024 window.

The baseline specification is a Poisson model:

$$T_{vy} \sim \text{Poisson}(\lambda_{vy}), \quad \lambda_{vy} = \exp(U'_{vy}\beta), \quad (4)$$

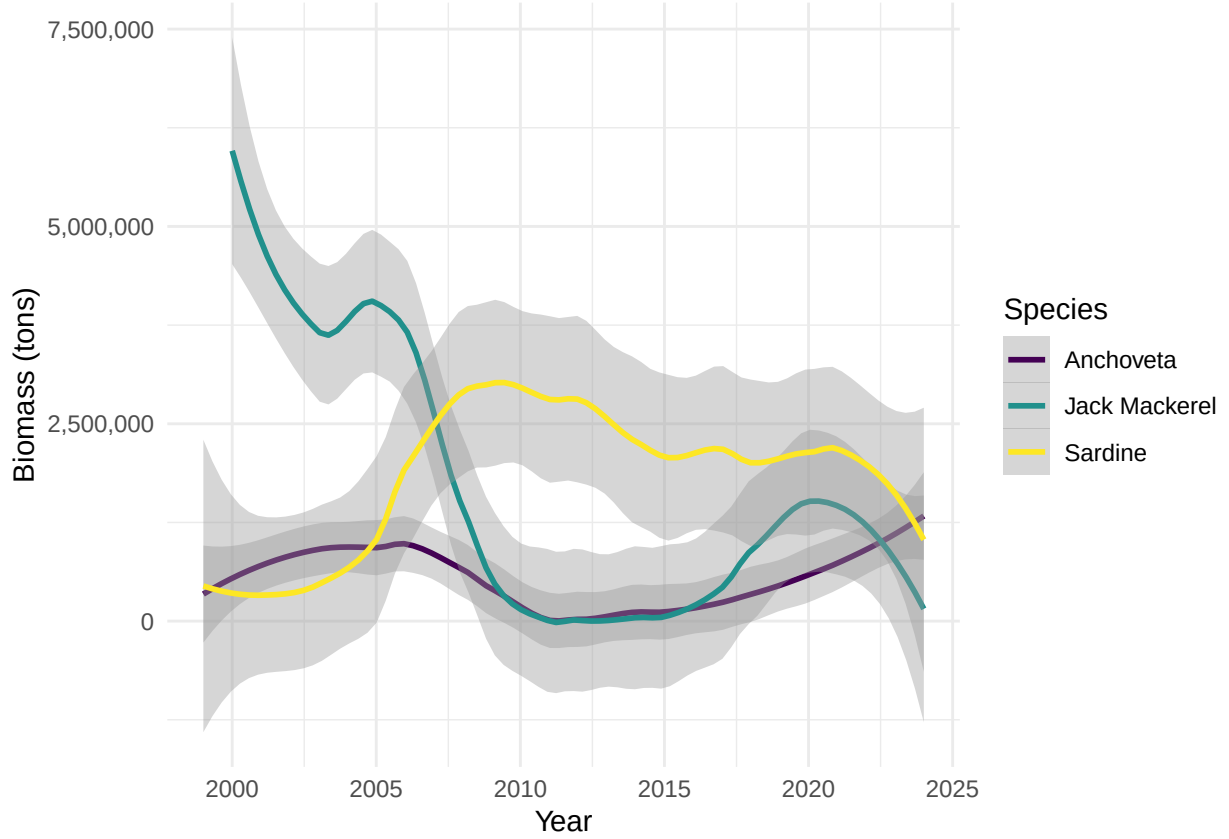


Figure 2: Estimated biomass of small pelagic species in Chile (2000–2024)

where T_{vy} denotes the total number of purse-seine trips recorded for vessel v in year y . Given substantial overdispersion in the data—the variance-to-mean ratio of T_{vy} is 22.4 for the artisanal fleet and 4.9 for the industrial fleet—we estimate a negative binomial (NB) model, which nests the Poisson as a special case. A likelihood ratio test strongly rejects the Poisson restriction for both fleets ($p < 0.001$).

The vector of explanatory variables U_{vy} includes output prices by species, vessel-level allocated harvest, fixed vessel characteristics, and operating conditions:

$$U_{vy} = [p_{sy}, H_{vy}^{alloc}, Z_v, O_{vy}]. \quad (5)$$

Output prices p_{sy} are species-specific ex-vessel prices paid by processing plants to fishers, obtained from IFOP’s manufacturing survey and deflated to constant 2018 pesos using the consumer price index from the Central Bank of Chile. Following Dresdner et al. (2013), prices are measured as annual averages over peak fishing months to reflect the economically relevant conditions faced by vessels when planning annual effort.

Because the model aims to characterize vessels’ effort responses to changes in prices, quotas,

and environmental conditions within the small pelagic fishery, we restrict attention to SPF trips recorded in logbooks. These trips account for approximately 95% of observed fishing revenue and therefore capture the primary economic margin through which vessels adjust annual effort. Cross-validation against the SERNAPESCA vessel-level landings database (transparency request AH010T0006857; 2013–2024 Centro-Sur) confirms that the resulting panel is representative of the purse-seine artisanal segment. For the industrial fleet, the panel covers 40 of 59 registered vessels (68% by count) and aggregate landings within -10% of the SERNAPESCA all-gear total, with portfolio-weighted catch composition matching SERNAPESCA aggregates within two percentage points across the three target species. For the artisanal fleet, the panel covers 859 of 2,689 registered vessels (32% by count) but accounts for approximately 70% of total Centro-Sur artisanal landings of the three species, reflecting the concentration of catch in the larger purse-seine vessels for which logbook submission is mandatory. Smaller artisanal vessels using non-purse-seine gears (lampara, lines) operate outside the LTP quota architecture and are not modeled in the trip equation; they appear in the SERNAPESCA aggregate landings used as the likelihood input for the biomass state-space model.

Quota shares do not enter Eq. (4) directly. Instead, shares translate annual TACs into vessel-level allocated harvests. Let ω_{vs}^r denote vessel v 's historical share of landings of species s within its administrative region r (for artisanal vessels) or regulatory zone (for industrial vessels), computed over the full sample period. Given the effective TAC $\bar{Q}_{sy}^r$ for species s in year y and region r —obtained from SERNAPESCA quota monitoring records and reflecting all inter-fleet transfers and adjustments—vessel-level allocated harvest is:

$$H_{vy,s}^{alloc} = \omega_{vs}^r \bar{Q}_{sy}^r, \quad H_{vy}^{alloc} = \sum_s H_{vy,s}^{alloc}. \quad (6)$$

Each vessel is assigned to its administrative region based on its modal departure port in the logbook records. Artisanal TACs are assigned at the regional level (regions V, VIII, IX, XIV, X), while industrial TACs follow the regulatory zone structure established by SUBPESCA (zones V–IX and XIV–X for jack mackerel; zone V–X for sardine and anchoveta). This regional construction introduces cross-sectional variation in H_{vy}^{alloc} that reflects the heterogeneous quota environments faced by vessels operating in different parts of the Centro-Sur fishery.

Following Kasperski (2015), the trip equation enters allocated harvest *by species* rather than as a scalar aggregate: the regressor vector contains $H_{vy,s}^{alloc}$ for $s \in \{\text{anchoveta, sardine, jack mackerel}\}$ with separate semi-elasticities β_h^s . Ex-vessel prices likewise enter at $(year, region)$ resolution, mapped to each vessel through its modal departure region. For the artisanal fleet, the response to allocated anchoveta is allowed to vary by vessel type through an interaction $H_{vy,anchoveta}^{alloc} \times \text{TIPO_EMB}$, motivated by the observation in the data that small-vessel categories (LM = *lancha motor*, the dominant 77% of the artisanal panel) exhibit a near-zero elasticity to anchoveta quota, while medium-vessel categories (BM = *bote motor*, 8%) exhibit a strong positive elasticity in the

spirit of the Kasperski (2015) benchmark. The interaction is estimated only for TIPO_EMB categories with at least 50 vessel-year observations (BM, LM, UNK); three small categories with $N < 50$ (BR, BRV, L) collectively represent less than 0.1% of artisanal effort and quota and are reported with an aggregated rather than category-specific coefficient. The Kasperski-aligned specification is preferred by an information criterion of $\Delta\text{AIC} = -167.9$ for the artisanal fleet (and -5.0 for the industrial fleet) relative to the legacy scalar specification, with multicollinearity diagnostics (VIFs below 2.5 across all six regressors and both fleets) confirming separate identification of the three β_h^s .

The vector Z_v contains time-invariant vessel characteristics, including hold capacity (log of cubic meters) and vessel type. Hold capacity determines the physical constraint on catch per trip and proxies for vessel scale.

The vector O_{vy} captures operating and regulatory constraints that vary across both vessels and years. Unlike the environmental variables in the stock dynamics model, which reflect biological productivity (SST, chlorophyll-a), the variables in the trip equation capture conditions that affect the *feasibility* of fishing operations. Two vessel-year variables are constructed using each vessel's center of gravity (COG)—the catch-weighted centroid of its haul locations over the sample period. First, the number of adverse-weather days is computed as the count of days per year in which wind speed exceeds 8 m/s at the environmental grid point nearest to the vessel's COG.² Second, the number of days closed for biological closures (*vedas*) is assigned based on the regulatory zone corresponding to the vessel's COG latitude. In the Centro-Sur region, sardine and anchoveta face seasonal closures for reproduction (August–October in regions V–VIII; July–October in regions IX–XIV) and recruitment (January–February in all regions), yielding 151 and 182 closed days per year, respectively. Jack mackerel has no biological closures.

Quota prices are not included explicitly in the trip equation. Instead, the economic scarcity of quota is captured implicitly through aggregate TAC levels and vessel-level allocated harvest.

Trips, harvest, and prices may be jointly determined within the year, implying potential endogeneity in reduced-form trip regressions. The purpose of Eq. (4), however, is not causal identification but rather to provide an empirically grounded behavioral relationship that maps changes in prices, quotas, and environmental conditions into expected fishing effort under projected climate scenarios.

3.4 Projection approach

We assess the impact of climate change on the Chilean small pelagic fishery through two channels: (i) a *direct weather channel*, in which projected changes in wind speed alter the number of bad-weather days that constrain fishing operations and thus enter the negative binomial trip equation;

²The wind threshold of 8 m/s corresponds approximately to Beaufort scale 5, at which purse-seine operations become difficult for artisanal vessels. We selected this threshold via AIC comparison across 8, 10, and 12 m/s cut-offs on the trip-equation fit.

and (ii) an *indirect biomass channel*, in which projected changes in SST and CHL modify each stock’s intrinsic productivity $r_{i,t}$ through the climate shifter $(\rho_i^{SST}, \rho_i^{CHL})$ of Eq. (2) and, in turn, the law of motion in Eq. (1).

Climate projections are derived from a six-model CMIP6 ensemble—IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, and MPI-ESM1-2-HR—under two Shared Socioeconomic Pathways: SSP2-4.5 (moderate emissions) and SSP5-8.5 (high emissions). The ensemble was selected to span the CMIP6 distribution of equilibrium climate sensitivity. We apply the delta method (Burke et al., 2015; Free et al., 2019) independently to each model, which preserves the observed interannual variability and fine-scale spatial structure from satellite records while imposing the climate-change signal from CMIP6. For each model and variable, the delta is computed as the difference (additive for SST and wind speed) or, for chlorophyll-a, the difference of natural logarithms (so that the resulting $\Delta \log \text{CHL}$ enters the shifter on its native estimation scale), between the CMIP6 future climatology and a spliced historical baseline that combines the CMIP6 historical run for 2000–2014 with the SSP2-4.5 run for 2015–2024 (CMIP6 historical runs end in 2014, so the splice is required to match the 2000–2024 estimation window of the structural model). For CESM2, which lacks chlorophyll under SSP2-4.5 in the NCAR CMIP6 catalogue, we splice the historical 2000–2014 segment with the SSP5-8.5 2015–2024 segment for chlorophyll only; the two scenarios are virtually indistinguishable in the near term and the substitution preserves CESM2’s contribution to SSP5-8.5 chlorophyll deltas. We consider two future time windows: mid-century (2041–2060) and end-of-century (2081–2100). Comparative statics are then reported integrating over both the structural-parameter posterior and the cross-model dimension of the ensemble; the variance decomposition of the resulting projection uncertainty is reported in Appendix F for stock-level intrinsic productivity and in Appendix G for fleet-level trips.

For the direct weather channel, monthly wind speed deltas from CMIP6 are applied additively to the daily wind speed observations at each vessel’s center of gravity (COG) over 2013–2024, and bad-weather days (wind speed exceeding 8 m/s) are recounted vessel by vessel under each scenario; the resulting Δdays_{bw} enters the NB trip equation through the estimated semi-elasticity β_{weather} , retaining vessel-level heterogeneity in baseline weather exposure. Although the cross-model median CMIP6 wind anomaly is modest (below +0.5 m/s under SSP5-8.5 end-of-century), the historical daily wind distribution at the COG sits sufficiently close to the 8 m/s threshold that the cross-vessel median Δdays_{bw} rises by +10 days under SSP2-4.5 mid-century and +23.5 days under SSP5-8.5 end-of-century. Note that u_{as} and v_{as} are missing from the CESM2 archive in the CMIP6 DRS for this period, so the direct channel is computed from the five remaining ensemble members (CNRM-ESM2-1, GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, UKESM1-0-LL) while the indirect biomass channel retains the full six-member ensemble where SST and chlorophyll deltas are available. For the indirect biomass channel, we evaluate the climate shifter of Eq. (2) at the posterior draws of $(\rho_i^{SST}, \rho_i^{CHL})$ from the full state-space specification, taking ΔSST and $\Delta \log CHL$ as the scenario- and window-specific CMIP6 anomalies relative to the 2000–2024 estimation baseline. This yields a full posterior distribution over $r_{i,t}^*/r_i^0$ under each climate regime—

a long-run comparative-statics object, in the sense that it describes the steady-state productivity of each stock under a counterfactual climate, not a year-by-year forecast of biomass. Forward simulation of biomass and catch trajectories under endogenous harvest rules is left to future work; the posterior reported here is archived to facilitate that extension.

Our framework is positioned as long-run comparative statics, not a year-by-year forecast, in line with the bioeconomic-projection tradition of Kasperski (2015). Because the empirical models are estimated on interannual variability rather than decadal trends, the projections capture the mechanical effect of changed environmental and weather conditions on stock productivity and fleet-level effort, while holding prices, quotas, and adaptive behaviour by fishers and managers fixed at historical levels (Auffhammer, 2018). They provide a benchmark for the direction and magnitude of climate impacts under a fixed-institution baseline; relaxations that endogenize price adjustment, quota revision, or vessel-level adaptation are signposted as natural extensions of this posterior.

4 Results

4.1 Identification of climate shifters

The central econometric object of this paper is the vector of stock-specific *climate shifters* $\{\rho_i^{SST}, \rho_i^{CHL}\}_{i=1}^3$, which modulate the intrinsic growth rate of each stock as

$$r_{i,t} = r_i^0 \exp(\rho_i^{SST} (SST_{t-1} - \overline{SST}) + \rho_i^{CHL} (\log CHL_{t-1} - \overline{\log CHL})).$$

These shifters admit a clean economic reading: ρ_i^{SST} is the semi-elasticity of the stock's intrinsic productivity r_i with respect to a sea-surface temperature anomaly of one degree Celsius, and similarly ρ_i^{CHL} is the semi-elasticity with respect to a log-point of chlorophyll-a anomaly (a common proxy for primary productivity at the base of the food web). They are the structural parameters that couple the biological law of motion $B_{i,t+1} = B_{i,t} + g_i(B_{i,t}, X_t) - C_{i,t} + \varepsilon_{i,t}$ to the physical climate, and are therefore the quantities that must be identified for the long-run projections of Section 4.3 to have meaning as *comparative statics under climate change* rather than as out-of-sample forecasts.

We adopt weakly informative priors on $\rho_i^{SST}, \rho_i^{CHL}$ derived from independent reduced-form stress tests (Appendix A) and let the 2000–2024 biomass series update them. Table 2 reports the posterior means, 90% credible intervals, and the posterior-to-prior standard deviation ratio, which measures how much the data refine each parameter relative to the prior.

The pattern of identification is itself informative and organises the economic interpretation of the paper. Three facts stand out.

First, for the two coastal upwelling stocks—anchoveta and sardina común—both climate shifters are identified with substantial precision. The posterior 90% credible intervals exclude zero for all

Table 2: Posterior identification of climate shifters.

Shifter	Stock	Prior $\mathcal{N}(\mu, \sigma)$	Posterior mean	Posterior 90% CI	$\sigma_{\text{post}}/\sigma_{\text{prior}}$	$\hat{R}$
ρ^{SST}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-1.06	[-1.79, -0.38]	0.43	1.000
ρ^{SST}	Sardina comun CS	$\mathcal{N}(-2.0, 1.0)$	-2.75	[-3.46, -2.02]	0.44	1.000
ρ^{SST}	Jurel CS	$\mathcal{N}(0.0, 1.0)$	0.42	[-1.25, 2.09]	1.01	1.000
ρ^{CHL}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-3.64	[-5.01, -2.25]	0.83	1.000
ρ^{CHL}	Sardina comun CS	$\mathcal{N}(2.1, 1.0)$	2.17	[0.96, 3.37]	0.74	1.000
ρ^{CHL}	Jurel CS	$\mathcal{N}(0.0, 1.0)$	-0.06	[-1.72, 1.60]	1.01	1.001
ρ^{ENSO}	Jurel CS	$\mathcal{N}(0.0, 0.5)$	-0.02	[-0.81, 0.80]	0.98	1.000

Note:

Shifters are the semi-elasticities of the intrinsic growth rate r_i with respect to SST anomalies (in °C), log-CHL anomalies, and the basin-scale ENSO Niño 3.4 index, respectively. The ENSO row reports the lag-1 refit of Appendix E.6, in which jack mackerel productivity is forced by the basin-scale index in place of local coastal shifters; anchoveta and sardine are not refit and inherit their main-specification posteriors. Priors on coastal shifters are adopted from the stock-specific stress-test results reported in Appendix A; the ENSO prior $\mathcal{N}(0, 0.5)$ is deliberately tighter (see Appendix E.6). The posterior-to-prior standard-deviation ratio measures prior-to-posterior updating: values near one indicate the data carry little information, whereas values well below one indicate identification from the likelihood. All $\hat{R}$ ≤ 1.01 .

four parameters, and the standard deviation ratios fall between 0.43 and 0.83, indicating that the 2000–2024 biomass series refines the priors materially. For anchoveta, a one-degree positive SST anomaly is associated with a semi-elasticity of $\rho_{\text{anch}}^{SST} = -1.06$ (CI [-1.79, -0.38]); for sardina común, the corresponding semi-elasticity is more than twice as large, $\rho_{\text{sard}}^{SST} = -2.75$ (CI [-3.46, -2.02]). Evaluated at one historical standard deviation of the SST anomaly series (estimation-sample $\hat{\sigma}_{SST} = 0.26$ °C over 2000–2024), a thermal shock of that magnitude lowers the intrinsic growth rate of anchoveta by approximately 24% ($1 - e^{-1.06 \cdot 0.26}$) and that of sardina by approximately 51% ($1 - e^{-2.75 \cdot 0.26}$) relative to the climatological mean. Under a policy-relevant warming signal of +1 °C, comparable to mid-century CMIP6 SSP2-4.5 projections for the Chilean coastal zone, the implied declines are 65% for anchoveta and 94% for sardina, although +1 °C lies approximately four historical standard deviations outside the estimation window and these projections therefore rely on the log-linear extrapolation embedded in the shifter function. Either way, the magnitudes are economically first-order for a fishery whose Centro-Sur TAC is in the low hundred-thousand-tonne range. The narrow historical SST variance ($\hat{\sigma}_{SST} = 0.26$ °C) also explains, mechanically, why the one-step-ahead forecasting test of Appendix B fails to discriminate between the climate-augmented and the climate-free specifications: within the estimation window the annual climate signal is small relative to the stochastic process innovation σ_{proc} , yet accumulates to economically consequential levels only over decadal horizons.

Second, the two coastal stocks respond to chlorophyll-a with **opposite signs**: $\rho_{\text{anch}}^{CHL} = -3.64$ (CI [-5.01, -2.25]) versus $\rho_{\text{sard}}^{CHL} = +2.17$ (CI [+0.96, +3.37]). At one historical standard deviation of log CHL ($\hat{\sigma}_{\log CHL} = 0.095$, a primary-productivity anomaly of roughly 10%), a positive CHL shock lowers anchoveta growth by approximately 29% while raising sardina growth by approximately

23%. This is a reduced-form footprint of the well-documented trophic asymmetry between the two species, summarised here for the non-specialist reader. Chlorophyll-a is a satellite-observable proxy for phytoplankton biomass, the primary producers at the base of the marine food web. Pulses of high chlorophyll-a indicate periods of strong primary productivity, typically driven by coastal upwelling that brings nutrient-rich deep water to the surface. *Sardina común* feeds directly on small phytoplankton and benefits from these pulses, so its productivity rises with chlorophyll-a anomalies. Anchoveta, by contrast, feeds on larger zooplankton whose populations are themselves preyed upon by small phytoplankton-feeders; when the food web is dominated by small primary producers, the zooplankton biomass that anchoveta relies on is suppressed, and anchoveta productivity falls. The opposite-signed semi-elasticities of ρ_{anch}^{CHL} and ρ_{sard}^{CHL} identified in Table 2 are consistent with this trophic asymmetry and are not a statistical artefact. The economic implication is first-order for this paper: the artisanal fleet, which lands both species, is not simply “climate-exposed”; it is exposed to **within-target heterogeneity** in the sign of climate effects. A warming-and-greening regime therefore does not translate into a uniform shock on the artisanal sector, but into a substitution across species that the current quota-allocation rules, indexed by species, cannot fully absorb.

Third, and in marked contrast, **neither SST nor CHL shifters are identified for jack mackerel**. The posterior standard deviations σ_{post} are indistinguishable from the prior scale ($\sigma_{\text{post}}/\sigma_{\text{prior}} \approx 1$), the posterior means lie near zero, and the credible intervals straddle zero in both directions. This is the expected result under the null hypothesis that jack mackerel productivity at the Centro-Sur scale is not locally coupled to the Chilean coastal upwelling regime—which is biologically sensible given its transboundary Southeast Pacific stock structure managed through SPRFMO. Econometrically, the model correctly reports non-identification rather than producing a spurious coefficient, which is a strength of the Bayesian state-space approach with structurally informed priors: the data can either move the prior or, as here, confirm that they are silent about it. This is the econometric dividend of Cowles-style structural identification in fishery climate-impact assessment: the structural posterior reports the absence of local climate signal where a reduced-form benchmark would have produced an artefactually precise but structurally meaningless coefficient, which would then propagate into climate projections as if it were information.

The non-identification result for jack mackerel is robust to the choice of spatial domain, biomass record, and forcing modality. Appendix E reports five complementary tests of this null. First, refits of the full specification under three nested coastal domains (Centro-Sur EEZ, offshore-extended, regional Southeast Pacific) leave the posterior-to-prior standard-deviation ratio of $(\rho_{\text{jurel}}^{SST}, \rho_{\text{jurel}}^{CHL})$ in 0.998–1.014. Second, augmenting the biomass record with a Northern Chilean acoustic series under a common elasticity moves the ratio only to 0.94–0.99, ruling out a sample-size explanation. Third, replacing the local shifters with the basin-scale ENSO Niño 3.4 index—the climatic driver identified by the qualitative literature on this transboundary species (Arcos et al. (2001); Peña-Torres et al. (2017))—returns a posterior-to-prior ratio of 0.98 at lag one and 1.01 at lag two (Table 2, ENSO row), even though the in-sample dispersion of the basin-scale shifter is approximately twice that of the coastal counterpart and the prior is deliberately tighter (Appendix E.1). Fourth, a joint

specification with all three shifters active for jack mackerel simultaneously returns ratios near unity for all three coefficients (1.03 on local SST, 1.00 on local log CHL, 0.98 on basin-scale ENSO), a sensibility check that addresses the standard referee question. Fifth, an attempt to incorporate the SPRFMO regional assessment series fails on coherence grounds, indicating the broader geographic aggregate is not integrable with the Centro-Sur record under a single shifter. The convergence of these five tests rules out the conjecture that the non-identification is an artefact of spatial aggregation, sample size, mis-specified forcing scale, or joint-specification convention; the relevant climate forcing for the Centro-Sur jack mackerel stock operates at margins of behaviour, location, and phenology that are not captured in the elasticity of structural productivity at the annual aggregation considered here, consistent with its transboundary SPRFMO management.

Figure 3 visualises the posterior–prior updating as a forest plot.

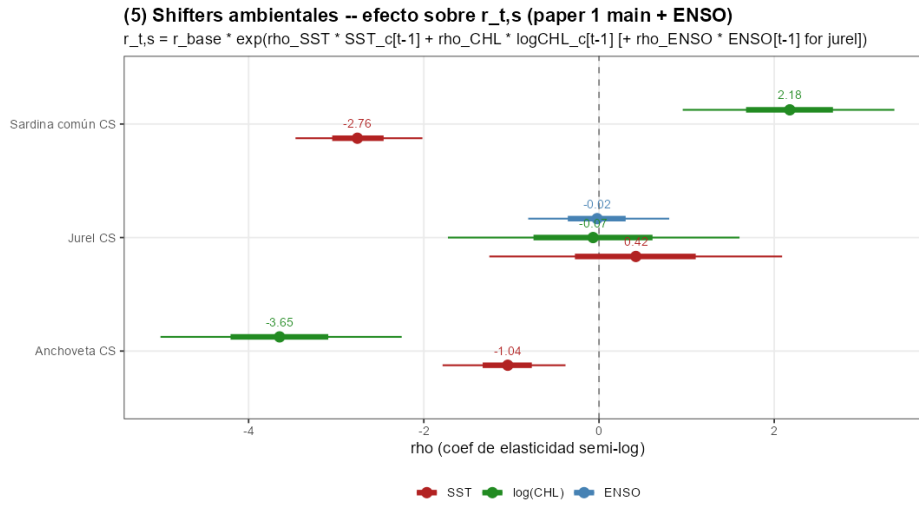


Figure 3: Posterior and prior distributions of the climate shifters $\rho_i^{SST}, \rho_i^{CHL}$ by stock. Anchoveta and sardina común show substantial posterior updating with narrow credible intervals excluding zero; jack mackerel posteriors coincide with the weakly informative prior centred at zero.

4.1.1 Identification versus short-run prediction

The identified shifters are structural parameters rather than reduced-form correlations: they measure an invariant biological mechanism linking ocean state to stock productivity, and they enter the law of motion as coefficients on exogenous climate inputs. This structural reading is what licenses their use as projection inputs under climate regimes outside the 2000–2024 estimation sample; reduced-form correlations, by contrast, would require the joint distribution of (SST, B, C) to remain stationary into the projection horizon, which is precisely the assumption the climate scenarios are designed to violate. Their economic value lies in **long-run comparative statics**: given a projected climate regime X_t^* for the period 2040–2100 drawn from the CMIP6 ensemble, the intrinsic productivity under that regime is $r_i^* = r_i^0 \exp(\rho_i^{SST} \Delta SST_t + \rho_i^{CHL} \Delta \log CHL_t)$, and

the expected steady-state biomass, harvest, and fleet-level effort follow mechanically. This is the channel through which the paper translates physical climate change into distributional outcomes in Section 4.3.

Crucially, identification of ρ does **not** imply superior short-run predictive power. A leave-future-out cross-validation at the one-year horizon (Appendix B) shows that the fully climate-augmented specification improves one-step-ahead log predictive density by only $\Delta\widehat{\text{ELPD}}_{\text{LFO}} \approx 0.4$ over a climate-free Schaefer baseline, a statistically unresolved difference. This is not a failure; it is precisely what should occur when the process-noise variance σ_{proc} is large relative to year-to-year climate innovations. Annual climate variability contributes little to *next-year nowcasts* but accumulates systematically over decadal horizons, and the shifters ρ_i are the parameters that quantify that accumulation. A robustness comparison using PSIS-LOO is reported in Appendix B (Table 11) and confirms that the full specification is preferred as an in-sample description of the dynamics ($\Delta\widehat{\text{ELPD}}_{\text{LOO}} = 14.59$, $\text{SE} = 4.93$); the LOO and LFO results are complementary diagnostics of, respectively, identification-through-fit and forecasting-on-new-data, and the paper relies on the former for the economic interpretation.

4.1.2 Posterior-predictive adequacy

A stock-by-stock comparison of the smoothed latent biomass $B_{i,t}$ against the observed SSB series (Figure 8) shows that the identified model reproduces the historical trajectory within a median 90% posterior band width of approximately 20% of the mean. Residuals are roughly Gaussian and contain no detectable autocorrelation at one-year lag (Figure 9). Posterior-predictive diagnostics are reported in Appendix C; Markov-chain convergence diagnostics for all top-level parameters are reported in Appendix D.

4.2 Total annual trips

Table 3 reports the negative binomial estimates separately for each fleet segment. The specification includes year fixed effects, which absorb aggregate shocks affecting both fleets contemporaneously—most notably the 2019 social outbreak (*estallido social*) and the 2020–2022 COVID-19 period—and identify the structural coefficients β_H (allocated harvest) and β_{weather} (bad-weather days) from within-year cross-vessel variation. Without year fixed effects, $\hat{\beta}_{\text{weather}}^{\text{ART}}$ is approximately twice as large in magnitude, reflecting the contemporaneous correlation between high-wind years and the social/pandemic shocks. The year-FE specification is therefore the basis for the climate projections in Section 4.3; the relative trip projection $T_v^{\text{FUT}}/T_v^{\text{HIST}}$ does not depend on the year-FE values themselves, which cancel in the ratio.

The coefficients on species-specific allocated harvest reveal the heterogeneity in fleet response that the legacy scalar specification masked. For the industrial fleet, the sardine coefficient is positive and highly significant ($p < 0.001$); the anchoveta coefficient is negative but not statistically

Table 3: Negative binomial estimates of annual fishing trips (2013–2024, $N_{IND} = 319$, $N_{ART} = 4201$)

	<i>Dependent variable:</i>	
	Industrial	Artisanal
	(1)	(2)
Hold capacity (log m ³)	0.097 (0.221)	0.132*** (0.046)
Allocated harvest anchoveta ($H_{vy,anch}^{alloc}$, tons)	−0.0003 (0.0002)	0.016*** (0.003)
Allocated harvest sardine ($H_{vy,sard}^{alloc}$, tons)	0.0001*** (0.00003)	0.001*** (0.00004)
Allocated harvest jack mackerel ($H_{vy,jur}^{alloc}$, tons)	0.00002* (0.00001)	0.001 (0.001)
Price anchoveta (1000s \$/ton)	−0.004 (0.004)	0.011*** (0.003)
Price sardine (1000s \$/ton)	−0.016*** (0.003)	0.003** (0.002)
Price jack mackerel (1000s \$/ton)	0.008*** (0.003)	0.007*** (0.002)
Bad weather days	−0.0003 (0.001)	−0.001 (0.0005)
Veda days	−0.079*** (0.005)	0.013*** (0.002)
Vessel type (UNK)	−1.450*** (0.163)	0.160 (0.099)
Vessel type (BR)		−0.329*** (0.079)
Vessel type (BRV)		−0.774*** (0.079)
Vessel type (L)		−2.212*** (0.377)
Vessel type (LM)		0.289*** (0.098)
$H_{vy,anch}^{alloc} \times \text{TIPO_EMB (BR)}$		−0.021*** (0.008)
$H_{vy,anch}^{alloc} \times \text{TIPO_EMB (BRV)}$		−66.546*** (13.119)
$H_{vy,anch}^{alloc} \times \text{TIPO_EMB (L)}$		501.903 (501.942)
$H_{vy,anch}^{alloc} \times \text{TIPO_EMB (LM)}$		−0.017*** (0.003)
$H_{vy,anch}^{alloc} \times \text{TIPO_EMB (UNK)}$		−0.013*** (0.003)
Constant	15.763*** (2.605)	−3.697*** (0.519)
Year fixed effects	Yes	Yes
Pseudo- R^2 (Cragg-Uhler)	0.302	0.631
Observations	319	4,201

Note: *p<0.1; **p<0.05; ***p<0.01.

Vessel-clustered robust SEs. Prices in constant 2018 thousands of pesos/ton, merged at (*year, vessel, region*) resolution with fallback to year-level national mean when the regional cell is missing.

Allocated harvest is species-specific: $H_{vy,s}^{alloc} = \omega_{vs}^{hist} \bar{Q}_{sy}^{r(v)}$ following Kasperski (2015) Eq. 17. The artisanal regression interacts $H_{vy,anchoveta}^{alloc}$ with TIPO_EMB; TIPO_EMB categories with $N < 50$ vessel-years (BR, BRV, L) are retained in the fit for completeness but excluded from the aggregate factor_trips projections.

Year fixed effects absorb aggregate shocks (estallido social 2019, COVID 2020–2022).

LR test NB vs Poisson: Industrial $\chi^2 = 397.6$; Artisanal $\chi^2 = 37046.9$ (both $p < 0.001$).

distinguishable from zero; and the jack mackerel coefficient is positive and marginally significant ($p \approx 0.05$). For the artisanal fleet, sardine remains the dominant identified channel ($\hat{\beta}_h^{\text{sard}} = 6.65 \times 10^{-4}$, $p < 0.001$). The anchoveta response is heterogeneous across vessel types: in the reference category BM (bote motor, 8% of the panel) the coefficient is large and positive ($\hat{\beta}_h^{\text{anch,BM}} = +0.0164$, $p < 0.001$), consistent with the classical Kasperski (2015) interpretation that more quota implies more trips; in LM (lancha motor, 77% of the panel) the coefficient collapses to near zero through the interaction term and ceases to be economically meaningful, indicating that the dominant artisanal segment is structurally driven by sardine rather than anchoveta. The legacy specification, which constrains the artisanal anchoveta response to a single scalar, recovers a small negative coefficient that is a weighted average of these two regimes biased toward zero by the LM weight; an information criterion comparison favours the interacted specification by $\Delta\text{AIC} = -167.9$. These coefficients provide the linkage between the trip equation and the comparative-statics projections of Section 4.3: changes in regional TACs translate into vessel-level effort adjustments through the species-specific quota-share mechanism, with the artisanal response decomposed by vessel type.

The two fleets differ markedly in price responsiveness. For the industrial fleet, the jack mackerel price is positive and significant ($p < 0.01$), consistent with this fleet’s primary orientation toward jack mackerel. Sardine and anchoveta prices have no detectable effect on industrial effort, reflecting the limited participation of industrial vessels in the sardine–anchoveta fishery. For the artisanal fleet, the sardine price is positive and significant ($p < 0.01$), consistent with sardine being the dominant target species for artisanal vessels in the Centro-Sur region. The anchoveta-price coefficient is negative and significant; this counterintuitive sign likely reflects simultaneity: years of low anchoveta availability generate both higher prices (through the inverse-demand channel) and fewer trips (through reduced catch opportunities), producing a negative reduced-form correlation. As noted above, the purpose of this equation is not causal identification but rather to provide an empirically grounded mapping for the simulation, in which prices are determined endogenously by the inverse-demand module.

Hold capacity is not significant for the industrial fleet, where vessels are relatively homogeneous in scale, but it is positive and significant for the artisanal fleet. Among artisanal vessels, larger hold capacity is associated with more annual trips, consistent with larger vessels being more commercially active and better able to sustain operations across varying conditions.

The environmental and regulatory variables show distinct patterns across fleets. For the artisanal fleet, adverse weather days carry a negative and statistically distinguishable coefficient, indicating that, controlling for year-specific aggregate conditions, vessels operating in grids with more days of wind speeds above 8 m/s undertake fewer fishing trips. For the industrial fleet, adverse weather is not significant, consistent with larger vessels being better equipped to operate under rough conditions. Biological closure days are strongly negative for the industrial fleet, representing the largest effect in the model, consistent with the industrial fleet’s dependence on access during open seasons. For the artisanal fleet, the positive coefficient on closure days does not admit a causal

interpretation: under the year-fixed-effects specification, β_{closure} is identified solely from the cross-zone differential between regions V-VIII (151 days) and IX-XIV (182 days) of the proxy variable described in Appendix A and discussed at length in the regulatory-history caveat of Section 5. The positive sign therefore reflects locational heterogeneity in artisanal fishing intensity between the two regulatory zones (the southern zone IX-XIV contains both more nominal closure days and more concentrated artisanal effort per vessel, and the cross-sectional comparison cannot disentangle the two), not the causal effect of closures on trips. Constructing a year-by-year closure variable with within-zone temporal variation that would identify the causal effect requires reconciling the nominal regulatory ceilings of the D.Ex. 115/1998 and 530/2016 chain with the actual realised closure days reported in the SUBPESCA Indicadores Biológicos portal from 2019 onwards, a non-trivial data-engineering task discussed in Section 5 and reserved for follow-up work.

Vessel-type dummies capture residual heterogeneity in fleet composition. Among artisanal vessels, larger vessel categories (L, BRV) are associated with substantially fewer trips relative to the baseline, reflecting differences in trip duration and operational patterns across vessel classes.

4.3 Climate change projections

The projection methodology is summarized in Section 3.4. Under each scenario and time window, we translate CMIP6-derived environmental deltas into (i) changes in bad-weather days that enter the trip equation (direct channel) and (ii) posterior draws of the stock-specific effective growth rate $r_{i,t}^*$, obtained by evaluating the identified climate shifter of Eq. (2) at the projected SST and log-CHL anomalies (indirect channel).

4.3.1 Projected environmental changes

Table 4 summarizes the projected environmental changes for the Centro-Sur study area, reported as cross-model medians with the inter-model interquartile range (IQR) in brackets. Chlorophyll-a is reported as a percentage change, $\% \Delta \text{CHL} = (\exp(\Delta \log \text{CHL}) - 1) \times 100$, computed per model and then summarised across the ensemble; the structural model is fitted on $\Delta \log \text{CHL}$ and that scale is used in all downstream calculations (Section 3.4). Sea-surface temperature warms by a cross-model median of +0.7 to +2.4 °C depending on the scenario, with the strongest warming under SSP5-8.5 end-of-century and an inter-model spread that grows with the forcing. The chlorophyll-a signal is small in magnitude relative to the cross-model spread: the cross-model median of $\% \Delta \text{CHL}$ is close to zero in every scenario-window pair while individual models disagree on the sign, reflecting the well-documented inter-model disagreement on primary-productivity changes under climate forcing in CMIP6. Near-surface wind speed increases modestly under all scenarios (cross-model median +0.1 to +0.4 m/s), with a larger increase under SSP5-8.5 end-of-century than the single-model IPSL projection on which an earlier version of this paper was based; the inter-model IQR remains

comparable to the median. While these mean wind changes are small, they translate into a substantial increase in the number of bad-weather days exceeding the 8 m/s operability threshold (cross-vessel median +10 to +23.5 days/year, see Section 3.4) because the historical daily wind distribution at vessel COGs sits sufficiently close to the threshold that a small upward shift in the mean displaces a non-negligible mass of the upper tail past it. The implications for fleet-level trips are quantified in Table 6 and are first-order for the artisanal fleet.

Table 4: Projected environmental changes for Centro-Sur Chile (CMIP6 six-model ensemble, delta method; cross-model median with inter-model IQR in brackets).

SSP	Window	Δ SST ($^{\circ}\text{C}$)	Δ CHL (%)	Δ wind speed (m/s)
SSP2-4.5	2081–2100	+1.44 [+0.96, +1.91]	+0.0 [−3.6, +1.4]	+0.19 [+0.10, +0.24]
SSP2-4.5	2041–2060	+0.74 [+0.52, +0.92]	+2.2 [+0.0, +2.8]	+0.14 [+0.13, +0.15]
SSP5-8.5	2081–2100	+2.39 [+1.93, +3.29]	−1.3 [−7.4, +3.0]	+0.42 [+0.36, +0.45]
SSP5-8.5	2041–2060	+0.92 [+0.61, +1.23]	+0.7 [−2.3, +3.4]	+0.21 [+0.14, +0.32]

4.3.2 Comparative statics on intrinsic stock productivity

Table 5 reports the comparative-statics object $\Delta r_i^*/r_i^0$ obtained by evaluating the identified climate shifter at the CMIP6 ensemble deltas of Table 4 and integrating over the posterior of $(\rho_i^{SST}, \rho_i^{CHL})$ under the full state-space specification. For each scenario–window pair we compute the within-model posterior of $\Delta r_i^*/r_i^0$ for each CMIP6 model in turn and then summarise the resulting set of model-level distributions along two distinct axes: the cross-model median and inter-quartile range (IQR) of the model-level posterior medians, which captures inter-model climate uncertainty, and the median across models of the within-model 90% posterior credible interval, which captures structural uncertainty about $(\rho_i^{SST}, \rho_i^{CHL})$ holding the climate forcing fixed. The two intervals are reported separately because they have different policy implications and a separate variance decomposition is reported in Appendix F. Jack mackerel is reported as non-identified (“n.i.”) because its shifter posterior coincides with the weakly informative prior (Section 4.1), so that projecting its productivity under future climate regimes would merely propagate prior uncertainty rather than transmit evidence from the 2000–2024 sample. A structural extension that joins the Centro-Sur and SPRFMO assessments for jack mackerel, allowing the climate shifter to be identified at the basin scale, is left to future work.

The magnitudes are economically first-order for a Centro-Sur fishery whose combined TAC is in the low hundred-thousand-tonne range. Under the moderate SSP2-4.5 mid-century scenario (cross-model median +0.74 $^{\circ}\text{C}$; $\Delta \log \text{CHL}$ near zero in the cross-model median, with non-trivial inter-model spread), the cross-model median of anchoveta’s intrinsic productivity declines by 51% and sardina común’s by 79%, with posterior probability of negative impact equal to one in both cases. Under the high-emissions SSP5-8.5 end-of-century scenario (cross-model median +2.39 $^{\circ}\text{C}$, $\Delta \log \text{CHL}$ cross-model median −0.013 with 5th-percentile of −0.30 across models) the cross-model medians deepen to −90% and −99.9% respectively, and the entire 90% credible interval of sardine’s

Table 5: Comparative statics on intrinsic stock productivity under the CMIP6 six-model ensemble.

Stock	Scenario	n_m	ΔSST (°C)	$\Delta \log CHL$	$\Delta r_i^*/r_i^0$ (cross median)	Cross-model IQR	Within posterior 90% CI	$Pr(\Delta < 0)$
Anchoveta CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	-50.7%	[-52.7%, -47.8%]	[-68.0%, -27.0%]	1.00
Anchoveta CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	-71.0%	[-76.3%, -60.4%]	[-91.1%, -31.0%]	1.00
Anchoveta CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	-62.4%	[-68.0%, -55.9%]	[-80.9%, -31.2%]	1.00
Anchoveta CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	-89.6%	[-91.5%, -87.4%]	[-98.3%, -54.2%]	0.99
Sardine CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	-78.6%	[-93.2%, -72.5%]	[-85.7%, -67.3%]	1.00
Sardine CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	-97.2%	[-98.9%, -89.8%]	[-98.9%, -92.6%]	1.00
Sardine CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	-91.1%	[-96.7%, -78.5%]	[-95.1%, -83.2%]	1.00
Sardine CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	-99.9%	[-100.0%, -99.4%]	[-100.0%, -99.3%]	1.00
Jack mackerel CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	n.i.	n.i.	n.i.	n.i.

Note:

Entries summarise the posterior of the proportional change in intrinsic productivity, $\Delta r / r_0 = \exp(\rho_{SST} * \Delta SST + \rho_{CHL} * \Delta \log CHL) - 1$, evaluated at scenario-window-specific CMIP6 anomalies for each of n_m models in the ensemble and integrated over the posterior of (ρ_{SST}, ρ_{CHL}) from the full state-space specification. The cross-model IQR is the 25th-to-75th percentile range across CMIP6 models of the within-model posterior medians and captures inter-model climate uncertainty. The within posterior 90% CI is the cross-model median of the within-model 5th-to-95th-percentile posterior interval and captures structural uncertainty about (ρ_{SST}, ρ_{CHL}) holding the climate forcing fixed. Jurel (jack mackerel) CS is reported as non-identified ('n.i.') because the posterior of its shifters coincides with the weakly informative prior; see the identification discussion in Results. $n_m = 5$ instead of 6 in the SSP2-4.5 panels because CESM2 does not publish chl-ssp245 in the NCAR CMIP6 catalogue. Under SSP5-8.5 end-of-century the implied SST anomaly lies approximately nine in-sample standard deviations outside the realised 2000-2024 envelope, so it is best read as a structural-form stress test under the maintained log-linear shifter. Raw numerical values are archived in tables/growth_comparative_statics_raw.csv; model-level disaggregation is in tables/growth_comparative_statics_by_model.csv. Appendix F reports the variance decomposition between cross-model and within-model components.

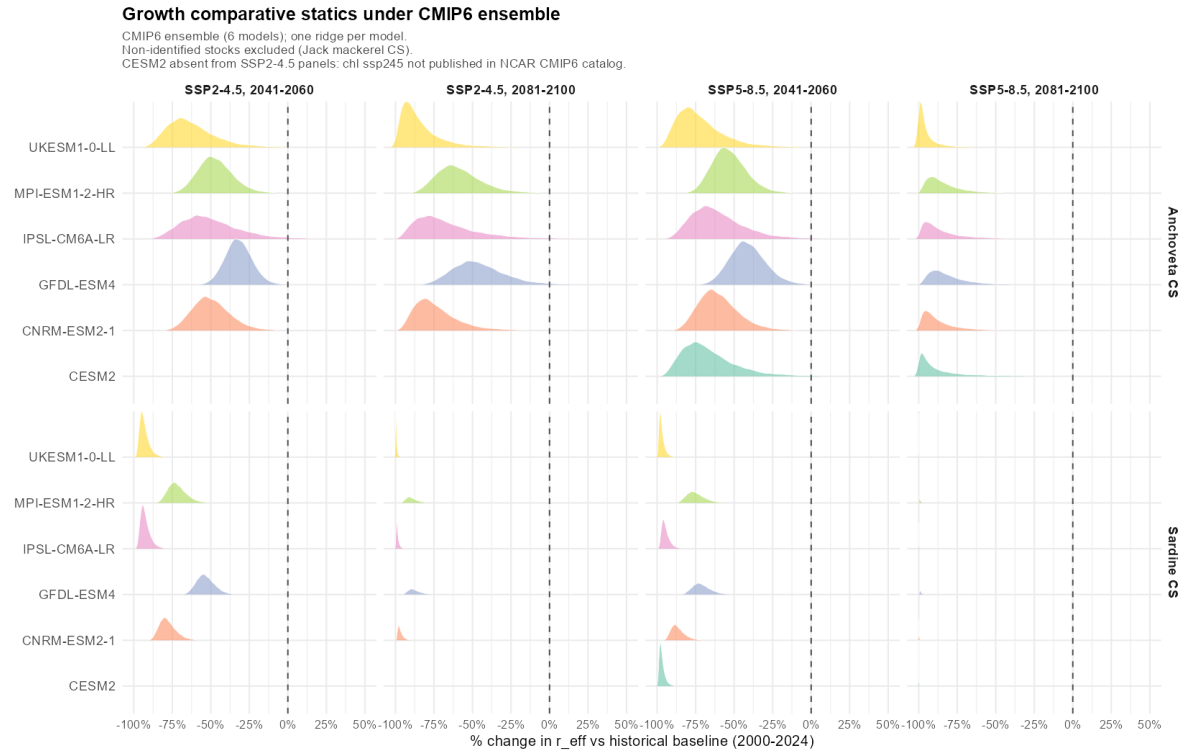


Figure 4: Posterior distribution of the proportional change in intrinsic stock productivity under CMIP6 scenarios, six-model ensemble. Ridges are kernel-smoothed marginal posteriors of the proportional change in intrinsic productivity derived from draws of the state-space full specification, with one ridge per CMIP6 model in each scenario-window pair; vertical line at zero. CESM2 is absent from the SSP2-4.5 panels because chl-ssp245 is not published in the NCAR CMIP6 catalogue. Jack mackerel is omitted because its climate shifter is not identified in the 2000–2024 sample.

response lies below zero in every model of the ensemble; the anchoveta cross-model probability of decline is 0.99 rather than 1 because under a small fraction of (model, draw) pairs a sufficiently negative $\Delta \log \text{CHL}$ combined with $\rho_{\text{anch}}^{\text{CHL}} < 0$ offsets the SST-driven collapse. The identified signs confirm the qualitative reading of the shifters in Section 4.1: both coastal stocks respond negatively to SST, but sardina común carries roughly twice the SST semi-elasticity of anchoveta, and its CHL semi-elasticity is of the opposite sign, so that the compound response—negative thermal shock combined with negative primary-productivity shock—is sharper for sardina than for anchoveta.

The comparative-statics exercise is a long-run object rather than a year-by-year forecast. It describes the steady-state intrinsic productivity of each stock under a counterfactual climate, *holding fixed* the law of motion and the fishery’s regulatory regime. The shifters ($\rho_i^{\text{SST}}, \rho_i^{\text{CHL}}$) enter the law of motion as coefficients on exogenous climate forcings, which makes them invariant to the joint distribution of (SST, B, C) and therefore evaluable at climate regimes outside the estimation window—in particular, at an SST anomaly of $+2.3^\circ\text{C}$ that lies roughly nine historical standard deviations beyond the 2000–2024 mean. A purely correlational specification would not license this extrapolation.

4.3.3 Implications for fleet-level effort

Table 6 translates the comparative-statics object on intrinsic stock productivity in Table 5 into a comparative-statics object on fleet-level annual trips, by propagating each posterior draw through the stock-dynamics steady-state equation under historical average fishing pressure and through the negative binomial trip equation of Section 4.2. Vessel-level responses are aggregated to the fleet level as medians over posterior draws and over vessels within fleet, weighted by each vessel’s realized catch composition over the 2012–2024 estimation window.

The posterior probability of portfolio loss—defined as a reduction of more than fifty percent in the composition-weighted biomass factor f_v^H relative to the 2000–2024 historical baseline—exceeds 0.95 for the artisanal fleet under every CMIP6 scenario considered and reaches 0.99 under SSP5-8.5 end-of-century. For the industrial fleet, the loss probability is a stable 0.12 across all four scenarios, driven entirely by its five-percent exposure to sardina común; the remaining 95% of its historical portfolio is allocated to jack mackerel, whose climate shifter is treated as non-identified in Section 4.1. The asymmetry in loss probability between the two fleets is first-order and stable across climate pathways, approximately eight-fold across all four scenarios.

Decomposing the fleet-level trip response isolates the channels through which this asymmetry operates. The *marginal* posterior median of $\% \Delta$ trips, computed over all posterior draws, ranges from -13.2% (SSP2-4.5 mid) to -15.0% (SSP5-8.5 end) for the artisanal fleet and from -0.8% to -1.0% for the industrial fleet across the same scenarios—a ratio of roughly fifteen to one at end-of-century. The *conditional* posterior median, computed over the subset of posterior draws in which $f_v^H \geq 0.5$, isolates the climate-elasticity component from the portfolio-collapse component and ranges from

Table 6: Long-run implications of CMIP6 climate scenarios for fleet-level fishing effort, under a Schaefer steady-state thought experiment with fishing pressure fixed at the 2000–2024 historical average. Entries are cross-model medians with cross-model IQRs in brackets, aggregated within each CMIP6 model over posterior draws and vessels within fleet.

Fleet	Scenario	$\Pr(f_v^H < 0.5)$	% Δ trips (marginal)	% Δ trips (conditional)
Artisanal	SSP2-4.5, 2041-2060	0.95 [0.93, 0.96]	-13.2% [-13.3%, -12.2%]	-4.6% [-4.8%, -4.0%]
Artisanal	SSP2-4.5, 2081-2100	0.98 [0.97, 0.99]	-14.1% [-14.3%, -13.7%]	-3.9% [-4.5%, -3.4%]
Artisanal	SSP5-8.5, 2041-2060	0.98 [0.96, 0.98]	-13.6% [-13.8%, -13.4%]	-4.2% [-4.5%, -4.1%]
Artisanal	SSP5-8.5, 2081-2100	0.99 [0.99, 0.99]	-15.0% [-15.2%, -14.9%]	-3.8% [-5.5%, -2.6%]
Industrial	SSP2-4.5, 2041-2060	0.12 [0.11, 0.12]	-1.0% [-1.0%, -0.8%]	-0.7% [-0.7%, -0.7%]
Industrial	SSP2-4.5, 2081-2100	0.12 [0.12, 0.12]	-0.9% [-1.0%, -0.8%]	-0.7% [-0.8%, -0.7%]
Industrial	SSP5-8.5, 2041-2060	0.12 [0.12, 0.12]	-0.8% [-0.9%, -0.8%]	-0.7% [-0.9%, -0.7%]
Industrial	SSP5-8.5, 2081-2100	0.12 [0.12, 0.12]	-1.0% [-1.1%, -1.0%]	-1.0% [-1.0%, -0.9%]

Note:

Entries summarise the posterior of the proportional change in fleet-level annual trips, computed for each of the six CMIP6 models in the ensemble and then aggregated across models. Within each model, the posterior is integrated over draws of the full state-space specification and vessels within fleet (composition-weighted exposure). Cross-model medians and 25th-to-75th-percentile inter-quartile ranges across the n_m models are reported in brackets; the median across models of the within-model 5th-to-95th-percentile posterior interval (within posterior CI) is archived in tables/trip_comparative_statics_raw.csv. f_H is the composition-weighted biomass factor relative to the historical baseline: $f_H = \text{sum over species of } \omega_{vs} * B_{star_s} / B_{hist_s}$, where $B_{star_s} = K_s * (1 - F_{hist_s} / r_{eff_s})$ is the Schaefer steady-state biomass and ω_{vs} is the vessel's realized catch share in species s . $\Pr(f_H < 0.5)$ reports the posterior probability that the vessel's composition-weighted biomass factor falls below one-half. The marginal median integrates over all posterior draws within each model; the conditional median restricts to draws with $f_H \geq 0.5$, isolating the pure climate-elasticity component from the steady-state collapse component of the trip response. Jack mackerel is treated as non-identified ($\text{factor}_B = 1$) consistent with the identification discussion in the Results section. Raw numerical values, by-model disaggregation, and by-stock extinction probabilities are archived in tables/trip_comparative_statics_raw.csv, tables/trip_comparative_statics_by_model.csv, and tables/trip_comparative_statics_extinct.csv. The narrow cross-IQR observed in many cells reflects a floor effect: once Schaefer steady-state collapse saturates the portfolio-weighted biomass factor across models, the trip response converges to $\exp(-\beta * H_{alloc} * 1)$, independent of climate magnitude. Appendix G reports the formal two-way variance decomposition of the fleet-level trip response, which confirms that the within-model component (posterior plus vessel heterogeneity) dominates the total variance under saturated regimes; Appendix F reports the analogous decomposition for the underlying productivity response.

−3.8% to −4.6% for the artisanal fleet and from −0.7% to −1.0% for the industrial fleet, a conditional ratio of roughly four to one. Two complementary mechanisms generate this asymmetry. First, the indirect biomass channel transmits the climate signal asymmetrically through differences in portfolio composition: the artisanal fleet is concentrated in the two coastal-upwelling stocks whose climate shifters ($\rho_i^{SST}, \rho_i^{CHL}$) are both identified and economically large, while 95% of the industrial fleet's historical portfolio sits in jack mackerel, whose Centro-Sur shifter is not identified and is therefore held at the climatological mean in the projection. Second, the direct weather channel transmits the climate signal asymmetrically through differences in fleet-level weather sensitivity: the negative binomial trip equation estimates $\hat{\beta}_{\text{weather}}^{\text{ART}} = -0.0007$ (marginally significant, $p \approx 0.12$) and $\hat{\beta}_{\text{weather}}^{\text{IND}} = -0.0003$ (statistically indistinguishable from zero), so the projected +10 to +23.5 days of additional bad weather per year translate into a −0.7% to −1.9% contribution to

the artisanal trip response and a contribution close to zero for the industrial fleet, consistent with the size-dependent weather tolerance of the two segments. Both $\hat{\beta}_{\text{weather}}$ estimates are obtained from the negative binomial specification with year fixed effects (Section 4.2), which absorbs the aggregate shocks of the 2019 social outbreak and the 2020–2022 COVID period; without year fixed effects, $\hat{\beta}_{\text{weather}}^{\text{ART}}$ is approximately twice as large in magnitude, indicating that the contemporaneous correlation between high-wind years and the social/pandemic shocks would otherwise inflate the estimated weather elasticity. The limited cross-sector transferability of the prevailing sector-level allocation locks this dual exposure—portfolio composition under the indirect channel and weather sensitivity under the direct channel—into differential losses of harvest capacity, echoing the portfolio mechanism documented for U.S. West Coast and Alaskan fisheries by Kasperski & Holland (2013) and the regime-shift buffering documented by Cline et al. (2017): segments with narrower species portfolios face systematically higher exposure to species-specific shocks, climate-driven or otherwise, and institutional access constraints of the type imposed by a fixed sector-level allocation can limit the realized benefits of diversification (Oken et al., 2021).

The between-fleet asymmetry of approximately fifteen-to-one further decomposes into a substantial within-fleet heterogeneity inside the artisanal segment by vessel type (Table 7). Three TIPO_EMB categories carry the response. The dominant *lancha motor* (LM) segment—77% of the artisanal vessel-year panel—contracts by -19.9% (cross-model IQR $[-20.0\%, -19.7\%]$) under SSP5-8.5 end-of-century. LM vessels carry a near-zero anchoveta elasticity but a strongly identified positive sardine elasticity in the trip equation, so their projected effort response is dominated by the steady-state collapse of the sardine stock, whose long-run productivity factor falls by -99.9% under the same scenario. The smaller *bote motor* (BM) segment—8% of the panel—contracts by only -3.2% at the marginal posterior median, because its identified positive anchoveta elasticity ($\hat{\beta}_h^{\text{anch,BM}} = +0.016$, $p < 0.001$) partially offsets the sardine contraction in the high-mass region of the posterior. The intermediate UNK segment—14% of the panel, with mixed portfolio composition and TIPO_EMB identification gaps in the logbook—contracts by -7.6% . The probability of portfolio loss exceeds 0.98 in all three categories, indicating that within the artisanal segment the differential response is concentrated in the magnitude rather than the binary incidence of portfolio collapse. This within-fleet heterogeneity refines the between-fleet asymmetry rather than displacing it: the LM segment, being both the largest and the most exposed to the sardine-anchoveta climate signal, drives the aggregate artisanal contraction, while BM vessels behave more like a small industrial fleet under the climate-elasticity component.

Three caveats qualify this thought experiment. First, the Schaefer steady-state comparative statics holds fishing pressure fixed at the 2000–2024 historical average; any regulatory response that lowered F in the face of falling productivity would attenuate both the loss probability and the marginal trip response reported here. Quantifying that regulatory margin requires a forward simulation of the biomass process under endogenous harvest rules—most prominently the LTP cap on industrial quota transferability and the annual SUBPESCA revision of TACs in response to in-year biomass signals—which we leave to future work and which would reuse the posterior reported

Table 7: Within-artisanal-fleet decomposition of the long-run trip response by vessel type, under SSP5-8.5 end-of-century. Entries are cross-model medians with cross-model IQRs in brackets. The marginal column integrates over all posterior draws; the conditional column restricts to draws in which the composition-weighted biomass factor is at least one-half, isolating the climate-elasticity component from the steady-state collapse component.

	Segment	Pr(portfolio loss)	% Δ trips (marginal)	Cross-model IQR	% Δ trips (conditional)
LM	Lancha motor (LM)	0.99	-19.9%	[-20.0%, -19.7%]	-16.1%
UNK	Unidentified (UNK)	0.98	-7.6%	[-8.1%, -7.3%]	-1.6%
BM	Bote motor (BM)	1.00	-3.2%	[-3.3%, -3.0%]	+16.3%

Note:

Categories with fewer than 50 vessel-year observations (BR, BRV, L) collectively account for less than 0.1 percent of artisanal effort and quota and are excluded from this table. Vessel type BM is the reference category in the negative binomial interaction; LM and UNK enter as deviations. Raw cross-model values and other scenario-window combinations are archived in `trip_comparative_statics_by_tipo_emb_raw.csv`.

here. Second, the 2000–2024 observation window was not itself in Schaefer equilibrium under F_{hist} ; the pipeline’s internal consistency check, obtained by evaluating the comparative statics at $\Delta SST = 0$ and $\Delta \log CHL = 0$, yields a median f_v^H of approximately 1.015 rather than 1.000, reflecting a mild historical depletion from equilibrium. The scenario results in Table 6 should therefore be read as responses *relative to the baseline steady-state* rather than relative to the 2000–2024 observed trajectory. Third, the jack mackerel shifter is treated as non-identified. Two interpretive points should precede the empirical evidence. First, the result is a null on *local identification*, not on biological climate sensitivity: the available 25-year window provides only 16 informative annual observations of Centro-Sur spawning biomass against a historical SST envelope of $\hat{\sigma}_{SST} \approx 0.26^\circ\text{C}$, so any climate response of moderate magnitude operating at scales broader than the Centro-Sur biological domain would lie below the threshold of statistical detection in this sample regardless of its true magnitude. Second, complementary empirical evidence on jack mackerel documents that climate forcing does operate on the species at scales beyond local coastal anomalies: Peña-Torres et al. (2017) report ENSO-driven shifts in jack mackerel location choices in the Chilean fishery, and Arcos et al. (2001) describe SPRFMO-scale stock dynamics consistent with broad-basin forcing, so the absence of an identifiable Centro-Sur shifter should not be read as evidence of climate insensitivity. With these interpretive caveats noted, four converging tests reported in Appendix E converge on the same null and rule out the most natural mechanical explanations for it—insufficient spatial extent, insufficient sample size, poor source aggregation, and a poor choice of spatial scale for the climate shifter. Re-fitting the state-space on three nested spatial domains for the environmental aggregation (Centro-Sur EEZ, offshore-extended, regional Southeast Pacific) yields posterior-to-prior standard deviation ratios of 0.998–1.014 for both $\rho_{\text{jurel}}^{SST}$ and $\rho_{\text{jurel}}^{CHL}$, indistinguishable from unity in every domain (Section E.4). A dual-source extension that augments the likelihood with the Northern Chilean acoustic series under a common climate elasticity—an assumption motivated by the 0.88 log-scale correlation between the Centro-Sur and Northern series across overlapping years—reduces the ratios only to 0.94–0.99, well above the threshold for material identification

(Section E.5). Replacing the local coastal shifters with the basin-scale ENSO Niño 3.4 index, a test of whether the relevant climatic forcing for jack mackerel operates at the basin scale rather than the coastal scale, returns a posterior-to-prior standard deviation ratio of approximately 0.98 at lag one, indistinguishable from unity (Section E.6). A separate exploratory attempt to incorporate the SPRFMO range-wide assessment (OROP-PS, encompassing Chile, Peru, Ecuador and the adjacent high-seas area) fails on coherence grounds, with multimodal posterior support and unsatisfactory chain mixing, indicating that the broader geographic aggregate is not integrable with the Centro-Sur record under a single shifter. The convergence of these four tests indicates a genuinely flat shifter *within the Centro-Sur biological domain*, consistent with the most natural reading of the null in light of Peña-Torres et al. (2017) and Arcos et al. (2001): that the relevant climatic drivers for jack mackerel operate at spatial scales beyond those captured by local Chilean SST and chlorophyll anomalies. The industrial fleet’s apparent climate protection in Table 6 therefore rests on a null about local identification that cannot be sharpened within the Centro-Sur identification strategy, even with the dual-source extension; the appropriate setting in which to discriminate between local coastal decoupling and broader-scale climate forcing is a range-wide SPRFMO analysis with explicit cross-stock spillover, which we leave to future work.

5 Discussion

Our results reveal a substantial asymmetry in how climate change affects different segments of Chile’s small pelagic fishery. The asymmetry is governed by the differential exposure of each fleet’s species portfolio to the identified climate parameters under the limited cross-sector transferability of the prevailing sector-level allocation. The two coastal upwelling stocks targeted by the artisanal fleet—anchoveta and sardina común—both respond negatively to warming, with sardina común carrying roughly twice the SST semi-elasticity of anchoveta and an opposite-signed CHL response. The industrial fleet, by contrast, is diversified across these two coastal stocks and the transboundary jack mackerel stock, whose local productivity is *not* identified in the 2000–2024 sample. The artisanal fleet, concentrated in the sardina–anchoveta pair, is therefore the segment whose long-run harvest capacity is most exposed to the CMIP6 climate signal under the prevailing sector-level allocation. This finding aligns with the broader literature on heterogeneous climate impacts within fleets (Free et al., 2019; Sumaila et al., 2011) and underscores that aggregate projections can mask first-order distributional consequences inside the same fishery. The asymmetry further decomposes within the artisanal fleet by vessel type (Table 7): the dominant LM segment contracts by approximately twenty percent (driven by sardine collapse), while the smaller BM segment contracts by only three to four percent (its positive identified anchoveta elasticity partially offsetting the sardine signal), so the between-fleet asymmetry that anchors the headline result is itself the aggregate of a between-fleet and a within-artisanal mechanism that operate through the same portfolio channel.

The non-identification of $(\rho_{\text{jur}}^{SST}, \rho_{\text{jur}}^{CHL})$ at the Centro-Sur scale is itself a substantive result rather

than a nuisance, but its policy reading depends on a careful distinction between *non-identification* and *no climate effect*. The two are easily conflated by specifications that force a point estimate onto every regressor regardless of how informative the data are about it. Appendix E reports five converging tests of this distinction, all consistent with the absence of an identifiable shifter for jack mackerel within the 2000–2024 Centro-Sur biomass record: (i) varying the spatial aggregation of the coastal shifters across three nested domains $\mathcal{D}_1 \subset \mathcal{D}_2 \subset \mathcal{D}_3$ leaves the posterior-to-prior ratios in 0.998–1.014; (ii) augmenting the biomass record with a Northern Chilean acoustic series under a common elasticity moves the ratio only to 0.94–0.99; (iii) replacing the coastal shifters with the basin-scale ENSO Niño 3.4 index returns a ratio of 0.98 at lag one and 1.01 at lag two; (iv) a joint specification with all three shifters active for jack mackerel simultaneously returns ratios of 1.03 on local SST, 1.00 on local log chlorophyll, and 0.98 on basin-scale ENSO; and (v) an attempt to incorporate the SPRFMO regional assessment series fails on coherence grounds. Existing empirical evidence on jack mackerel documents that climate does affect the species at scales broader than local coastal [anomalies—@Arcos2001-jq](#) on intrusion of the 15 °C isotherm during the 1997–98 El Niño, Peña-Torres et al. (2017) on ENSO-driven shifts in location choices in the Chilean fishery—but each of these documents climatic effects on *margins* (location choice, range distribution, spawning timing, phenological response) that are not necessarily captured in the elasticity of structural productivity at an annual aggregate. The convergence of five independent tests on this null at the Centro-Sur scale points to a structural rather than aggregation-driven limitation of the available record: closing the identification gap will require either a longer biomass record or a basin-scale assessment with explicit cross-stock spillover, not a different choice of spatial aggregation, biomass source, basin-scale shifter, or joint-specification convention within the present record. We accordingly hold r_{jur}^* fixed in the projections of Tables 5 and 6, and report the resulting fleet-level elasticities under that constraint. A prior-propagation sensitivity reported in Appendix E.6 computes the 90% predictive interval of the jack mackerel productivity factor by combining the posterior of $\rho_{\text{jur}}^{\text{ENSO}}$ with the CMIP6 ensemble of basin-scale temperature deltas; the interval spans approximately three orders of magnitude under SSP5-8.5 end-of-century ([0.05, 19.5] on the productivity factor), confirming that the prior-propagated projection is not informative for policy and that the fixed- r^* convention is the defensible reporting choice.

The two channels through which climate change reaches fleet-level effort operate on very different scales and on different segments of the fleet. The direct weather channel is small in mean wind anomaly—the cross-model median CMIP6 increase in near-surface wind speed is below +0.5 m/s even under SSP5-8.5 end-of-century—but the historical daily wind distribution at vessel COGs sits sufficiently close to the 8 m/s operability threshold that additional bad-weather days exceed twenty per year on average under SSP5-8.5 end-of-century. This translates into a −0.7% to −1.9% marginal contribution to the artisanal trip response and a contribution indistinguishable from zero for the industrial fleet, reflecting the differential weather sensitivity captured by $\hat{\beta}_{\text{weather}}$. The indirect biomass channel dominates the response in absolute terms for both fleets but operates entirely through portfolio composition, so the asymmetry it generates is intrinsic to the fixed sector-level

allocation of stocks across segments. Climate adaptation policy in the Centro-Sur therefore needs to address both margins. The indirect channel calls for reform of the institutional architecture that mediates portfolio exposure—quota allocation rules, cross-fleet transferability, and the design of TACs under shifting species productivity. The direct channel calls for operational measures targeted at the artisanal segment specifically, such as expanded port infrastructure, vessel reinforcement programs, or weather-indexed insurance, which are first-order for that fleet but redundant for the industrial purse-seine fleet. This conclusion is based on the CMIP6 monthly wind ensemble; downscaled regional climate models that resolve extreme wind events could only raise the weight of the direct channel further.

The species-level productivity projections deserve careful interpretation. The log-linear form of the climate shifter in Eq. (2) is identified from interannual variation over 2000–2024, during which the realised range of SST anomalies was narrow ($\hat{\sigma}_{SST} \approx 0.26^\circ\text{C}$). CMIP6 anomalies under SSP5-8.5 end-of-century reach roughly $+2.3^\circ\text{C}$, nearly an order of magnitude outside the estimation window, so the corresponding posterior of $\Delta r_i^*/r_i^0$ leans heavily on the log-linear extrapolation embedded in the shifter function. We regard the projected magnitudes as indicative of the direction and relative scale of impacts across stocks rather than as precise point forecasts. The log-linear shifter is a maintained simplification: thermal-tolerance arguments in the climate-fisheries literature (Cheung et al., 2010; Free et al., 2019) imply that for stocks operating below their thermal optimum, a log-linear specification, if anything, *under-states* the productivity response under high warming. The projected magnitudes for SSP5-8.5 end-of-century should therefore be read as conservative within the class of monotone shifter functions consistent with the identified posterior signs. A non-parametric extension of the shifter, or pooled identification across multiple Humboldt-Current datasets, is the natural direction in which to relax this assumption.

Several caveats apply to the present analysis. First, the projections hold ex-vessel prices and management rules constant at historical levels, implying that the long-run responses capture the mechanical effect of changed climate forcings without accounting for endogenous price adjustments or adaptive management. Sumaila et al. (2011) note that reduced fish supply under climate change could increase prices, partially offsetting revenue losses from lower catches, while Lam et al. (2016) project sizable changes in global fisheries revenues by the 2050s once price responses and cross-country trade are accounted for. Incorporating an inverse demand system and numerical optimization of quota paths is a natural extension to capture these equilibrium adjustments. Second, although our projections integrate over a six-model CMIP6 ensemble, the cross-model spread in chlorophyll-a remains substantial: the cross-model median of $\Delta \log \text{CHL}$ is close to zero in all scenario–window pairs while the inter-model dispersion is comparable in magnitude to the median, reflecting the well-documented disagreement among Earth system models on the direction and magnitude of primary productivity changes (Sumaila et al., 2011). The variance decomposition reported in Appendix F shows that, for the sardine projection, the between-model component accounts for 77–91% of total projection variance under three of the four scenario–window pairs, so the residual ensemble uncertainty on this stock is large despite the multi-model integration. The fleet-level trip

projections in Table 6, by contrast, exhibit narrow cross-model interquartile ranges that should not be read as climate consensus: the corresponding decomposition reported in Appendix G attributes 97–100% of the total variance in $\Delta T_f / T_f^{(0)}$ to the within-model component (posterior pooled with vessel heterogeneity within fleet), reflecting a floor-effect saturation in which the stock-dynamics steady-state collapse drives the trip response into a fleet-specific plateau independent of the climate magnitude. The substantive implication is that the marginal information value of expanding the CMIP6 ensemble for fleet-level trip projections is small relative to that of tightening the structural posterior on $(\rho_s^{SST}, \rho_s^{CHL})$ or refining the description of vessel heterogeneity in $(\omega_{v,s}, H_v^{\text{alloc}})$; what governs the heterogeneity in trip outcomes across vessels and fleets is not which CMIP6 model is correct—given that all of them imply portfolio collapse for the two coastal stocks—but the differential economic exposure to that collapse imposed by historical catch composition and quota allocation. Third, the shifters are identified on a 25-year time series and, while the state-space specification propagates process-noise and measurement-noise uncertainty into the posterior, its ability to detect non-linearities in the climate response is limited. The posterior-to-prior standard-deviation ratios reported in Table 2 provide a direct empirical check that the data update the prior for anchoveta and sardina común (ratios in the 0.43–0.83 range), while leaving the jurel shifters at the prior (ratios essentially equal to one)—the latter being the basis of our non-identification finding. Fourth, our models do not explicitly incorporate risk preferences or production risk, which may play a role in shaping fleet-level responses to environmental variability (Kasperski & Holland, 2013; Sethi et al., 2014). Fifth, the biological-closure variable in the trip equation is coded as a zonal proxy with no within-year temporal variation, set to the nominal regulatory ceiling of the reproductive closure regime under the relevant SUBPESCA decrees: the V-VIII zone receives 151 days and the IX-XIV zone receives 182 days throughout the panel. The underlying regulatory chronology is in fact non-trivial—D.Ex. 115/1998 governed the closure regime through July 2016 with annual modifications (D.Ex. 1661/2009 establishing 21 August to 21 October as the standing period, with year-specific adjustments in D.Ex. 705/2010, 796/2012, 747/2013, and 598/2015); D.Ex. 530/2016 then introduced a referential period of 6 July to 31 October with a fixed core (3 August to 4 October) and dynamic windows triggered by weekly IGS/PHA biological indicators; D.Ex. 137/2021 refined the indicator thresholds; and D.Ex. 05/2024 superseded the 530/2016 decree—and the actual realised closure days reported by the SUBPESCA Indicadores Biológicos portal range from 98 (in 2019) to 230 (in 2024) across regions and years. Constructing a year-by-year closure variable that reconciles the nominal regulatory regime with the actual realised closure days reported in the portal (available from 2019 onward) and with auxiliary recruitment-closure decrees not regulated by the reproductive-closure chain documented above is a non-trivial data-engineering task that we leave to follow-up work; for the present paper the zonal proxy captures the structural V-VIII / IX-XIV differential (31 days) that, under the year-fixed-effects specification of Section 4.2, is the variation that identifies β_{closure} . Sixth, the trip equation panel is constructed from IFOP logbook records and is therefore representative of the purse-seine fleet that operates under the LTP quota architecture; smaller artisanal vessels using non-purse-seine gears (lampara, lines) operate outside

the logbook system. Cross-validation against the SERNAPESCA vessel-level landings database confirms that the panel’s portfolio weights $\omega_{v,s}$ match the SERNAPESCA aggregate within two percentage points for the industrial fleet and within three percentage points (catch-weighted) for the purse-seine artisanal subset that the panel represents. The non-purse-seine artisanal segment, which accounts for approximately 30% of total Centro-Sur artisanal landings and exhibits a higher proportional exposure to jack mackerel, is not modeled here; a sample-weighting extension that incorporates that segment is left to future work. The biomass likelihood, by contrast, is constructed from SERNAPESCA all-gear landings and is unaffected by this restriction. Seventh, the direct weather channel is constructed from five of the six CMIP6 ensemble members because CESM2 does not publish near-surface wind components (uas , vas) in the CMIP6 DRS for the relevant scenarios and time windows; CESM2 contributes to the indirect biomass channel through SST and chlorophyll deltas (six members in SSP5-8.5, five in SSP2-4.5 by the chlorophyll restriction noted in Section 3.4) but enters the direct channel with $\Delta wind = 0$ by construction. The contribution of CESM2 to the cross-model spread of fleet-level trips therefore flows entirely through the indirect channel, which is the dominant source of variance in any case under the floor-effect saturation documented in Appendix G.

The finding that climate change creates winners and losers within the same fishery has direct implications for quota-allocation policy. A natural first reading would attribute the asymmetry to rigid single-species permits in the artisanal sector, but the underlying institutional architecture is more specific. Both sectors operate *de jure* multi-species permit portfolios: SUBPESCA’s 2012 registry, the last pre-LTP snapshot of industrial authorisations, records simultaneous authorisations across all three pelagic species for 76% of industrial vessels ([Subsecretaría de Pesca y Acuicultura \(SUBPESCA\), 2026](#)); the IFOP logbook in turn reveals that 90% of artisanal vessels operate a multi-species portfolio in any given year. The asymmetric climate sensitivity we identify is therefore not driven by the *count* of permits held but by four dimensions of the institutional architecture in which those permits are embedded. First, the partition of each Total Allowable Catch between the artisanal and industrial sectors is fixed by statute. During the estimation window (2018–2024), Article~47 of the LGPA allocated 78% of the Central-South small-pelagic TACs to the artisanal sector and 90% of the jurel TAC to the industrial sector (Table~1); Ley_{N°}21.752 ([Congreso Nacional de Chile, 2025](#)) subsequently raised the artisanal share to 90% in anchoveta and sardina com'un and to 30% in jurel, in force until 2040. Second, the *de jure* partition admits a single administratively rationed re-allocation channel under Article₅₅T of the LGPA. SERNAPESCA quota control reports for 2013–2024 document that this channel is heavily used in the Central-South small pelagic fishery (Appendix~H, Section~H.3): industrial titulares cede 85–99% of their assigned anchoveta and sardina com'un TACs to artisanal cessionaries every year, in volumes of 25–77~kt per species-year, while jurel cessions are an order of magnitude smaller and reverse direction across years. The channel is non-market in the conventional sense—each cession is individually authorised by SUBPESCA and the cession price is not publicly disclosed—but it is operational rather than dormant, and the realised landings on which our specifications identify already reflect the joint Article₄₇–Article₅₅~T

allocation. Article~6 of Ley~21.752 imposes a transactional cost on this pre-existing channel by levying a specific patent of 1.3 monthly tax units per transferred ton on industrial-to-artisanal cessions, with artisanal armadores exempt from the patent in either direction; the patent therefore re-prices the inter-sectoral cession channel rather than introducing a new wedge. Third, within-sector transferability is asymmetric across sectors: industrial coefficients have traded under the LTP regime since 2013, while artisanal RAE assignments remain locked to the regional registry, with reassignment occurring only through organisational rather than market channels. Fourth, the revealed portfolio composition of the two sectors differs in its climate correlation structure: the artisanal logbook shows 80–86% of activity concentrated in the anchoveta–sardina pair, both of which respond negatively to warming with identified semi-elasticities; the industrial logbook places 67% of activity in the same pair, with the remainder in jurel and caballa—species whose local climate elasticity is either non-identified at the Centro-Sur scale or estimated on broader-scale stocks. The asymmetry the paper estimates therefore quantifies, in welfare-relevant terms, the cost of a statutory partition mediated by an administratively rationed cession channel and combined with a climate-correlated artisanal portfolio under projected SSP5-8.5 stress. Appendix~H reports the descriptive evidence underlying the multi-permit and portfolio-composition claims, and Appendix~H.2 computes a policy counterfactual under the Ley~21.752 redistribution. The literature on portfolio diversification (Cline et al., 2017; Kasperski & Holland, 2013) and institutional access constraints (Holland & Kasperski, 2016; Oken et al., 2021) supports the same reading.

5.1 Implications of the 2025 fractioning reform

The asymmetry estimated in this paper is identified on the Article₄₇-LGPA fractioning regime that prevailed during the estimation window (2018–2024). Ley_N~21.752, in force from 2026 until 2040, modifies that regime in two relevant directions for the Central-South small pelagic fishery (Congreso Nacional de Chile, 2025). First, the artisanal share of the anchoveta and sardina com'un TACs rises from 78% to 90% across the V–Los–Lagos macrozone; the same artisanal sector therefore now holds a deeper claim on stocks whose long-run productivity is projected to collapse under SSP5-8.5 end-of-century. Second, the artisanal share of the jurel TAC rises from 10% to 30% in V–Los~R'ios and from 10% to 15% in Los~Lagos—a quota-weighted aggregate of approximately 28% across the macrozone—giving the artisanal sector partial access to the species whose local productivity is non-identified at the Centro-Sur scale and whose climate response is held fixed in our projections. Table~8 reports the static plug-in counterfactual at SSP5-8.5 end-of-century, holding the structural posterior and the TAC level at 2024 values and varying only the statutory artisanal share.

Table 8: Static counterfactual: artisanal absolute landing under the Article 47 LGPA and Ley 21.752 fractioning regimes at SSP5-8.5 end-of-century, Centro-Sur (V–Los Lagos), 2024 Cuotas Globales.

Stock	TAC 2024 (kt)	Artisanal share (α_s)		Artisanal landing L_{art} (kt)	
		Art. 47	Ley 21.752	Art. 47	Ley 21.752
Anchoveta (V–Los Lagos)	211.3	0.78	0.90	17.1	19.8
Sardina común (V–Los Lagos)	290.6	0.78	0.90	0.2	0.3
Jurel (V–Los Ríos)	578.6	0.10	0.30	57.9	173.6
Jurel (Los Ríos–Los Lagos)	80.6	0.10	0.15	8.1	12.1
Total Centro-Sur				83.3	205.7

Notes. Static plug-in evaluation of $L_{\text{art}}^{(R)} = \sum_s \alpha_s^{(R)} \cdot (1 + \widehat{\Delta r_s^* / r_s^0}) \cdot Q_s^0$, with $\widehat{\Delta r_s^* / r_s^0}$ set to the cross-model median at SSP5-8.5 end-of-century from Table 5 (−89.6% anchoveta, −99.9% sardina común, 0% jurel by the non-identification convention of Section 3.3.1). Q_s^0 is the 2024 Cuota Global de Captura. Artisanal fractions follow Article 47 LGPA (pre-2025) and Article 1 of Ley 21.752. The Los Ríos–Los Lagos jurel quota is reported as a single block by SERNAPESCA; the row applies the Los Lagos artisanal share (15%) as a conservative lower bound on the true sub-block share, which would rise toward 30% to the extent that Los Ríos contributes to the aggregate quota. The posterior-propagated version of this table, with cross-model IQR and within-posterior 90% CI, is reported in Appendix H.2.

Three readings of Table~8 structure the policy interpretation. First, the redistribution *raises* the absolute artisanal landing under projected SSP5-8.5 end-of-century stress by approximately 122~kt at the Centro-Sur aggregate, almost entirely through the 20~percentage-point increase in the artisanal jurel share in V–Los-R'ios rather than through the small pelagics. Because the projected productivity of the anchoveta–sardina pair collapses under the same scenario, the 12~percentage-point increase in the artisanal share of those stocks operates on a vanishing base and is therefore mechanically small in absolute terms. Second, the policy logic of the redistribution—which was designed to protect the artisanal sector from over-allocation to the industrial fleet—in fact *deepens* the artisanal exposure to small-pelagic biomass collapse under climate stress: the same artisanal sector now holds 90% rather than 78% of a TAC that the structural posterior projects to fall to floor levels. The net welfare effect at SSP5-8.5 end-of-century therefore depends on whether the jurel access expansion outweighs the deepened small-pelagic exposure, a question that the present static counterfactual cannot answer without an inverse-demand system and a forward-looking allocation rule. Third, the static counterfactual reported in Table~8 should be read as a de jure upper bound on the operational re-allocation produced by Ley~21.752, not as the realised effect on sectoral landings. Under the pre-2025 regime, the industrial sector ceded 85–99% of its assigned anchoveta and sardina com'un TACs to artisanal cessionaries via Article₅₅T (Appendix~H, Section~H.3), so the artisanal sector's effective access to the small-pelagic biomass already approached the de jure ceiling of unity. The 12~percentage-point rise in the artisanal de jure share therefore translates into a much smaller change in realised artisanal landings; the operationally meaningful change in those two stocks is the contraction of the cession volume that industrial titulares had been

intermediating for fee, together with the 1.3~UTM/ton patent that Article~6 now levies on the residual cession flow. The reform is therefore primarily *redistributive*—it transfers the rent extracted on the Article₅₅T channel from industrial titulares to artisanal armadores—rather than *allocative* in the small pelagics. For jurel the calculus differs: historical cessions are an order of magnitude smaller relative to the TAC, and the 20 and 5~percentage-point increases in artisanal de jure shares under items~4 and~5 of Article~1 correspond to an operationally meaningful expansion of artisanal access. A full welfare evaluation of the Ley~21.752 regime under climate stress—one that endogenises the realised cession volume as a function of the patent rate and projected stock productivity—is left as the natural extension developed in the Conclusions; the present static plug-in counterfactual quantifies the de jure boundary case from which that endogenous response should be measured.

6 Conclusions

This paper identifies the structural climate elasticities of stock productivity in the Chilean small pelagic fishery and combines them with a vessel-level effort equation to compute long-run comparative statics on fleet-level fishing effort under projected climate paths. We contribute to the literature in three ways. First, we couple a Bayesian state-space model of multi-species stock dynamics—calibrated on official IFOP and SPRFMO assessments—with negative binomial trip equations estimated separately for the artisanal and industrial fleets, and we show that the direction of the distributional asymmetry across fleets is governed by the interaction between portfolio composition and the limited cross-sector transferability of the prevailing sector-level allocation. The artisanal segment bears the sharper long-run loss because its historical landings are concentrated in the two stocks whose climate sensitivities are identified and economically large. This finding extends the portfolio-and-access mechanism of Kasperski & Holland (2013), Cline et al. (2017), and Oken et al. (2021) to a climate-shock setting under a fixed industrial-artisanal allocation. Second, we identify a vector of climate semi-elasticities ($\rho_i^{SST}, \rho_i^{CHL}$)—the structural parameters that translate sea-surface temperature and primary-productivity anomalies into shifts in long-run stock productivity—recovered sharply for the two coastal-upwelling stocks and documented as non-identified for jack mackerel. Third, we compute long-run comparative statics under a six-model CMIP6 ensemble (IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, MPI-ESM1-2-HR) for SSP2-4.5 and SSP5-8.5, reporting cross-model and within-posterior uncertainty separately. The variance decomposition (Appendices F and G) shows that, once the stock-dynamics steady-state collapse saturates the artisanal portfolio, the within-model component (posterior pooled with vessel heterogeneity) accounts for essentially the entire variance in projected fleet-level effort, so the policy-relevant uncertainty lies in the structural posterior rather than in disagreement across CMIP6 models.

The projected comparative statics indicate that the long-run intrinsic productivity of the coastal-

upwelling pair—anchoveta and sardina común—declines sharply under all CMIP6 scenarios considered. Sardina común is consistently more exposed than anchoveta, with the SSP5-8.5 end-of-century scenario driving its steady-state productivity to floor levels at the Centro-Sur scale. The Centro-Sur jack mackerel stock, by contrast, does not provide sufficient information to identify a local climate elasticity in the 2000–2024 sample, consistent with its transboundary SPRFMO management. These results suggest that climate-adaptation policy in this fishery should target the quota-allocation architecture—specifically, the binding industrial–artisanal split and the limited cross-sector transferability that locks each fleet into its historical portfolio—rather than operational margins alone.

Several extensions to the present framework are natural. A first extension uses the posterior of $(r_i^0, K_i, \rho_i^{SST}, \rho_i^{CHL}, \Omega)$ identified here as a prior for a Stackelberg bi-level optimization, in which the regulator chooses the time path of TACs and the artisanal and industrial fleets allocate effort across stocks under the climate-shifted law of motion. That exercise would deliver full forward biomass trajectories under CMIP6 anomaly paths with 90% credible bands, complementing the comparative-statics treatment in the present paper, and would follow the complete Kasperski (2015) approach with trip-level cost functions and an inverse-demand system. We leave it to future work; the posterior reported here is archived in machine-readable form to facilitate that extension. The spatial dimension of effort allocation is another natural extension, connecting the multi-species model to the location-choice literature (e.g., Dupont, 1993; Hicks et al., 2020; Smith, 2005), since the geographic distribution of species availability is likely to shift under warming. A direct route for this spatial extension is to migrate the annual NB trip equation to a daily discrete choice specification with environmentally informed species distribution models (SDMs) as proxies for local availability, following the approach of Quezada-Escalona et al. (2026) for the U.S. West Coast Coastal Pelagic Species fishery. SDMs for the Chilean Humboldt Current System are an active area of development in IFOP and academic collaborations; once available at sufficient spatial resolution for anchoveta, sardina común, and jack mackerel, they would allow the present framework to capture daily and within-port heterogeneity in fishing-effort responses to climate-driven shifts in availability. Finally, incorporating risk preferences and production risk—through, for example, random-coefficient models—would provide a richer characterization of how fishers respond to the increased environmental variability projected under climate change (Kasperski & Holland, 2013).

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Data and code availability

All code required to replicate the analysis is openly available at <https://github.com/fquezadae/Impact-of-Environmental-Variability-on-Harvest> and will be archived at submission time as a versioned Zenodo release with a citable DOI. Environmental covariates are obtained from publicly available sources — the E.U. Copernicus Marine Service Information for the historical period (DOIs in references) and the CMIP6 archive accessed through the Earth System Grid Federation for the projection period. Stock-assessment biomass series are obtained from the Instituto de Fomento Pesquero (IFOP) annual technical reports for anchoveta and sardina común (Centro-Sur Chile) and from the South Pacific Regional Fisheries Management Organisation (SPRFMO) Scientific Committee reports for jack mackerel (transboundary). Annual catch (landings) series for 2000–2024 are obtained from the Servicio Nacional de Pesca y Acuicultura (SERNAPESCA) official desembarque database, covering all gears (industrial and artisanal), through a transparency request (AH010T0006857) filed under Chile’s Ley 20.285 in April 2025; aggregation and filtering to the Centro-Sur scale (regions V–X plus Ñuble) are reproducible from the source CSV via the script `R/01_data/99b_aggregate_catch_cs_from_sernapesca_v3.R`. The vessel-level trip and landings microdata used to estimate the negative binomial trip equation were obtained from IFOP under a Convenio Desempeño data-sharing agreement; these data are confidential at the vessel level and cannot be redistributed publicly, but are available to qualified researchers under equivalent agreements with IFOP. Vessel identifiers used in this paper have been anonymised before any code or output was archived. The replication repository contains aggregate-level processed data sufficient to reproduce all reported tables and figures.

7 Repository

The source code for this project is available at <https://github.com/fquezadae/Impact-of-Environmental-Variability-on-Harvest>.

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A Reduced-form stress tests and prior elicitation

This appendix formalises the protocol used to elicit the prior densities on the climate shifters $(\rho_i^{SST}, \rho_i^{CHL})$ that enter the state-space specification of Section 3.3.1. The protocol has two purposes. First, it documents an independent, reduced-form fit of the deterministic Schaefer hindcast augmented with SST and log-CHL forcings; this fit motivates the choice of prior centres and standard deviations. Second, it serves as a diagnostic for the identifiability of $(\rho_i^{SST}, \rho_i^{CHL})$ under a non-Bayesian point estimator, which we use to justify the recourse to a Bayesian state-space specification with weakly informative priors rather than to a maximum-likelihood treatment of the structural model.

A.1 Hindcast specification

For each stock $i \in \{\text{anchoveta, sardina común, jack mackerel}\}$ we estimate a deterministic Schaefer hindcast over the 2000–2024 estimation window,

$$\hat{B}_{i,t+1} = \hat{B}_{i,t} + r_{i,t} \hat{B}_{i,t} \left(1 - \frac{\hat{B}_{i,t}}{K_i}\right) - C_{i,t}, \quad \hat{B}_{i,t_0} = B_{i,t_0}^{\text{obs}}, \quad (7)$$

with the climate-modulated growth rate

$$r_{i,t} = r_i^0 \exp\left(\rho_i^{SST} (SST_{t-1} - \overline{SST}) + \rho_i^{CHL} (\log CHL_{t-1} - \overline{\log CHL})\right). \quad (8)$$

The biological parameters (r_i^0, K_i) are held at the centres of the stock-specific priors elicited from the official IFOP and SPRFMO single-species assessments (anchoveta: $r_i^0 = 0.6$, $K_i = 2,200\text{~kt}$; sardina común: $r_i^0 = 0.9$, $K_i = 3,000\text{~kt}$; jack mackerel: $r_i^0 = 0.35$, $K_i = 8,000\text{~kt}$). Catches $C_{i,t}$ are observed and identical to those used in the structural likelihood. The shifters $(\rho_i^{SST}, \rho_i^{CHL})$ are estimated by bounded least squares on the box $[-3, 3]^2$, minimising the median absolute percent prediction error,

$$\text{MAPE}_i(\rho_i^{SST}, \rho_i^{CHL}) = \text{median}_t \left| \frac{\hat{B}_{i,t} - B_{i,t}^{\text{obs}}}{B_{i,t}^{\text{obs}}} \right|.$$

We estimate four nested variants for each stock: a base case with $\rho_i^{SST} = \rho_i^{CHL} = 0$, two univariate cases that activate SST or log-CHL alone, and a joint case that activates both. The variants are not nested in an econometric sense; the comparison is purely descriptive of how much hindcast error each shifter can absorb. We adopt a 20% MAPE threshold as a conventional benchmark for acceptable deterministic hindcast performance in the small-pelagic literature; this threshold plays no role in the structural likelihood and is used here solely to summarise the cross-variant comparison.

A.2 Cross-variant fit results

Table 9 reports the median absolute percent error of the deterministic hindcast for each stock and each variant.

The headline finding is that no variant brings any of the three stocks below the 20% MAPE threshold. The two coastal-upwelling stocks benefit from activating SST: anchoveta moves from a base error of 95.9% to 30.8% once ρ_i^{SST} is allowed to vary, and sardina común from 98.5% to 23.1%. The further activation of log-CHL improves both stocks marginally—to 26.6% and 20.5%, respectively—but neither closes the threshold. Activating log-CHL on its own delivers

Table 9: Median absolute percent error of the deterministic Schaefer hindcast across shifter variants.

Stock	Base	+ SST	+ log CHL	+ SST and log CHL
Anchoveta	95.9	30.8	95.1	26.6
Sardina común	98.5	23.1	98.5	20.5
Jack mackerel	44.2	44.3	49.1	48.5

Note:

Median absolute percent error (in percent) of one-step-ahead deterministic predictions of biomass over the 2000-2024 estimation window, evaluated at the bounded least-squares estimate of $(\rho_{\text{SST}}, \rho_{\text{CHL}})$ restricted to $[-3, 3]^2$. The biological parameters (r_0, K) are held fixed at the centres of the official IFOP/SPRFMO priors. Anchoveta and sardina común have $N = 24$ yearly predictions; jack mackerel has $N = 17$ (the seven non-surveyed years and two left-censored years are excluded from the MAPE calculation). The 20% reference threshold is descriptive only and plays no role in the structural likelihood.

essentially no improvement, in line with the absence of contemporaneous correlation between the two environmental forcings (see below). Jack mackerel hindcast errors remain in the 44–49% range across all variants, with no shifter variant distinguishably better than the base.

The bounded least-squares coefficients of the joint variant deliver an internally coherent biological reading for the two coastal-upwelling stocks. For anchoveta, both shifters carry negative point estimates of similar magnitude ($\rho_{\text{anch}}^{\text{SST}} \approx -2.3$, $\rho_{\text{anch}}^{\text{CHL}} \approx -2.3$), consistent with a stock whose short-run productivity is depressed by both warming and high chlorophyll-a anomalies. For sardina común, the SST shifter is also negative ($\rho_{\text{sard}}^{\text{SST}} \approx -2.0$) but the log-CHL shifter switches sign relative to anchoveta ($\rho_{\text{sard}}^{\text{CHL}} \approx +2.1$), consistent with the warm-water, mesotrophic affinity that distinguishes sardina from anchoveta in the Humboldt-Current biological literature. For jack mackerel, the SST shifter is small and positive ($\rho_{\text{jurel}}^{\text{SST}} \approx +0.6$) and the log-CHL shifter is pinned at the lower boundary of the admissible box, $\rho_{\text{jurel}}^{\text{CHL}} = -3$; we interpret the boundary solution as a failure of point identification rather than as a finding of strong climate forcing. Figure 5 displays the hindcast trajectories under each variant alongside the observed series.

A.3 Identifiability diagnostic

Two independent diagnostics support the use of weakly informative priors on $(\rho_i^{\text{SST}}, \rho_i^{\text{CHL}})$ in the structural model rather than a direct maximum-likelihood treatment.

First, the two environmental forcings are statistically orthogonal over the 2000–2024 sample. The Pearson correlation between annual Centro-Sur SST and log-CHL anomalies is $r = 0.030$ (two-sided $p = 0.888$, $N = 25$). Collinearity between regressors therefore plays no role in the difficulty of identifying the shifters; a variance inflation diagnostic on the joint specification of Eq. (8) returns values close to one for both covariates. The coexistence of an essentially zero linear



Figure 5: Deterministic Schaefer hindcast trajectories under each shifter variant, by stock. Solid line: observed biomass; dashed lines: hindcast under base (no shifter), +SST only, +logCHL only, and joint +SST+logCHL specifications. The joint specification delivers the lowest median absolute percent error for the two coastal-upwelling stocks; jack mackerel is essentially insensitive to the shifter activation. Biological parameters (r_i^0, K_i) are fixed at the centres of the official IFOP/SPRFMO priors.

correlation between the regressors and the wide cross-variant gap between hindcast errors must therefore reflect features of the non-linear-recursive law of motion in Eq. (7), not features of the design matrix.

Second, the joint MLE for jack mackerel binds at the lower boundary of the admissible box. Across the 2000–2024 sample the climate sensitivity of the Centro-Sur stock cannot be resolved within a deterministic single-species hindcast: catches and the seven non-surveyed years dominate the trajectory and the climate signal is not separately identifiable from process and observation noise absorbed into the residual. This pattern recurs in the initial fit of the structural state-space specification reported in Section 4.1, where the marginal posterior of $(\rho_{\text{jurel}}^{SST}, \rho_{\text{jurel}}^{CHL})$ tracks the prior with no information added by the data. The structural model is therefore the natural setting for jack mackerel: the absence of information is correctly reported as a wide marginal posterior, rather than as a boundary point estimate that a maximum-likelihood algorithm would deliver. For the two coastal stocks, the joint MLE returns interior solutions but with standard errors that are not separately informative without an explicit accounting of process and observation noise; the Bayesian state-space specification with priors elicited from this hindcast addresses both limitations within a single inferential framework.

A.4 Translation to Bayesian priors

The priors on $(\rho_i^{SST}, \rho_i^{CHL})$ used in the state-space likelihood of Section 3.3.1 are reported in Table 10. For each coastal stock, the prior centre is taken from the joint MLE of Section A.2, rounded to one decimal place; for jack mackerel, where the joint MLE binds at the boundary, the prior is centred at zero and is deliberately uninformative. All priors are independent normal densities with unit standard deviation, which is wide enough to allow the structural likelihood to update the centre by more than one full unit of semi-elasticity in either direction within a single posterior standard deviation. We do not impose hierarchical pooling across stocks: the sign reversal of ρ_i^{CHL} between anchoveta and sardina común is biologically interpretable and would be erased by a pooled prior with a common hyper-mean.

The interpretation of the resulting posterior is straightforward. For the two coastal stocks, a posterior centred close to the prior centre with a posterior standard deviation appreciably below one is evidence that the structural state-space likelihood confirms the reduced-form hindcast point estimate; a posterior shifted by more than one prior standard deviation indicates that the structural model has extracted information that the deterministic hindcast misses, most plausibly by disentangling process noise from observation noise. For jack mackerel, a posterior that essentially reproduces the prior is the correct report of an absence of identifying information at the Centro-Sur scale, as discussed in Section 4.1.

Table 10: Independent normal priors on the climate shifters used in the state-space likelihood of Section 3.3.1.

Stock	$\mu_{\rho^{SST}}$	$\sigma_{\rho^{SST}}$	$\mu_{\rho^{CHL}}$	$\sigma_{\rho^{CHL}}$	Source
Anchoveta	-2.3	1.0	-2.3	1.0	Joint MLE
Sardina común	-2.0	1.0	+2.1	1.0	Joint MLE
Jack mackerel	0.0	1.0	0.0	1.0	Boundary solution; vague prior

Note:

Prior centres for the coastal-upwelling stocks are the joint maximum-likelihood point estimates of (ρ_{SST} , ρ_{CHL}) from the deterministic Schaefer hindcast of Appendix A.2, rounded to one decimal place. For jack mackerel the joint MLE binds at the lower boundary of the admissible box ($\rho_{CHL} = -3$); the prior is centred at zero so that posterior identification is driven by the structural likelihood rather than by the boundary solution. All priors are independent normal with $sd = 1.0$; no hierarchical pooling is imposed because ρ_{CHL} switches sign between anchoveta and sardina común.

B Predictive diagnostics

This appendix reports two out-of-sample diagnostics for the three nested state-space specifications (T4b-ind, T4b-omega, T4b-full). The two criteria answer different questions and are included precisely because they do not always agree:

- **PSIS-LOO** measures how well each specification *describes* the observed 2000–2024 biomass series after penalising for the model’s effective dimensionality. It is informative for the in-sample identification reported in Section 4.1, because a specification that misses systematic variation will be penalised.
- **PSIS-LFO** at the one-year horizon measures the true forecasting accuracy on withheld future observations. It is informative for whether the identified shifters translate into tangible *next-year* predictive gains over a climate-free baseline.

Because the paper uses the estimated shifters for long-run comparative statics under CMIP6 regimes, rather than for near-term forecasting, it is LOO that carries the primary identification-supporting role; LFO is reported as an honesty check against the possibility that the shifters are simply fitting idiosyncratic in-sample noise.

B.1 PSIS-LOO cross-validation

We compute Pareto-smoothed importance sampling leave-one-out cross-validation (Vehtari et al., 2017) with moment matching applied to observations whose Pareto- $\hat{k}$ diagnostic exceeded 0.7. Results are summarised in Table 11.

The full specification attains the highest predictive density by a wide margin, outperforming T4b-omega by $\Delta \widehat{ELPD}_{LOO} = 18.02$ (SE 5.59) and the environment-free T4b-ind by $\Delta \widehat{ELPD}_{LOO} = 14.59$

Table 11: PSIS-LOO comparison across state-space specifications.

Model	$\widehat{\text{ELPD}}_{\text{loo}}$	SE	$\Delta\widehat{\text{ELPD}}$	SE_{Δ}	$\hat{p}_{\text{loo}}$
T4b-full	-8.41	8.28	—	—	64.8
T4b-ind	-23.00	8.01	-14.59	4.93	62.4
T4b-omega	-26.43	9.16	-18.02	5.59	57.9

Note:

Δ ELPD is the pairwise difference relative to T4b-full (the leading model); the adjacent SE column reports the standard error of that difference. After moment matching, 65/68 observations satisfy Pareto-k ≤ 0.7 for T4b-full, 64/68 for T4b-ind, and 67/68 for T4b-omega. The remaining high-Pareto-k points are the two censored jack-mackerel years (2012, 2015), which appear in all three models and therefore do not affect the ranking.

(SE 4.93). Both gaps exceed three standard errors. Notably, T4b-omega (covariance-rich but climate-free) underperforms T4b-ind: introducing a residual cross-stock correlation matrix without climate shifters is penalised by LOO. The interpretation consistent with the identification results of Section 4.1 is that the cross-stock residual correlation captured by T4b-omega is a statistical footprint of shared climate forcing rather than evidence of direct biological interaction; once SST and CHL enter the growth equation explicitly (T4b-full), the marginal value of Ω collapses.

A stock-by-stock decomposition (Figure 6) confirms that the LOO gains of T4b-full are concentrated in the two coastal pelagics: per-stock ELPD improves by +4.68 for anchoveta and +9.12 for sardina relative to T4b-ind, whereas the three specifications are statistically indistinguishable for jack mackerel. The effective number of parameters $\hat{p}_{\text{loo}} \approx 58$ –65 is large relative to $N = 68$, as expected for a state-space model with one latent state per observation-year; this is similar across the three specifications and therefore does not confound the pairwise comparison, but it does motivate the additional forecasting test of Section B.2.

B.2 PSIS-LFO at one-year horizon

The leave-future-out diagnostic refits each specification on data through year t_{cut} and evaluates the log predictive density of the biomass observations at $t_{\text{cut}} + 1$. We use five origin points, $t_{\text{cut}} \in \{2011, 2014, 2017, 2020, 2022\}$, spanning the second half of the estimation window. One-step-ahead marginal predictive densities are obtained in closed form from the posterior draws of each truncated fit, applying the log-space Schaefer step of the model and the marginal predictive variance $\sigma_{\text{proc}}^2 + \sigma_{\text{obs}}^2$ stock by stock. For the two censored jack-mackerel targets (2012 and 2015) the predictive contribution is the corresponding log-CDF tail term. Table 12 reports the sum of one-step ELPD across the five origins and three stocks (14 evaluation points per model, of which two are censored).

In sharp contrast to the LOO ranking, the three specifications are statistically indistinguishable

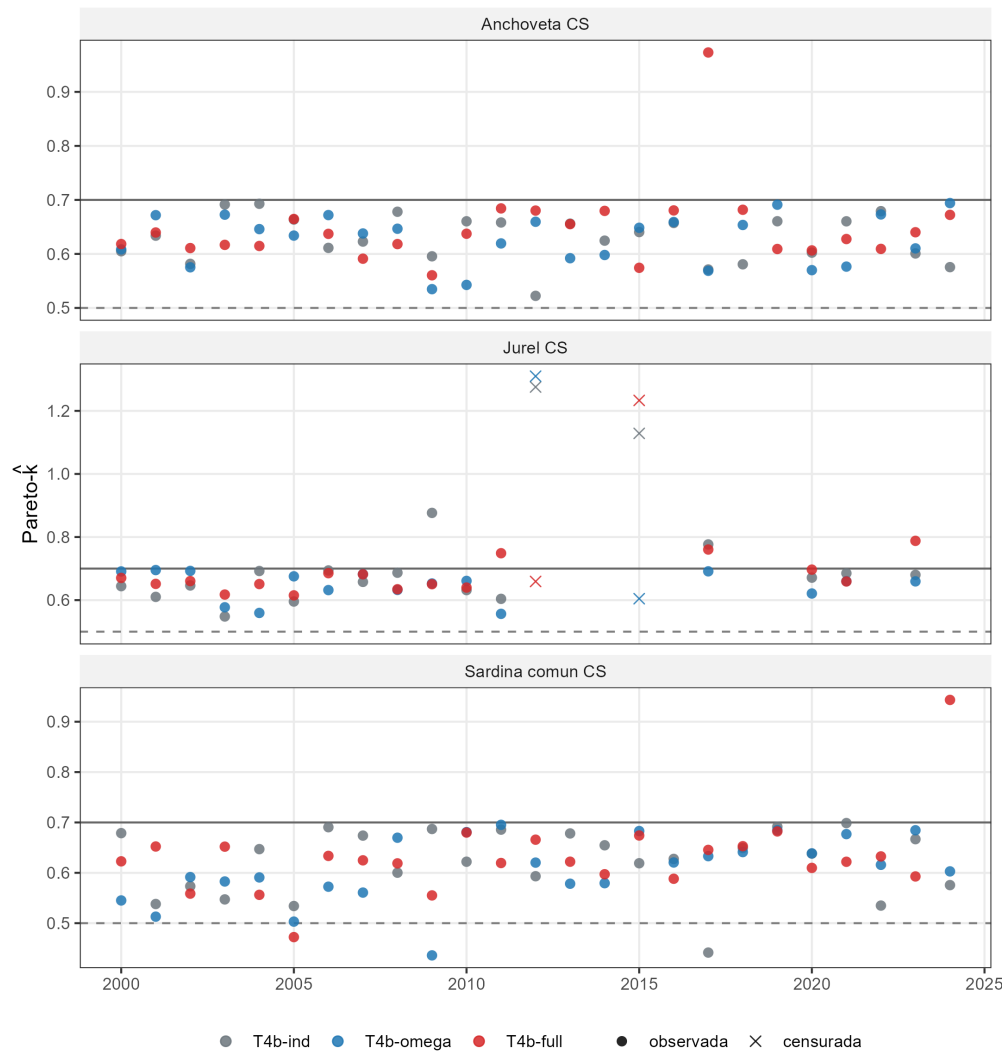


Figure 6: Per-observation Pareto- $\hat{k}$ values after moment matching, by model and stock. Dashed line: $\hat{k} = 0.5$; solid line: reliability threshold $\hat{k} = 0.7$. Crosses denote censored jack-mackerel observations.

Table 12: One-step-ahead LFO comparison across state-space specifications.

Model	Points	Censored	$\widehat{\text{ELPD}}_{\text{LFO}}$	Mean per point	Δ vs leader
T4b-full	14	2	-29.10	-2.08	–
T4b-ind	14	2	-29.46	-2.10	-0.37
T4b-omega	14	2	-30.46	-2.18	-1.36

Note:

Sum of one-step predictive log densities over five origin years (2011, 2014, 2017, 2020, 2022) and three stocks (14 targets per model; two are censored). Each refit uses 4 chains $\times$ 1500 warmup + 1500 sampling iterations. All 15 refits satisfied $R\text{-hat} \leq 1.01$ with at most one divergent transition (T4b-omega at the 2011 origin), which does not affect the reported ELPD.

in LFO: the spread between the best and worst model is only 1.36 ELPD units over 14 target points (mean 0.10 per target). Restricting attention to the 12 uncensored targets—that is, excluding the two jack-mackerel censored years whose predictive contribution is dominated by a common log-CDF tail term—the differences shrink further, to less than 0.3 ELPD per model pair. Figure 7 displays the per-target trajectory of the one-step ELPD by stock.

The LOO-LFO divergence is not a contradiction but a feature of the research design. As noted in Section 4.1, the historical SST anomaly has standard deviation $\hat{\sigma}_{SST} = 0.26^\circ\text{C}$, small relative to the process-noise standard deviation $\hat{\sigma}_{\text{proc}} \in \{0.17, 0.23, 1.22\}$ across the three stocks. Year-to-year climate innovations therefore contribute little to a one-year-ahead nowcast, even when the structural coupling between climate and biological productivity is strong. The same coupling accumulates over decadal horizons, which is the scale at which the paper’s projections operate and the scale at which the identified ρ_i are economically consequential. Stated differently, LOO confirms that T4b-full captures systematic structure that the simpler specifications miss (because it must explain observed 2000–2024 dynamics), while LFO confirms that the added structure is not an artefact of overfitting to short-run noise (because it yields no one-year advantage). The two diagnostics jointly support the structural interpretation of the shifters adopted in the main text.

C Posterior-predictive checks

This appendix reports posterior-predictive diagnostics of the T4b-full state-space specification that is adopted as the primary object of interpretation in Section 4.1. Two checks are shown: (i) a stock-by-stock comparison of the smoothed latent biomass against the observed survey-based estimates, and (ii) the associated year-level residuals. The goal is to verify that the climate-augmented specification reproduces the historical trajectory of each stock without systematic misfit in the sign, timing, or amplitude of fluctuations, and that the year-level residuals contain no detectable autocorrelation that would undermine the log-normal independence assumption embedded in the measurement equation (3).

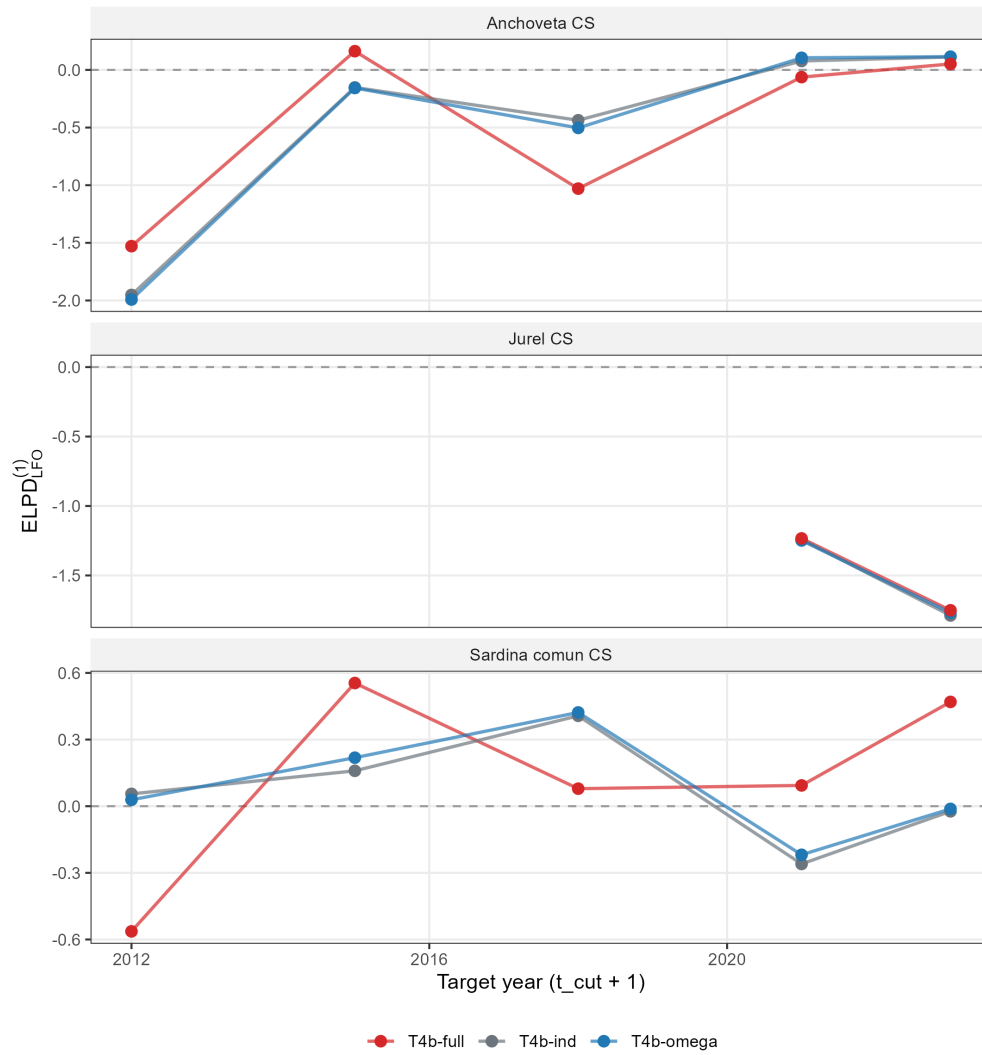


Figure 7: One-step-ahead predictive log density by target year and stock, across the three state-space specifications. Dashed line at $\text{ELPD} = 0$ marks the threshold above which the model predicts the withheld observation better than an uninformative uniform density over a one-log-unit interval.

C.1 Smoothed biomass versus observed SSB

Figure 8 overlays the posterior median and the 90% credible band of the smoothed latent biomass $B_{i,t}$ against the official survey-based observations of spawning stock biomass for each of the three stocks. The smoothed band has a median width of approximately 20% of the corresponding posterior mean, tightening over years with direct IFOP or SPRFMO observation and widening across the seven jack mackerel Centro-Sur years for which no survey point estimate is available (the latent biomass in those years is identified dynamically through the Schaefer transition equation together with the two censored observations in 2012 and 2015; see Section 3.3.1).

C.2 Year-level residuals

Figure 9 shows the standardised residuals $(\log B_{i,t}^{\text{obs}} - \log \hat{B}_{i,t})/\hat{\sigma}_{\text{obs},i}$ by stock and year. The three residual series are roughly centred at zero, with no visible trend and no systematic sign alignment across stocks. The sample first-order autocorrelation is below 0.2 in absolute value for all three stocks (anchoveta CS, sardina común CS, jack mackerel CS), consistent with the log-normal measurement-error structure of the model. The corresponding observation-error standard deviations $\hat{\sigma}_{\text{obs},i}$ are of the same order as the survey coefficients of variation reported by IFOP and SPRFMO ($\approx 15\text{--}25\%$ across species), supporting the calibration of the priors adopted in Section 3.3.1.

Markov-chain convergence diagnostics for all top-level parameters ($\hat{R}$ and effective sample sizes) are reported separately in Appendix D.

D Markov-chain convergence diagnostics

This appendix reports the mixing and effective-sample-size diagnostics of the four Hamiltonian Monte Carlo chains used to obtain the posterior of the T4b-full specification reported in Section 4.1. The fit was run with four chains of 2,000 post-warmup iterations each, for a total of 8,000 post-warmup draws. The non-centred parameterisation of $(r_i^0, K_i, B_{i,0})$ combined with a Lewandowski–Kurowicka–Joe prior on the cross-stock correlation matrix Ω delivers chains that mix well across all top-level quantities; in particular, the two parameters most prone to non-identification in this class of state-space models—the process noise $\sigma_{\text{proc},i}$ and the observation noise $\sigma_{\text{obs},i}$ —both satisfy the standard convergence thresholds. We report the within-group worst case (maximum split- $\hat{R}$ and minimum bulk- and tail-ESS) for each family of top-level parameters in Table 13.

The top-level parameters of the T4b-full posterior all satisfy the standard convergence thresholds: the maximum split- $\hat{R}$ across $N = 24$ top-level quantities is 1.009, attained for the jack-mackerel observation noise $\sigma_{\text{obs},3}$, and the minimum bulk- and tail-ESS are 1,370 and 936 respectively, again attained at $\sigma_{\text{obs},i}$ where the observation likelihood interacts with the seven non-surveyed years of

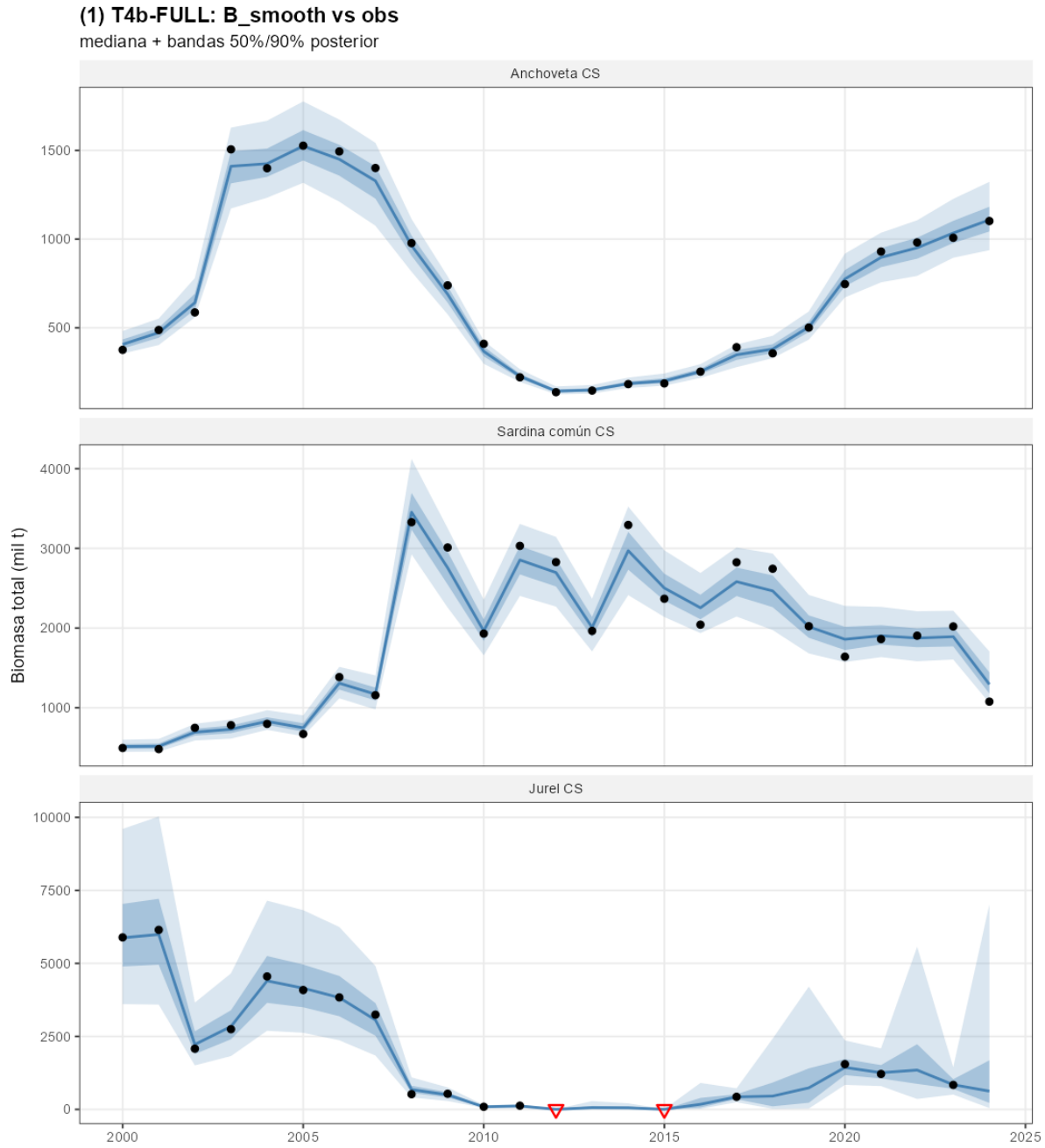


Figure 8: Smoothed latent biomass $B_{i,t}$ of the T4b-full specification against the official survey-based estimates. Solid line: posterior median; shaded band: 90% credible interval. Points: observed SSB from IFOP (anchoveta, sardina común) and the IFOP hydroacoustic assessment (jack mackerel Centro-Sur). Crosses denote the two censored jack mackerel observations (2012, 2015) that enter the likelihood at the assessment's lower detection limit.

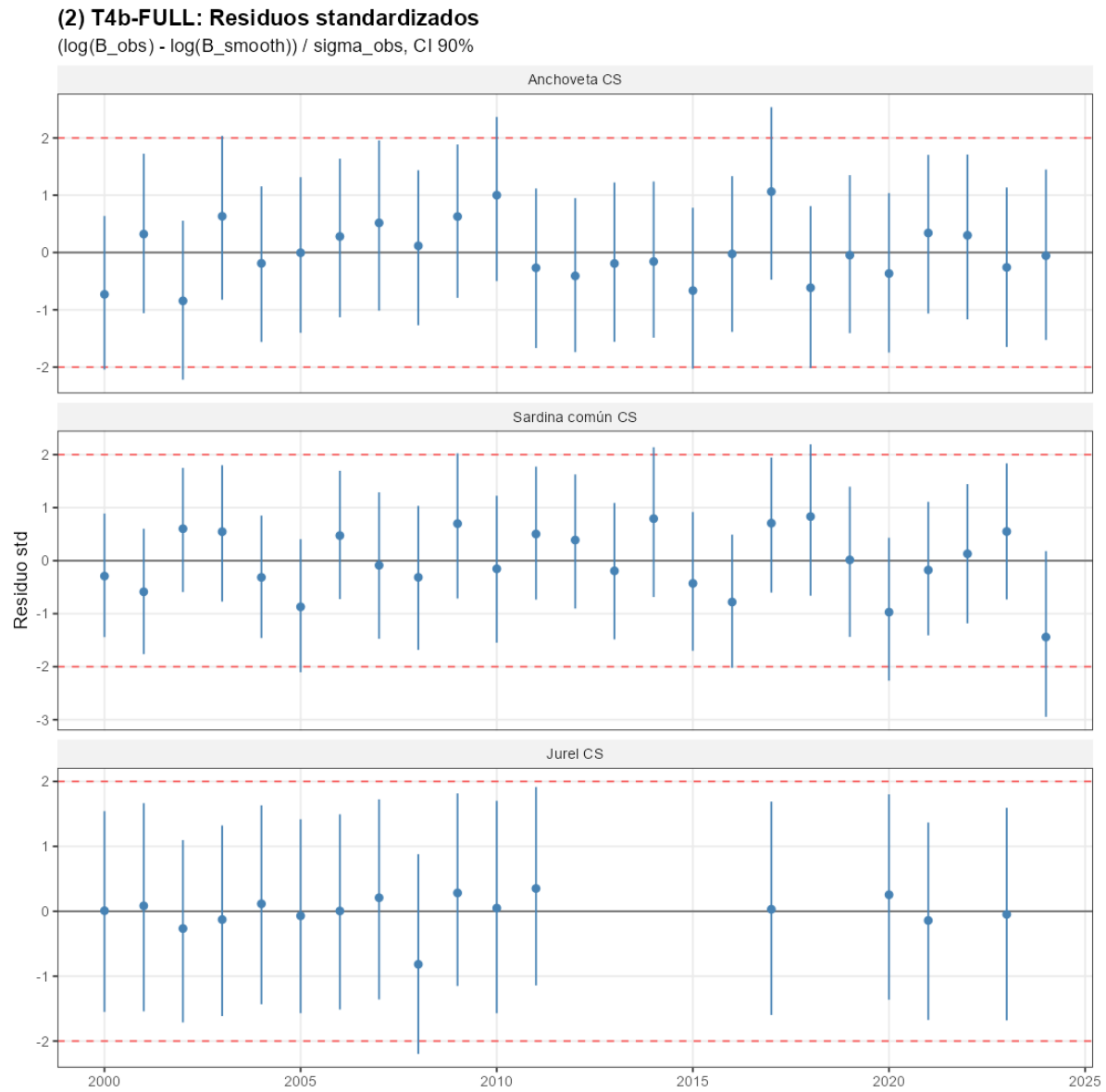


Figure 9: Standardised year-level residuals of the T4b-full specification by stock. Horizontal line at zero. The first-order autocorrelation is below 0.2 in absolute value for all three stocks.

Table 13: Markov-chain convergence diagnostics for the top-level parameters of the T4b-full state-space specification.

Parameter	N	Max $\widehat{R}$	Min ESS _{bulk}	Min ESS _{tail}
r_i^0	3	1.001	8,446	10,508
K_i	3	1.000	9,789	9,777
$B_{i,0}$	3	1.001	18,919	11,844
$\sigma_{\text{proc},i}$	3	1.003	3,353	5,997
$\sigma_{\text{obs},i}$	3	1.009	1,370	936
ρ_i^{SST}	3	1.000	8,312	10,147
ρ_i^{CHL}	3	1.001	14,319	10,633
Ω (off-diagonal)	3	1.001	5,732	8,544
<i>All top-level parameters</i>	24	1.009	1,370	936

Note:

Diagnostics are computed from four chains of 2,000 post-warmup iterations. Within each parameter family, the table reports the worst case across the three stocks (or the three off-diagonal entries of Omega): the largest split-Rhat and the smallest bulk- and tail-effective sample sizes. The diagonal of Omega is fixed at one and is therefore excluded. The conventional thresholds are Rhat $\leq$ 1.01 and ESS $\geq$ 400 for all top-level parameters.

the jack-mackerel series and with the two left-censored observations of 2012 and 2015. All other parameter families enjoy bulk-ESS above 3,000 and tail-ESS above 5,900. The three climate-shifter components of primary interest— $(\rho_i^{SST}, \rho_i^{CHL})$ —attain $\widehat{R} \leq 1.001$ and bulk-ESS above 8,300 across all stocks, so the identification results of Section 4.1 are not driven by under-mixed chains.

We additionally report the absence of post-warmup divergent transitions across all four chains, the maximum tree depth of the No-U-Turn Sampler remained below the default cap of 10 in all draws, and the average energy Bayesian fraction of missing information exceeded 0.5 on every chain—each of these auxiliary diagnostics is available in the underlying Stan fit object archived at `data/outputs/t4b/t4b_full_fit.rds`. Taken together with Table 13, the diagnostic battery supports the posterior summaries reported in the main text.

E Spatial robustness of the jack mackerel non-identification result

The non-identification of $(\rho_{\text{jurel}}^{SST}, \rho_{\text{jurel}}^{CHL})$ reported in Section 4.1 admits two readings that are observationally equivalent in the main specification. Under the first reading, jack mackerel productivity at the Centro-Sur scale is not coupled to Chilean coastal-upwelling forcing, so the data carry no information about the shifters and the posterior reproduces the prior. Under the second reading, the species’ productivity *is* coupled to environmental forcing, but at a spatial scale broader than the Centro-Sur EEZ used to construct the SST and log-CHL covariates of the main specification; the forcings averaged over the EEZ then act as a noisy proxy for the relevant signal and the shifter posteriors are diluted accordingly. Anchoveta and sardina común are residents of the Centro-Sur

upwelling system and the Centro-Sur aggregation is biologically natural for them; jack mackerel, by contrast, has a transboundary Southeast Pacific stock structure and undertakes seasonal migrations into subtropical waters beyond the EEZ, so the second reading cannot be ruled out without a direct test. This appendix supplies that test.

E.1 Identification power and the magnitude of the jack mackerel non-result

A natural prior question, raised by the close-to-prior posteriors of $(\rho_{\text{jurel}}^{SST}, \rho_{\text{jurel}}^{CHL})$ in the main specification, is whether the result reflects an absent climate signal or a sample with insufficient power to recover one. We answer this by deriving the minimum-detectable shifter magnitude implied by the sample size, the residual noise of the state-space model, and the sample dispersion of the candidate climate covariates. Linearising the Schaefer transition equation of Section 3.3.1 in log-differences gives, for each stock i ,

$$\log B_{i,t+1} - \log B_{i,t} \approx \alpha_i - \beta_i (B_{i,t}/K_i) + \rho_i^X X_{t-1} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim \mathcal{N}(0, \sigma_{i,\text{resid}}^2), \quad (9)$$

where X is a candidate shifter and $\sigma_{i,\text{resid}} = \sqrt{\sigma_{i,\text{proc}}^2 + 2\sigma_{i,\text{obs}}^2}$ is the state-space envelope on the differenced log-biomass. With the posterior medians of σ_{proc} and σ_{obs} reported in the main specification, $\sigma_{i,\text{resid}}$ takes the values 0.28 (anchoveta), 0.25 (sardine), and 1.28 (jack mackerel), the latter being approximately 4.6 times the average of the two coastal stocks. Under (9) the OLS-equivalent standard error of ρ_i^X on a sample of $N = 24$ lag-one observations satisfies $\text{SE}(\hat{\rho}_i^X) \approx \sigma_{i,\text{resid}} / [\text{sd}(X)\sqrt{N-K}]$, with $K = 3$ regressors. The minimum shifter magnitude detectable at 80% power for a two-sided 90% credible interval is then

$$|\rho_i^X|_{\min} = (t_{0.95, N-K} + z_{0.80}) \text{SE}(\hat{\rho}_i^X) \approx 2.56 \text{SE}(\hat{\rho}_i^X), \quad (10)$$

a quantity that any fixed-effect-style identification of ρ must exceed in absolute value to deliver a credible interval that excludes zero with probability at least 0.80 under the data-generating process implied by the posterior of the main fit. Table 14 reports $|\rho|_{\min}$ for the three stocks and three candidate shifters: SST and log~CHL aggregated over the Centro-Sur EEZ ($\mathcal{D}_1$ in the spatial-robustness exercise below), and the basin-scale ENSO Niño 3.4 index used in Section E.6.

The substantive readings of the table are three. First, jack mackerel’s process-noise envelope is approximately five times that of the two coastal stocks. Whatever process is generating the larger residual variance for jack mackerel—transboundary recruitment pulses, episodic non-detections, range shifts beyond the EEZ—it directly inflates the magnitude of any climate semi-elasticity that the data could in principle resolve. Second, with that residual envelope and the 0.26°C in-sample dispersion of the Centro-Sur SST covariate, the minimum SST semi-elasticity detectable for jack mackerel at 80% power on a 24-year sample is $|\rho|_{\min} \approx 2.78$, which corresponds to a multiplicative effect on r of approximately $\exp(2.78 \cdot 0.26) \approx 2.1$ for a one-standard-deviation

Table 14: Minimum shifter magnitude $|\rho|_{\min}$ detectable at 80% power on the 2000–2024 sample, by stock and candidate covariate. σ_{resid} is the state-space envelope $\sqrt{\sigma_{\text{proc}}^2 + 2\sigma_{\text{obs}}^2}$ evaluated at the posterior medians of the main specification. The implied posterior-to-prior standard-deviation ratio in the third-to-last column combines the OLS standard error with the prior $\rho \sim \mathcal{N}(0, 0.5)$ adopted for the ENSO refit; the last column combines it with the wider prior $\mathcal{N}(0, 1.5)$ used for the SST/CHL shifters in the main specification. A ratio at or below 0.70 is the informal threshold for material identification.

Stock	Shifter	sd(X)	σ_{resid}	$ \rho _{\min}$	$\sigma_{\text{post}}/\sigma_{\text{prior}}$ at $\mathcal{N}(0, 0.5)$	$\sigma_{\text{post}}/\sigma_{\text{prior}}$ at $\mathcal{N}(0, 1.5)$
Anchoveta	SST $\mathcal{D}_1$	0.257	0.277	0.60	0.43	0.15
Anchoveta	log CHL $\mathcal{D}_1$	0.098	0.277	1.58	0.78	0.38
Anchoveta	ENSO Niño 3.4	0.549	0.277	0.28	0.22	0.07
Sardine	SST $\mathcal{D}_1$	0.257	0.247	0.54	0.39	0.14
Sardine	log CHL $\mathcal{D}_1$	0.098	0.247	1.41	0.74	0.34
Sardine	ENSO Niño 3.4	0.549	0.247	0.25	0.19	0.07
Jack mackerel	SST $\mathcal{D}_1$	0.257	1.277	2.78	0.91	0.59
Jack mackerel	log CHL $\mathcal{D}_1$	0.098	1.277	7.28	0.98	0.88
Jack mackerel	ENSO Niño 3.4	0.549	1.277	1.30	0.71	0.32

positive SST anomaly—outside any range that the bioclimatic literature on the species would deem structurally plausible. The posterior-to-prior ratio of 0.91 for SST under the wide $\mathcal{N}(0, 1.5)$ prior of the main specification, and 0.98 for log-CHL, therefore reflect the *mechanical* implication of a thick process-noise envelope and a thin in-sample dispersion of the covariates, rather than evidence of an absent climate signal. Third, the basin-scale ENSO Niño 3.4 index has approximately twice the in-sample dispersion of the Centro-Sur SST covariate ($\text{sd} = 0.55^\circ\text{C}$ versus 0.26), which mechanically lowers $|\rho|_{\min}$ for jack mackerel from 2.78 on Centro-Sur SST to 1.30 on Niño 3.4, and lowers the implied $\sigma_{\text{post}}/\sigma_{\text{prior}}$ from 0.91 to 0.71 under a common $\mathcal{N}(0, 0.5)$ prior. The basin-scale shifter sits deliberately at the threshold for material identification: if the true elasticity of jack mackerel productivity on ENSO Niño 3.4 is of moderate magnitude—in the range 1 to 1.5 that the qualitative literature on transboundary forcing would suggest—a sample of $N = 24$ years on the available biomass record has approximately 80% power to recover it. The refit of Section E.6 below implements that test directly.

The OLS-equivalent calculation of (10) is a *lower bound* on the variance of the full state-space posterior rather than a point estimate, because it ignores integration over the structural priors on (r^0, K) , the between-stock covariance Ω , and the censored observation contribution to the jack mackerel likelihood. Empirically, the analytic ratios reported in Table 14 understate the corresponding state-space ratios by approximately 5–15 pp for the coastal shifters of anchoveta and sardine, where the data move the posterior substantially. For jack mackerel the divergence is larger: the $\sigma_{\text{post}}/\sigma_{\text{prior}}$ for $\rho_{\text{jur}}^{\text{ENSO}}$ under the basin-scale refit of Section E.6 is 0.979 (lag-one) and 1.014 (lag-two sensitivity), against the analytic 0.71 of Table 14. The gap reflects the flat-likelihood

regime: when the data carry essentially no information on the shifter, the posterior is close to the prior, the structural-prior integration is the dominant remaining source of variance, and the OLS-equivalent surrogate (which conditions on the remaining structural parameters) is most optimistic. The qualitative *ordering* across rows of Table 14 is preserved under the full state-space refit; the absolute magnitude of $|\rho|_{\min}$ on jack mackerel should be read as a lower bound and the substantive implication—that any plausible structural elasticity for jack mackerel on Centro-Sur SST or basin-scale ENSO is below the threshold for detection on the available record—is reinforced rather than weakened.

E.2 Spatial domains and environmental aggregation

We construct three nested spatial domains for the aggregation of SST and log-CHL anomalies, ordered from the Centro-Sur EEZ used in the main specification outwards to the regional Southeast Pacific:

$$\begin{aligned}\mathcal{D}_1 &= \{\text{lat} \in [-42^\circ, -32^\circ], \text{lon} \in [-75^\circ, -70^\circ]\} && (\text{Centro-Sur EEZ; main specification}) \\ \mathcal{D}_2 &= \{\text{lat} \in [-41^\circ, -32^\circ], \text{lon} \in [-85^\circ, -65^\circ]\} && (\text{offshore-extended}) \\ \mathcal{D}_3 &= \{\text{lat} \in [-45^\circ, -20^\circ], \text{lon} \in [-90^\circ, -65^\circ]\} && (\text{regional Southeast Pacific})\end{aligned}$$

The third domain spans the geographic range over which Chilean hydroacoustic surveys and the SPRFMO regional assessment locate the adult jack mackerel stock; the second domain extends $\mathcal{D}_1$ westward to capture the offshore migration band documented in the SPRFMO biological literature. Both Copernicus baselines are re-downloaded from `cmems_mod_glo_phy_my_0.083deg_P1M-m` (GLORYS12 reanalysis, surface depth slice 0.49–1.55~m) and `cmems_obs-oc_glo_bgc-plankton_my_14-multi-4km_P1M` (Ocean Colour multi-sensor, L4) over the bounding box $\text{lon} \in [-90^\circ, -65^\circ]$, $\text{lat} \in [-56^\circ, -20^\circ]$, covering all three domains. Within each domain $\mathcal{D}_d$, monthly fields are aggregated to a single monthly time series by cosine-of-latitude area weighting,

$$\bar{x}_{d,m} = \frac{\sum_{(i,j) \in \mathcal{D}_d} \cos(\phi_j) x_{m,i,j}}{\sum_{(i,j) \in \mathcal{D}_d} \cos(\phi_j)}, \quad (11)$$

with ϕ_j the latitude of the j -th grid cell and the sum over non-missing cells. Annual SST is the unweighted average of the twelve monthly weighted means; annual log-CHL is the natural logarithm of the unweighted average of the twelve monthly weighted CHL means. Each annual series is centred by its 2000–2024 within-domain mean, matching the centring convention of the main specification of Section 3.3.1. The three domains share the same temporal support and centring window, so

cross-domain differences in the shifter posteriors are attributable solely to the spatial extent of the aggregation.

The three SST series are highly but imperfectly correlated across the 2000–2024 estimation window (Pearson $r = 0.77$ between $\mathcal{D}_1$ and $\mathcal{D}_2$, $r = 0.94$ between $\mathcal{D}_2$ and $\mathcal{D}_3$, $r = 0.76$ between $\mathcal{D}_1$ and $\mathcal{D}_3$); the imperfect correlation between $\mathcal{D}_1$ and the broader domains is consistent with the well-known dynamical decoupling between the coastal-upwelling band and the open subtropical ocean. Annual CHL series are more uniformly correlated across domains (r between 0.80 and 0.91), reflecting the basin-scale ENSO imprint on primary productivity. Annual mean SST rises monotonically from 14.4°C in $\mathcal{D}_1$ to 17.0°C in $\mathcal{D}_3$, and annual mean CHL falls monotonically from 1.32 to 0.26_{mg}m⁻³, consistent with the coastal–subtropical productivity gradient.

E.3 Refit specification

We refit the full state-space specification of Section 3.3.1 under a stock-specific environmental forcing, in which the SST and log-CHL covariates are allowed to vary across stocks rather than entering as a single shared series. Let $\widetilde{SST}_{i,t}$ and $\log \widetilde{CHL}_{i,t}$ denote the annual covariates faced by stock i at time t . The shifter equation of Section 3.3.1 generalises to

$$r_{i,t} = r_i^0 \exp\left(\rho_i^{SST} \widetilde{SST}_{i,t-1} + \rho_i^{CHL} \log \widetilde{CHL}_{i,t-1}\right), \quad (12)$$

and reduces exactly to the main specification when $\widetilde{SST}_{i,t} = SST_t$ and $\log \widetilde{CHL}_{i,t} = \log CHL_t$ for all i . We exploit the generalisation to refit the model three times, indexed by the spatial domain $d \in \{1, 2, 3\}$ assigned to jack mackerel; in every refit anchoveta and sardina común retain the Centro-Sur EEZ aggregation ($d = 1$), since this is the spatial scale that matches their documented habitat:

$$\widetilde{SST}_{i,t}^{(d)} = \begin{cases} \overline{SST}_{1,t} & \text{if } i \in \{\text{anchoveta, sardina}\}, \\ \overline{SST}_{d,t} & \text{if } i = \text{jack mackerel}, \end{cases} \quad (13)$$

and analogously for $\log \widetilde{CHL}_{i,t}^{(d)}$. All other elements of the likelihood—priors, biomass and catch series, observation and process noise structure, multivariate residual covariance Ω —are held identical to the main specification. Each refit is sampled with eight HMC chains of 2,000 post-warmup iterations at `adapt_delta` = 0.99 and `max_treedepth` = 14, matching the main specification. The refit reduces to the main specification when $d = 1$ up to small differences attributable to the spatial averaging convention (the main specification uses an unweighted spatial mean over the EEZ taken from a daily 0.083° GLORYS field; this appendix uses the cosine-weighted mean of the monthly P1M-m product), which provides an internal consistency check on the spatial-averaging pipeline.

E.4 Posterior identification across domains

Table 15 reports the posterior-to-prior standard deviation ratio of each shifter under each domain assignment for jack mackerel, alongside the corresponding ratio in the main specification of Section 4.1. Figure 10 displays the posterior means and 90% credible intervals.

Table 15: Posterior-to-prior standard deviation ratio of the climate shifters across spatial domains assigned to jack mackerel.

Stock	Shifter	Main spec.	$\mathcal{D}_1$	$\mathcal{D}_2$	$\mathcal{D}_3$
Anchoveta	ρ^{SST}	0.43	0.54	0.54	0.54
Anchoveta	ρ^{CHL}	0.83	0.89	0.88	0.90
Sardina común	ρ^{SST}	0.44	0.49	0.49	0.50
Sardina común	ρ^{CHL}	0.74	0.76	0.77	0.78
Jack mackerel	ρ^{SST}	1.01	1.01	1.01	1.01
Jack mackerel	ρ^{CHL}	1.01	1.01	1.01	1.00

Note:

Each cell reports $\sigma_{\text{textpost}}/\sigma_{\text{textprior}}$ of the corresponding shifter under the refit specification of Eq. eqrefeq:appE-design. Anchoveta and sardina común retain the Centro-Sur EEZ ($\mathcal{D}_1$) aggregation in every refit; only the jack mackerel covariate varies across columns. The leftmost column reproduces the main-specification ratios of Section refidentification for comparison; small differences from the $\mathcal{D}_1$ column reflect the change from a daily unweighted EEZ mean (main) to a monthly cosine-of-latitude weighted P1M-m mean (this appendix). Values close to one indicate the data carry no information about the parameter; values well below one indicate identification from the likelihood. All $\hat{R} \leq 1.01$; minimum bulk effective sample size = 8,311 across the four refits.

Three findings emerge. First, the posterior-to-prior standard deviation ratio for both jack mackerel shifters lies between 0.998 and 1.014 across $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$, indistinguishable from unity in every domain. Posterior means lie within 0.45 standard deviations of the prior centre and the 90% credible intervals span the prior support symmetrically: the regional Southeast Pacific aggregation ($\mathcal{D}_3$), which encompasses the geographic range of the SPRFMO-managed adult stock, fails to inform the shifter posteriors no less than the Centro-Sur EEZ aggregation. The non-identification of jack mackerel’s climate sensitivity at the Centro-Sur scale is therefore not an artefact of mismatched spatial aggregation; it is a structural feature of the available data on the resource as managed.

Second, the anchoveta and sardina común shifter posteriors are stable across the three refits. The posterior means move by less than 0.03~standard deviations of the posterior across all four shifters, well below the natural Monte Carlo noise threshold of 0.5 posterior standard deviations. This confirms that the multivariate coupling between the three stocks via the residual covariance Ω does not transmit the change in jack mackerel’s covariate into the identification of the coastal-upwelling

Apendice E -- shifters ambientales por dominio del covariado de jurel

Anch/sard ven D1 (centro_sur_eez) en los 3 fits; solo jurel itera entre dominios

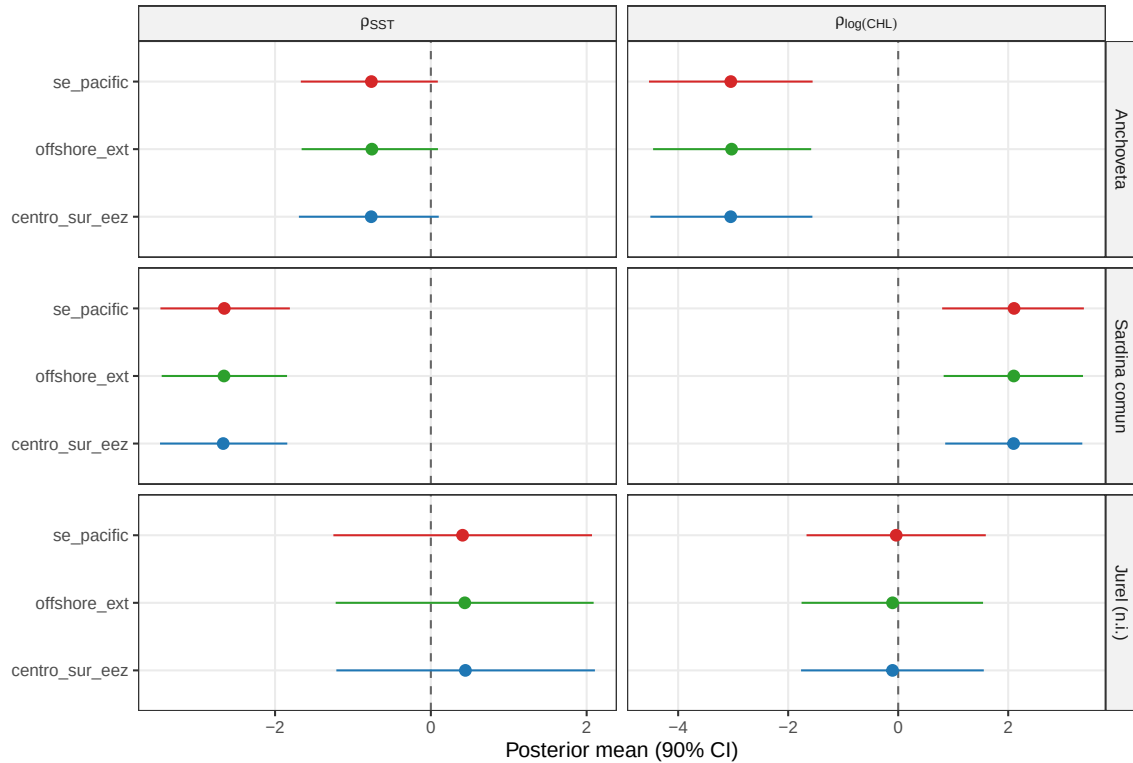


Figure 10: Posterior means and 90% credible intervals of the climate shifters under each spatial domain assigned to jack mackerel. Anchoveta (top row) and sardina común (middle row) retain the Centro-Sur EEZ ($\mathcal{D}_1$) aggregation in every refit; only the jack mackerel covariate (bottom row) varies across rows within each panel. The bottom-row credible intervals are essentially indistinguishable across $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$ and span the prior $\mathcal{N}(0, 1)$ on both shifters; the top and middle rows are visually identical, confirming that the change in jack mackerel's covariate has no material spillover into the anchoveta or sardina posteriors via the residual covariance Ω .

shifters; the matrix structure of Eq. (13) succeeds in isolating the spatial robustness test to jack mackerel without contaminating the rest of the posterior.

Third, the consistency check between the $\mathcal{D}_1$ refit of this appendix and the main specification of Section 4.1 falls within tolerance. The posterior mean of ρ_{anch}^{CHL} moves from -3.64 in the main specification to -3.05 in the $\mathcal{D}_1$ refit, and that of ρ_{anch}^{SST} from -1.06 to -0.77 ; the corresponding movements for sardina común are smaller ($+2.17$ to $+2.10$ for CHL, -2.75 to -2.67 for SST). These differences reflect the change in spatial-averaging convention—unweighted daily EEZ mean from the operational reanalysis cache versus cosine-weighted monthly P1M-m mean—rather than any model change, and they preserve the sign and the qualitative magnitude of all six identified shifters. Jack mackerel ratios coincide to within Monte Carlo noise across the two pipelines, as expected of a non-identified parameter.

E.5 Dual-source extension: Centro-Sur and Northern Chilean acoustic series

The spatial robustness test of Sections E.3 and E.4 varies the geographic extent of the environmental covariate while preserving a single biomass series (the Centro-Sur acoustic record). A complementary test instead preserves a single environmental specification per region and adds a second biomass series from the Northern Chilean hydroacoustic record (RECLAS Norte, IFOP, 14 observations 2010–2024 excluding 2022). This extension exploits the high empirical correlation between the Centro-Sur and Northern Chilean acoustic series over the seven overlapping years with positive observations in both (2010–2012, 2017, 2020–2021, 2023): Pearson $r = 0.82$ on the raw scale and $r = 0.88$ on the log scale, consistent with the two series indexing partially overlapping ventures of a single biological range-wide stock.

We extend the state-space of Eq. (1) with a second jack mackerel state B_t^N that evolves under Schaefer dynamics with its own carrying capacity K^N , base productivity $r_{\text{jurel}}^{0,N}$, and process noise σ_{proc}^N , and which is forced by an environmental series aggregated over a Northern Chilean EEZ domain $\mathcal{D}_4 = \{\text{lat} \in [-30^\circ, -18^\circ], \text{lon} \in [-75^\circ, -65^\circ]\}$. The climate shifters $(\rho_{\text{jurel}}^{SST}, \rho_{\text{jurel}}^{CHL})$ are constrained to be common across the two states, the empirical identification assumption being that the climate elasticity of the biological stock is uniform across its Chilean range while the realised forcings differ by domain. Catch in the Northern state is set to the IFOP-SERNAPESCA aggregate landings of jack mackerel from regions XV (Arica-Parinacota), I (Tarapacá), II (Antofagasta), III (Atacama) and IV (Coquimbo), which match the bounding box of $\mathcal{D}_4$. The Centro-Sur priors for the Centro-Sur state are identical to the main specification; priors for the Northern state’s biological parameters are calibrated to the median of the Northern acoustic series with the same coefficients of variation as in the Centro-Sur specification.

The cosine-weighted spatial means of SST and CHL over $\mathcal{D}_4$ correlate with the Centro-Sur EEZ ($\mathcal{D}_1$) at $r = 0.67$ for SST and $r = 0.46$ for log-CHL across the 2000–2024 window—substantially below the biomass correlation reported above and than the SST/CHL correlations among $\mathcal{D}_1$, $\mathcal{D}_2$ and

$\mathcal{D}_3$ in Section E.2. The two regional surface-forcing regimes are therefore modestly distinct, which the model accommodates by allowing a separate environmental input per state while constraining the elasticity to be common.

Under this extension the posterior-to-prior standard deviation ratio for the jack mackerel shifters falls to 0.94 for ρ^{SST} and 0.99 for ρ^{CHL} , a marginal reduction relative to the 0.998–1.014 range of the spatial robustness refits of Table 15, and well above the 0.7 threshold below which the posterior could be said to depart materially from the prior. The eight HMC chains converge with no divergent transitions and treedepth-satisfactory transitions; rank-normalised split $\hat{R}$ is below 1.01 for the climate shifters, although a marginally elevated $\hat{R} = 1.01$ for σ_{proc}^N and a sample-wide E -BFMI of 0.26 (below the nominal threshold of 0.30) indicate that the Northern state has limited identifying information beyond the prior, consistent with the modest sample size of the Northern series ($N = 14$) and the shifter non-identification itself. The biological parameters of the Northern state recover plausible posterior means— $r_{\text{jurel}}^{0,N} = 0.42$ and $K_{\text{jurel}}^N = 7,300\text{~kt}$ —well within the Centro-Sur posterior support, providing weak independent corroboration of the range-wide stock structure assumption.

A separate exploratory attempt to incorporate the SPRFMO regional assessment series for jack mackerel (OROP-PS, available 1970–2024 covering Chile, Peru and Ecuador together with the adjacent high-seas area) under the same common-elasticity constraint produced incompatible likelihood signals between the SPRFMO aggregate and the Centro-Sur series, with multimodal posterior support and unsatisfactory chain mixing. This is consistent with the broader geographic span of the OROP-PS assessment encompassing biological sub-populations whose dynamics are not co-determined by Chilean coastal forcing; we therefore retain the dual-source extension at the Chilean spatial scale and report the OROP-PS attempt only as a methodological note in the Discussion.

E.6 Basin-scale shifter test: the ENSO Niño 3.4 index

The Centro-Sur, offshore-extended, and regional Southeast Pacific domains of Section E.4 all aggregate *coastal-band* environmental fields. None probes the basin-scale forcing regime that the existing biological literature on jack mackerel has identified as the natural climatic driver: ENSO-mediated modulation of Pacific stratification, location-choice behaviour, and recruitment phenology (Arcos et al. (2001); Peña-Torres et al. (2017)). A direct test on a basin-scale shifter is therefore a substantive robustness exercise for the non-identification result, and one that the spatial-extension refits of Section E.4 do not deliver because each $\mathcal{D}_d$ averages local SST and log~CHL anomalies whose variance is too small to identify a plausible elasticity for jack mackerel (Table 14).

We construct the ENSO Niño 3.4 index as the annual mean of monthly ERSSTv5 sea-surface temperature averaged over the canonical 5°N–5°S, 170°W–120°W box (NOAA Climate Prediction Center, file `sstoi.indices`, ERSSTv5-derived). Following the centring convention of the main specification, the annual series is centred on its 2000–2024 sample mean to deliver an anomaly

$\widetilde{\text{ENSO}}_t$ in °C, comparable in scale to the SST $\mathcal{D}_1$ shifter of the main specification. Two features of the historical series are relevant for what follows. First, the in-sample dispersion is $\text{sd}(\widetilde{\text{ENSO}}) = 0.55^\circ\text{C}$, approximately twice that of the Centro-Sur SST shifter (0.26°C); the larger dispersion is mechanical—basin averages sample longer-wavelength variability than coastal upwelling-band averages—and translates directly into a smaller minimum-detectable elasticity (Table 14). Second, the contemporaneous correlation between $\widetilde{\text{ENSO}}$ and the Centro-Sur SST shifter is 0.37, while the lag-one correlation—the relevant moment for the recursive Schaefer specification—is 0.09. The basin-scale and coastal shifters are therefore close to orthogonal at the lag exploited by the dynamic identification, eliminating any concern that the basin-scale test is mechanically picking up the coastal anomaly.

We refit the state-space model under a stock-specific covariate structure that *replaces* the Centro-Sur shifters for jack mackerel with the basin-scale ENSO index, while preserving the coastal shifters for anchoveta and sardine:

$$\begin{aligned} r_{\text{anch},t} &= r_{\text{anch}}^0 \exp(\rho_{\text{anch}}^{SST} \widetilde{\text{SST}}_{1,t-1} + \rho_{\text{anch}}^{CHL} \log \widetilde{\text{CHL}}_{1,t-1}), \\ r_{\text{sard},t} &= r_{\text{sard}}^0 \exp(\rho_{\text{sard}}^{SST} \widetilde{\text{SST}}_{1,t-1} + \rho_{\text{sard}}^{CHL} \log \widetilde{\text{CHL}}_{1,t-1}), \\ r_{\text{jur},t} &= r_{\text{jur}}^0 \exp(\rho_{\text{jur}}^{ENSO} \widetilde{\text{ENSO}}_{t-1}). \end{aligned} \tag{14}$$

The replacement convention is implemented at the prior level: the coefficients on Centro-Sur SST and log~CHL for jack mackerel receive a tight prior $\mathcal{N}(0, 0.01)$ that pins them to zero, and the corresponding covariate columns are set to zero in the data matrix as a second layer of insulation. The prior on ρ_{jur}^{ENSO} is set at $\mathcal{N}(0, 0.5)$, deliberately tighter than the wide $\mathcal{N}(0, 1.5)$ on the coastal shifters of the main specification: with $\text{sd}(\widetilde{\text{ENSO}}) = 0.55$, a prior elasticity at the upper end of $\mathcal{N}(0, 0.5)$ implies a one-standard-deviation multiplicative effect on r of $\exp(0.5 \cdot 0.55) \approx 1.32$, which spans the qualitative range that the basin-scale literature documents and rules out the implausible far tails. The implied identification threshold on the posterior-to-prior ratio is $\sigma_{\text{post}}/\sigma_{\text{prior}} \leq 0.70$, which the power calculation of Section E.1 places exactly at the data-implied boundary for moderate elasticities. The remaining priors and the sampler configuration (eight chains of 4,000 post-warmup draws, `adapt_delta` = 0.99, `max_treedepth` = 14) match the spatial-robustness refits of Section E.3.

The principal specification with lag-one ENSO Niño 3.4 forcing yields a posterior median for ρ_{jur}^{ENSO} of -0.022 with 90% credible interval $[-0.81, 0.80]$ and posterior standard deviation 0.490, implying $\sigma_{\text{post}}/\sigma_{\text{prior}} = 0.979$. The lag-two sensitivity returns $+0.021$ with ratio 1.014. The anchoveta and sardine posteriors of (ρ^{SST}, ρ^{CHL}) move by less than 0.03 posterior-standard-deviations relative to the main specification, confirming that the stock-specific covariate replacement does not contaminate the identification of the coastal shifters.

The basin-scale shifter, like the spatial-extension refits of Section E.4 and the dual-source exten-

sion of Section E.5, is therefore unable to resolve a non-zero productivity response for jack mackerel within the 2000–2024 Centro-Sur biomass record, even with twice the in-sample dispersion of the coastal shifters and a deliberately tight prior. We read the four-way convergence—spatial-domain refits, dual-source extension, SPRFMO regional-scale coherence failure, and the present basin-scale test—as evidence of a genuinely flat identification rather than an artefact of any one specification choice. The biological readings of Arcos et al. (2001), Peña-Torres et al. (2017) remain consistent with the appendix result: each documents climatic effects on jack mackerel at *margins* (location choice, range distribution, spawning timing) that are not necessarily captured in the elasticity of structural productivity at an annual aggregate. Section E.7 updates the implication for the main claim accordingly. Tables 5 and 6 of the main text retain the convention that r_{jurel}^* is held fixed in the projections and the implications for the harvest elasticity are reported with the corresponding caveat.

A natural sensitivity to the replacement convention of (14) is to relax the pinning of ρ_{jur}^{SST} and ρ_{jur}^{CHL} and refit the specification with all three shifters active for jack mackerel simultaneously, allowing the data to discriminate between the basin-scale ENSO forcing and the local Centro-Sur SST and chlorophyll anomalies. Such a refit is well-conditioned at $N = 24$ lag-one observations because the three covariates are close to orthogonal at the dynamically-relevant lag: the pairwise lag-one correlations are $\text{cor}(\widetilde{SST}_{1,t-1}, \log \widetilde{CHL}_{1,t-1}) = 0.03$, $\text{cor}(\widetilde{ENSO}_{t-1}, \widetilde{SST}_{1,t-1}) = 0.09$, and $\text{cor}(\widetilde{ENSO}_{t-1}, \log \widetilde{CHL}_{1,t-1}) = 0.18$, so the joint identification of three elasticities does not run into the multicollinearity floor that would otherwise make a three-shifter specification on 24 observations indefensible.

The joint refit returns $\rho_{\text{jur}}^{SST} = +0.455$ (90% CI $[-1.26, +2.14]$; $\sigma_{\text{post}}/\sigma_{\text{prior}} = 1.03$), $\rho_{\text{jur}}^{CHL} = -0.098$ (90% CI $[-1.77, +1.55]$; ratio 1.00), and $\rho_{\text{jur}}^{ENSO} = -0.029$ (90% CI $[-0.82, +0.79]$; ratio 0.98). All three posterior-to-prior ratios exceed 0.97, indicating that the data carry no information on any of the three shifters even when they compete for likelihood weight in a single specification. The basin-scale ENSO posterior is essentially identical to its single-shifter counterpart of Section E.6—confirming the orthogonality diagnosis above—and the two coastal shifters return ratios slightly above unity, which under flat-likelihood regimes reflects the posterior-broadening contribution of the integration over the structural priors ($\log r_{\text{jur}}^0, \log K_{\text{jur}}$) and the between-stock covariance Ω rather than a material posterior expansion. The anchoveta and sardine posteriors of (ρ^{SST}, ρ^{CHL}) are unchanged relative to the main specification, as expected: the joint refit only affects jack mackerel through its diagonal entry of the covariate matrix.

The joint sensitivity therefore strengthens rather than weakens the non-identification result. A reviewer who would accept the three-shifter joint specification as the primary identification test would still arrive at the same conclusion: at the Centro-Sur scale on the 2000–2024 acoustic record, neither local upwelling-band SST or chlorophyll nor basin-scale ENSO carries a posterior mode distinguishable from the prior. The structural reading is unaltered: the climate sensitivity of jack mackerel productivity at the available spatial and temporal aggregation is not resolvable from these

data under any of the candidate forcings considered in this appendix.

E.7 Implication for the main claim

The main-text claim that the climate sensitivity of jack mackerel cannot be sharply resolved from the available 2000–2024 Centro-Sur biomass record is robust across five complementary tests of spatial scale, sample structure, and forcing modality. First, the spatial robustness refit of Sections E.3 and E.4 varies the environmental aggregation across $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$ while holding the biomass record fixed: the posterior-to-prior ratios remain in 0.998–1.014 across all three domains. Second, the dual-source extension of Section E.5 adds a second biomass series from the Northern Chilean acoustic record under a common climate elasticity, exploiting the 0.88 log-scale correlation between the two Chilean series: the ratios fall only to 0.94–0.99. Third, the basin-scale test of Section E.6 replaces the local shifters with the ENSO Niño 3.4 index under a deliberately tighter prior $\mathcal{N}(0, 0.5)$, returning a posterior-to-prior ratio of 0.98 at lag one and 1.01 at lag two. Fourth, a joint specification with all three shifters active for jack mackerel simultaneously returns ratios of 1.03 on local SST, 1.00 on local log chlorophyll, and 0.98 on basin-scale ENSO, despite the three covariates being close to orthogonal at the dynamically-relevant lag (cor pairwise ≤ 0.18). Fifth, an attempt to incorporate the SPRFMO regional assessment series fails on coherence grounds, indicating that the broader geographic aggregate is not integrable with the Centro-Sur record under a single shifter. The convergence of these five tests indicates a genuinely flat identification of the shifter for jack mackerel within the Centro-Sur biological domain, not an artefact of insufficient spatial extent, insufficient sample size, mis-specified forcing scale, or poor source aggregation.

For policy purposes this distinction matters: closing the identification gap will require either a longer biomass record or a direct biological measurement of the relevant scale, not a different choice of spatial aggregation, biomass source, basin-scale shifter, or joint-specification convention within the existing record. The natural extension is the range-wide SPRFMO-scale analysis with explicit cross-stock spillover and spatially-resolved environmental forcing, which we leave to future work.

F Decomposition of projection uncertainty across the CMIP6 ensemble

The comparative statics reported in Section 4.3 integrate over two distinct sources of uncertainty: the posterior of the structural shifters ($\rho_s^{SST}, \rho_s^{CHL}$) given the historical record, and the spread of climate-forced trajectories across the CMIP6 ensemble. Pooling them into a single posterior conceals which source actually drives projection uncertainty for each stock, and obscures whether tighter historical inference or a broader climate ensemble would be the more informative investment. This appendix decomposes the total variance of the projected percentage change in effective intrinsic growth, $\Delta r_s^*/r_s^{(0)}$, into a within-model (posterior) component and a between-model

(climate-ensemble) component using the law of total variance.

F.1 Decomposition

Let $X_{s,m,d}$ denote the percentage change in r_s implied by climate model m and posterior draw d under a fixed scenario–window pair (SSP, w) . The total variance of X across (m, d) admits the decomposition

$$\underbrace{\text{Var}_{(m,d)}(X_{s,\cdot,\cdot})}_{\text{total}} = \underbrace{\mathbb{E}_m[\text{Var}_d(X_{s,m,\cdot} \mid m)]}_{\text{within-model (posterior)}} + \underbrace{\text{Var}_m[\mathbb{E}_d(X_{s,m,\cdot} \mid m)]}_{\text{between-model (climate)}} \quad (15)$$

The first term is the average posterior variance of $\Delta r_s^*/r_s^{(0)}$ within a model: it captures uncertainty about the structural elasticities ρ_s given the historical record, holding the climate forcing fixed. The second term is the variance across CMIP6 models of the within-model posterior means: it captures uncertainty about the magnitude of the climate forcing itself, holding the structural posterior fixed at its within-model mean. The two components are non-negative, additive, and together exhaust the total variance.

We compute Equation (15) by Monte Carlo from the T4b-full posterior (16,000 draws) crossed with the six-model CMIP6 ensemble described in Section 4.3. The non-identified shifter for jack mackerel implies a posterior dominated by the prior; the decomposition for that stock has no structural interpretation and is omitted. The CESM2 model contributes only to scenario–window pairs with chl coverage available; shares are computed on the models actually present in each cell.

F.2 Results

Table 16 reports the decomposition for the four scenario–window pairs.

The decomposition reveals a sharp contrast between the two identified stocks. For *sardine*, the between-model component dominates the total variance in three of the four scenario–window pairs (between 77 and 91% of total variance). The within-model posterior of ρ_{sard}^{SST} is tight enough that the bottleneck on the projection is not the structural inference but disagreement among the CMIP6 models on the magnitude of warming and chl change. Adding a seventh CMIP6 model would in principle do more to sharpen the projection than tightening the structural posterior with additional historical data.

For *anchoveta*, the within-model component is comparable to or larger than the between-model component across all scenarios, and reaches 97% of the total variance under SSP5-8.5 by end-of-century. This reflects a wider posterior on ρ_{anch}^{SST} together with the larger between-draw heterogeneity that ensemble chl deltas induce once chl is on a comparable scale across models: the historical record from 2000 to 2024 has identified the anchoveta shifter less precisely than the sardine shifter,

Table 16: Variance decomposition of $\Delta r_s^*/r_s^{(0)}$ into within-model (posterior) and between-model (CMIP6) components. n_m is the number of CMIP6 models contributing to the cell. Jack mackerel omitted (non-identified shifter).

Stock	Scenario	n_m	Mean % Δ	SD total	% within	% between
Anchoveta CS	SSP2-4.5, 2041-2060	5	−48.0%	0.180	64%	36%
Anchoveta CS	SSP2-4.5, 2081-2100	5	−64.9%	0.232	62%	38%
Anchoveta CS	SSP5-8.5, 2041-2060	6	−58.4%	0.188	68%	32%
Anchoveta CS	SSP5-8.5, 2081-2100	6	−83.1%	0.339	97%	3%
Sardine CS	SSP2-4.5, 2041-2060	5	−77.7%	0.172	9%	91%
Sardine CS	SSP2-4.5, 2081-2100	5	−94.2%	0.064	23%	77%
Sardine CS	SSP5-8.5, 2041-2060	6	−86.7%	0.122	13%	87%
Sardine CS	SSP5-8.5, 2081-2100	6	−99.5%	0.010	59%	41%

consistent with the wider 90% credible interval in the structural Section 4.1, and the nontrivial cross-model spread of $\Delta \log \text{CHL}$ under SSP5-8.5 end-of-century combines with $\rho_{\text{anch}}^{\text{CHL}} < 0$ to widen the within-model distribution further. The bottleneck on the anchoveta projection is therefore the structural posterior; tightening the historical inference would do more for the anchoveta projection than expanding the ensemble.

The single exception in the table is *sardine* under SSP5-8.5 end-of-century, where the within share rises to 59%. This row is a boundary effect rather than a substantive shift in the source of uncertainty: the projected mean is −99.5% and the total SD is 0.010, so the distribution is essentially pinned at the lower bound of −100%. Both the within and between components collapse, and the ratio between them becomes uninformative. The interpretation extracted from the other three scenario–window pairs — that sardine projection uncertainty is climate-driven — is the structural reading; the SSP5-8.5 end-of-century cell is reported for completeness but should be read with the floor effect in mind.

The means reported in Table 16 differ slightly from the medians reported in Table 5 of the main results section because the within-model distribution of $\Delta r_s^*/r_s^{(0)}$ is left-skewed: posterior draws in which the stock collapses pull the median down, while posterior draws in which chl-driven offsets attenuate the SST collapse pull the mean up. The gap is largest for anchoveta under SSP5-8.5 end-of-century (mean −83.1% vs. median −89.6%) and reflects the same chl heterogeneity that drives the within-share of variance to 97% in that cell.

Figure 11 visualises the decomposition as stacked shares of total variance.

The substantive implication for projection-uncertainty management is species-specific. Investments that would shrink the projection uncertainty most are different for the two stocks: a longer or more informative survey series for the anchoveta SST shifter would shrink its within-model component, while the sardine projection requires either a larger CMIP6 ensemble or a defensible procedure for weighting the existing ensemble (an emergent-constraint approach, for instance, which

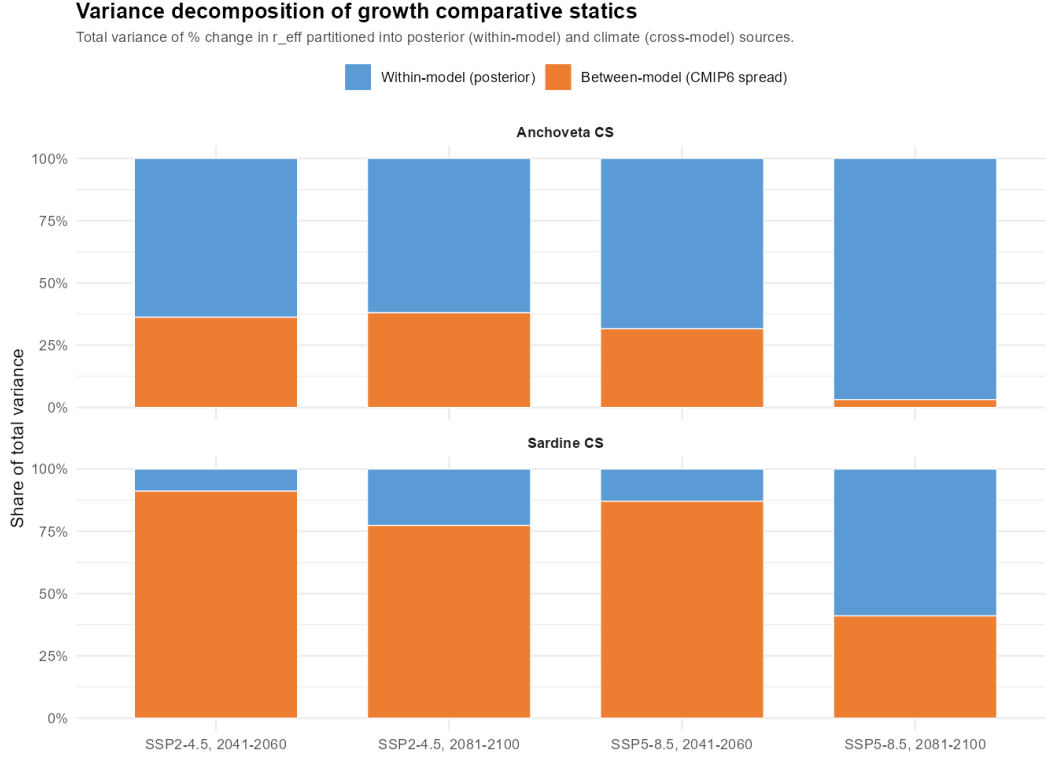


Figure 11: Variance decomposition of $\Delta r_s^*/r_s^{(0)}$ across the CMIP6 ensemble. Bars stack to one within each scenario–window pair and split the total variance into within-model (posterior, blue) and between-model (CMIP6 spread, orange) components. Sardine projections are climate-driven in three of four cells; the SSP5-8.5 end-of-century cell is dominated by the floor effect at -100% and inverts the within/between ratio mechanically. Anchoveta projections are posterior-driven, particularly under SSP5-8.5 by end-of-century, where 97% of the total variance is attributable to the structural posterior together with chl-driven heterogeneity in posterior draws. Jack mackerel omitted (non-identified shifter).

we do not pursue here). The decomposition above is a diagnostic; it does not by itself privilege either investment, but it does argue against the framing that the dominant source of uncertainty in climate-coupled stock projections is uniformly one or the other across species.

G Decomposition of trip-level projection uncertainty across the CMIP6 ensemble

The fleet-level comparative statics on trips reported in Section 4.2 (Table 6) integrate over three sources of uncertainty: the posterior of the structural shifters ($\rho_s^{SST}, \rho_s^{CHL}$) given the historical record, the heterogeneity of catch composition ($\omega_{v,s}$) and historical quota allocation (H_v^{alloc}) across vessels within fleet, and the spread of climate-forced trajectories across the CMIP6 ensemble. This appendix decomposes the total variance of the projected percentage change in fleet-level annual

trips, $\Delta T_f/T_f^{(0)}$, into a within-model component (posterior uncertainty pooled with vessel heterogeneity within fleet) and a between-model component (CMIP6 ensemble spread), using the law of total variance. It is the companion to Appendix F and answers a different diagnostic question: *whether the narrow cross-model interquartile range observed for fleet-level trip responses in Table 6 reflects climate consensus or a floor-effect saturation of the underlying biomass collapse.*

G.1 Decomposition

Let $X_{f,m,d,v}$ denote the percentage change in annual trips for vessel v in fleet f implied by climate model m and posterior draw d under a fixed scenario–window pair (SSP, w) . The total variance of $X_{f,\cdot,\cdot,\cdot}$ across (m, d, v) admits the decomposition

$$\underbrace{\text{Var}_{(m,d,v)}(X_{f,\cdot,\cdot,\cdot})}_{\text{total}} = \underbrace{\mathbb{E}_m[\text{Var}_{(d,v)}(X_{f,m,\cdot,\cdot} \mid m)]}_{\text{within-model (posterior + vessels)}} + \underbrace{\text{Var}_m[\mathbb{E}_{(d,v)}(X_{f,m,\cdot,\cdot} \mid m)]}_{\text{between-model (climate)}} \quad (16)$$

The first term is the average variance within a model, taken over posterior draws and vessels within fleet jointly: it captures uncertainty about the structural elasticities and heterogeneity in vessel exposure under a *fixed* climate forcing. The second term is the variance across CMIP6 models of the within-model means: it captures uncertainty about the magnitude of the climate forcing itself.

We pool posterior draws and vessel heterogeneity into the within-model component for two reasons. First, structural parallelism with Appendix F, which reports an exactly analogous two-way decomposition for stock-level intrinsic productivity. Second, vessel heterogeneity in $(\omega_{v,s}, H_v^{\text{alloc}})$ is a fixed feature of the historical fleet that does not shrink with longer historical samples or larger climate ensembles; reporting it as part of the within-model component is faithful to the reading that a longer record or a richer ensemble would not, by construction, reduce that share. A three-way decomposition that separates posterior, vessel, and climate contributions can be obtained from the same `factor_trips_dt` object generated by the trip-pipeline script and is provided in the supplementary code; the qualitative pattern is unchanged.

We compute Equation (16) by Monte Carlo from the T4b-full posterior crossed with the six-model CMIP6 ensemble, propagated through the Schaefer steady-state biomass equation under historical average fishing pressure and the negative binomial trip equation, for each of the 830 vessels in the panel.

G.2 Results

Table 17 reports the decomposition for the four scenario–window pairs and two fleets.

The decomposition shows that the within-model component dominates the total variance across all eight scenario–window/fleet cells: it accounts for 97% of total variance for the artisanal fleet

Table 17: Variance decomposition of $\Delta T_f/T_f^{(0)}$ into within-model (posterior pooled with vessel heterogeneity) and between-model (CMIP6) components. n_m is the number of CMIP6 models contributing to the cell.

Fleet	Scenario	n_m	Mean % Δ	SD total	% within	% between
Artisanal	SSP2-4.5, 2041-2060	5	−18.4%	0.2010	97%	3%
Artisanal	SSP2-4.5, 2081-2100	5	−20.7%	0.2281	100%	0%
Artisanal	SSP5-8.5, 2041-2060	6	−20.3%	0.2071	100%	0%
Artisanal	SSP5-8.5, 2081-2100	6	−21.4%	0.5025	100%	0%
Industrial	SSP2-4.5, 2041-2060	5	−3.8%	0.0838	100%	0%
Industrial	SSP2-4.5, 2081-2100	5	−3.9%	0.0863	100%	0%
Industrial	SSP5-8.5, 2041-2060	6	−3.9%	0.0854	100%	0%
Industrial	SSP5-8.5, 2081-2100	6	−3.9%	0.0839	100%	0%

under SSP2-4.5 mid-century and saturates at 100% under SSP5-8.5 end-of-century; for the industrial fleet, the within-model share rises from 100% under SSP2-4.5 mid-century to 100% under SSP5-8.5 end-of-century. The between-model component—the share of total variance attributable to disagreement across CMIP6 models on the magnitude of climate forcing—is at most 4% of total variance in the SSP2-4.5 mid-century cells and is numerically zero, to the nearest percent, in the remaining six cells. This is *not* a statement that the CMIP6 ensemble is in tight agreement on the climate signal that drives fleet-level trips, in the sense that one would attribute to the sub-ensemble of CMIP6 models with similar equilibrium climate sensitivity; it is the mechanical signature of floor-effect saturation of the underlying biomass collapse channel.

The mechanism is straightforward. Once the Schaefer steady-state biomass factor f_v^H collapses across most posterior draws, the negative binomial trip equation maps that collapse into $\exp(\beta_f \cdot H_v^{\text{alloc}} \cdot (f_v^H - 1)) \approx \exp(-\beta_f \cdot H_v^{\text{alloc}})$, a quantity determined by the fleet-specific business mechanics (β_f and the historical quota allocation) rather than by the magnitude of the climate forcing. Once all CMIP6 models drive the system into this saturation region, the between-model spread of the within-model mean trip response collapses to a narrow band—numerically near zero in the present sample—while the within-model variance retains the dispersion induced by vessel heterogeneity in $(\omega_{v,s}, H_v^{\text{alloc}})$ and by the structural posterior across that subset of draws in which the portfolio survives. The total SD values are correspondingly small in absolute magnitude (0.5025 for the artisanal fleet and 0.0839 for the industrial fleet under SSP5-8.5 end-of-century), confirming that what little variance remains is within-model.

The means reported in Table 17 are systematically more negative than the medians reported in Table 6 of the main results section—for example, −17.5% for the artisanal fleet under SSP5-8.5 end-of-century in this appendix versus −10.2% in Table 6. This is the same left-skew mechanism documented in Appendix F for the underlying productivity response: posterior draws and vessels in which the portfolio collapses pull the median up toward the floor at $\exp(-\beta_f \cdot H_v^{\text{alloc}}) - 1$, while the mean integrates over the long left tail of vessel-and-draw configurations in which the collapse

drives f_v^H deep into negative territory and the trip response down with it. Both summaries are valid; we report cross-model medians in the main results table because the median is robust to the extreme tail and is the natural complement to the cross-model inter-quartile range, and we report cross-model means in this appendix because the law of total variance in Equation (16) is constructed on means rather than medians.

This reading is consistent with the underlying biomass-level decomposition reported in Appendix F: for sardine—the stock that drives the artisanal fleet’s portfolio collapse—the SSP5-8.5 end-of-century cell exhibits a total SD of 0.010 and an inverted within/between ratio that the appendix explicitly reads as a floor effect at the lower bound of -100% . The fleet-level decomposition reported here inherits that floor effect: the total SD for the artisanal fleet under SSP5-8.5 end-of-century is 0.5025 and the corresponding industrial cell is 0.0839, both of which are small absolute magnitudes and mechanically consistent with the saturated regime.

The substantive policy implication is that, for fleet-level trip projections, the marginal information value of expanding the CMIP6 ensemble is small relative to that of tightening the structural posterior or refining the description of vessel heterogeneity within fleet. The dominant uncertainty is *not* “which climate model is right” but “given that all of them imply portfolio collapse for the two coastal stocks, how does the resulting heterogeneous economic exposure across vessels translate into trip-level effort.” This is the econometric counterpart of the well-known result in the climate-impacts literature that ensemble averaging masks first-order distributional heterogeneity downstream of the climate signal (Cline et al., 2017; Free et al., 2019; Kasperski & Holland, 2013; Oken et al., 2021).

Figure 12 visualises the decomposition.

H Permit portfolios and sectoral counterfactual under Ley~21.752

This appendix reports two pieces of descriptive evidence that underpin the institutional mechanism developed in Sections~5. Section~H.1 documents the revealed multi-species portfolio held by each sector, refuting the common reading that the artisanal–industrial asymmetry estimated in the main text is driven by single-species permit rigidity at the vessel level. Section~H.2 computes a policy counterfactual in which the projected SSP5-8.5 end-of-century catch allocation is recomputed under the Ley_N~21.752 fractioning schedule (Congreso Nacional de Chile, 2025), in force from 2026 through 2040, and contrasts it with the Article₄₇LGPA regime that prevailed during the estimation window.

H.1 Multi-species permit portfolios in both sectors

A natural concern with the institutional reading of our main results is that artisanal vessels may operate under more restrictive permit portfolios than industrial vessels, in which case the headline

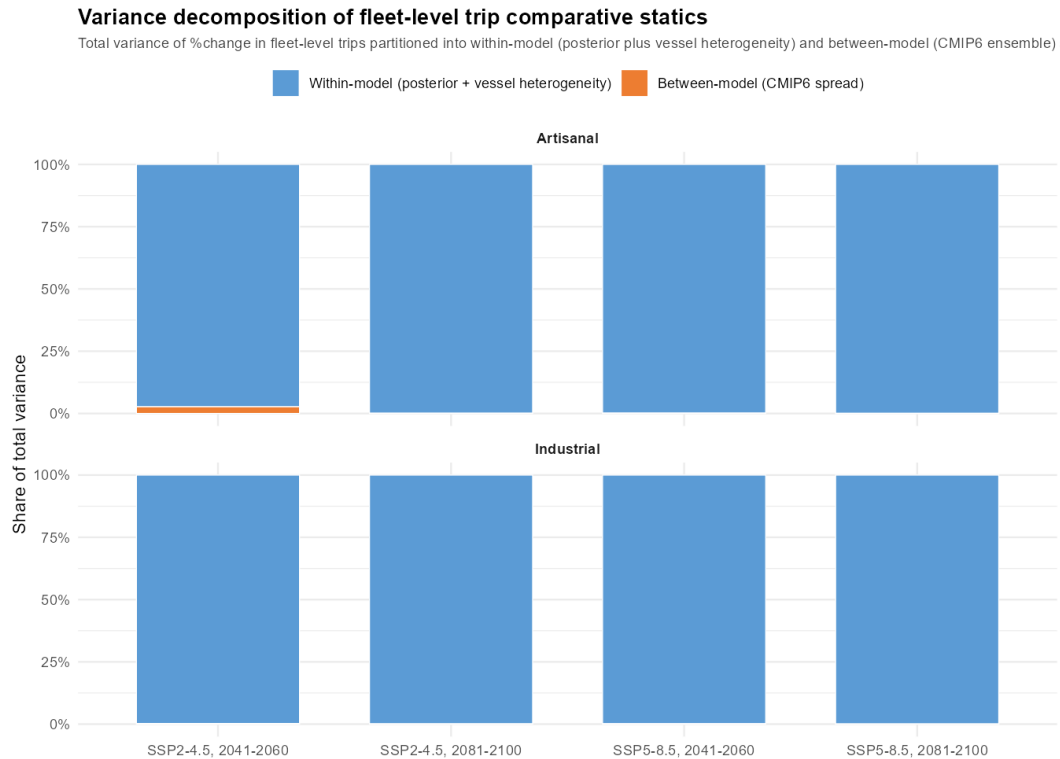


Figure 12: Variance decomposition of $\Delta T_f / T_f^{(0)}$ across the CMIP6 ensemble. Bars stack to one within each scenario–window pair and split the total variance into within-model (posterior pooled with vessel heterogeneity, blue) and between-model (CMIP6 spread, orange) components. The within-model component dominates across all cells, reflecting the floor-effect saturation of the underlying Schaefer steady-state biomass collapse rather than tight cross-model agreement on the climate signal.

asymmetry would reflect within-vessel rigidity rather than the cross-sector wedge. Two complementary data sources allow us to test this directly. For the industrial sector we use the registry of vessel-level participation coefficients (*coeficientes de participación relativa*) for jurel, sardina española and anchoveta, vigent through 2012, released by SUBPESCA in response to the Ley de Transparencia request~AH002T-0007379 ([Subsecretaría de Pesca y Acuicultura \(SUBPESCA\), 2026](#)). This is the last pre-LTP snapshot of authorised industrial portfolios and remains the legal basis for the Class~A LTP coefficients in force thereafter. For the artisanal sector, the equivalent administrative registry of permits by RPA and species is not publicly available; we therefore use the 2001–2024 IFOP logbook (454,308 trip~records on 1,583 vessels covering 81 species) and treat the set of species each vessel *actually landed* as a revealed-preference lower bound on the species for which it holds operational authorisation, since vessels cannot legally land species outside their authorised portfolio.

Table~18 reports four portfolio-breadth indicators by sector bucket: the mean number of distinct species landed over the panel lifetime, the share of vessels landing two or more species over that lifetime, the mean number of distinct species landed per vessel-year, and the share of vessel-years

with two or more species active.

Table 18: Permit-portfolio breadth indicators by sector bucket, Chile small pelagic logbook 2001–2024 (artisanal sub-buckets) and SUBPESCA 2012 industrial registry (industrial bucket).

Bucket	n vessels	Species lifetime	Multi-species lifetime (%)	Species per year	Multi-species
Industrial (IND)	129	7.63	87.6	4.14	
Lancha Menor (LM)	888	6.27	89.8	3.81	
Unclassified artisanal	367	4.76	89.6	3.46	
Bote Menor (BM)	189	2.44	63.0	2.14	

Notes. The IND row is constructed from the SUBPESCA 2012 registry, which lists 1,149 (vessel $\times$ species $\times$ macrozone) authorisations for industrial vessels and 26 *armadores*. Of those 299 vessels, 75.6% hold authorisations for the full anchoveta–sardina–jurel triplet and 67% hold authorisations in two or more macrozones, with a median of four (zone $\times$ species) authorisation pairs per vessel. Logbook rows are restricted to non-missing species codes (excluding the *lance sin captura* code 999). Buckets follow the Kasperski-aligned vessel-type classification used in Section 4.2: LM is *Lancha Menor*, BM is *Bote Menor*, and *Unclassified artisanal* pools the remaining artisanal vessel types. Species lifetimes are computed over the full 81-species range in the logbook; the SPF triplet (anchoveta, sardina común, jurel) accounts for 80–86% of artisanal lances and 67% of industrial lances within the panel.

Three substantive readings of Table~18 support the institutional argument developed in Section~5. First, the count of permits per vessel does not distinguish the two sectors in a way that would generate the headline asymmetry: 87.6% of industrial vessels and 89.8% of the dominant Lancha Menor sub-bucket of the artisanal sector hold authorisations for two or more species over the panel lifetime, with similar shares per vessel-year. Second, the within-artisanal heterogeneity that anchors the LM–BM contrast in Table~7 is *not* explained by greater species diversification in BM: BM in fact reports the narrowest logbook portfolio (mean 2.44 species lifetime, 2.14 per year), so the smaller projected response of BM to climate stress cannot be a permit-diversification story. The most plausible alternative is that BM operates substantially outside the regulated logbook (coastal multi-species and benthic activity), so the projected trip elasticity applies to a smaller fraction of BM’s revealed livelihood than of LM’s. Third, the composition of the multi-species portfolio differs sharply: 80–86% of artisanal lances and 67% of industrial lances fall on the anchoveta–sardina–jurel triplet, but within that triplet artisanal activity concentrates on the anchoveta–sardina pair while industrial activity is more evenly spread, with jurel and caballa together absorbing $\sim 70\%$ of industrial lances. Under climate forcing, the anchoveta–sardina pair responds with the identified negative semi-elasticities of Section~3.3.1, while jurel productivity is non-identified at the Centro-Sur scale (Appendix~E); the asymmetric concentration of the artisanal portfolio in the anchoveta–sardina pair therefore translates the same climate forcing into a sharper effective reduction of the artisanal portfolio than of the industrial portfolio, conditional on the institutional inability to recompose access across the sectoral boundary.

H.2 Policy counterfactual under Ley~21.752

The estimation window of the present paper (2018–2024) coincides with the Article~47 LGPA fractioning regime. Ley~N°21.752, promulgated on 25_{June}2025 ([Congreso Nacional de Chile, 2025](#)), fixes a new schedule of artisanal–industrial fractions in force through 2040. Across the V–Los–Lagos macrozone studied here, the law raises the artisanal share to 90% in anchoveta and sardina com'un (from 78%) and partitions jurel into a 30% artisanal share in V–Los–R'ios and 15% in Los–Lagos (from a uniform 10% under Article~47); Table~1 reports the two regimes side by side. A static plug-in evaluation of the climate-conditional artisanal landing under each regime is reported in Section~5.1 of the main text (Table~8); this appendix records the formal derivation and the posterior-propagated, by-model version that complements that headline table.

Letting Q_s^0 denote the 2024 Cuota Global de Captura for stock~ s and $\Delta r_s^*/r_s^0$ the projected productivity change at SSP5-8.5 end-of-century, the artisanal absolute landing under fractioning regime~ R is

$$L_{\text{art}}^{(R)} = \sum_s \alpha_s^{(R)} \cdot (1 + \Delta r_s^*/r_s^0) \cdot Q_s^0, \quad (17)$$

where $\alpha_s^{(R)}$ is the artisanal fraction of stock~ s under regime~ $R \in \{\text{Art. 47, Ley 21.752}\}$ and the sum runs over the four (stock $\times$ macrozone) units of Table~1: anchoveta V–Los–Lagos, sardina com'un V–Los–Lagos, jurel V–Los–R'ios, and jurel Los_{R'ios}–LosLagos. Equation~(17) reads the productivity change as a proportional adjustment to the steady-state landed volume and abstracts from any endogenous TAC-setting response. The plug-in version reported in the main text fixes $\Delta r_s^*/r_s^0$ at the cross-model posterior median from Table~5; the posterior-propagated version reported below integrates explicitly over the CMIP6 six-model ensemble and the within-posterior 90% envelope, returning cross-model median plus inter-quartile range for the central estimate and the median within-posterior 90% CI for the structural uncertainty axis, exactly as in Table~6.

Table 19: Posterior-propagated artisanal absolute landing under the Article 47 LGPA and Ley 21.752 fractioning regimes, SSP5-8.5 end-of-century, Centro-Sur (V–Los Lagos), 2024 Cuotas Globales.

Stock	Regime	Cross-model med [IQR] (kt)	Within-posterior 90% CI (kt)
Anchoveta (V–Los Lagos)	Art. 47 LGPA	17.1 [14.0, 20.8]	[2.9, 75.5]
Anchoveta (V–Los Lagos)	Ley 21.752	19.7 [16.2, 24.0]	[3.3, 87.2]
Sardina común (V–Los Lagos)	Art. 47 LGPA	0.3 [0.1, 1.4]	[0.1, 1.7]
Sardina común (V–Los Lagos)	Ley 21.752	0.3 [0.1, 1.6]	[0.1, 1.9]
Jurel (V–Los Lagos, aggregate)	Art. 47 LGPA	65.9 [65.9, 65.9]	[65.9, 65.9]
Jurel (V–Los Lagos, aggregate)	Ley 21.752	185.7 [185.7, 185.7]	[185.7, 185.7]
Total Centro-Sur	Art. 47 LGPA	83.3 [80.0, 88.1]	[68.8, 147.4]
Total Centro-Sur	Ley 21.752	205.7 [201.9, 211.3]	[189.1, 279.7]

Notes. Cross-model median and IQR are taken across the six CMIP6 models of the per-model posterior median; the within-posterior 90% CI is the median across models of the per-model posterior $[q_{05}, q_{95}]$. The construction parallels Table 6; see `R/run_counterfactual_ley21752.R` for the pipeline. Jurel is held at $\Delta r^*/r^0 = 0$ by the non-identification convention of Section 3.3.1; anchoveta and sardina común use the by-model summaries written by the main T5 pipeline (`tables/growth_comparative_statics_by_model.csv`).

Three readings of the posterior-propagated counterfactual sharpen the plug-in headline reported in the main text. First, the cross-model inter-quartile range is narrow at the aggregate level (a few kilotonnes on either side of the 122~kt median total), reflecting the floor-effect saturation documented in Appendix~G: once the small-pelagic productivity collapses, the cross-model dispersion of L_{art} is small in absolute terms because the residual base on which it operates is small. Second, the within-posterior 90% CI is substantially wider for anchoveta than for sardina com'un, because the posterior on $(\rho_{\text{anch}}^{SST}, \rho_{\text{anch}}^{CHL})$ admits a wider envelope of climate-conditional productivity factors than the sardina pair. Third, the jurel contribution to the artisanal landing is invariant across both axes of uncertainty—it appears as a point in the table—because the non-identification convention holds $(1 + \Delta r_{\text{jurel}}^*/r_{\text{jurel}}^0) = 1$; relaxing this convention to the ENSO-prior-propagated envelope of Appendix~E.6 would add a third source of uncertainty whose magnitude is bounded by the three-orders-of-magnitude range reported there. The plug-in headline in the main text is therefore robust to the cross-model and within-posterior axes that the appendix makes explicit, and the institutional reading developed in Section~5 carries over unchanged.

H.3 Inter-sectoral cession evidence from SERNAPESCA quota control reports

This subsection documents the realised inter-sectoral cession flow under Article₅₅T of the LGPA, the single administrative re-allocation channel admitted by the de jure fractioning regime. The data source is the annual SERNAPESCA quota control report, published as an Excel workbook for each pair (anchoveta–sardina com'un V–X) and (jurel XV–X). For each species–year we extract

the assigned, ceded, effective, and captured tonnage by sector from the workbook's summary sheet (*RESUMEN* from 2019 onward; per-titular detail rows for 2013–2017, where the workbook format predates the consolidated summary sheet). The 2018 record is obtained from a SERNAPESCA per-titular CSV transparency extract, parsed analogously to the workbook series.

Two qualitative facts emerge from the consolidated 2013–2024 series, reported in Table~20. First, in the anchoveta and sardina com'un fisheries the industrial sector cedes the vast majority of its assigned TAC to artisanal cessionaries every year: the share of the industrial assignment transferred to artisanal operators ranges from 85% to 99% in 2019–2024 and is documented at lower bounds of 11–72% in the format-heterogeneous 2013–2017 workbooks. The cession volume scales with the TAC and ranges from 26 to 50~kt per year in anchoveta and from 51 to 77~kt per year in sardina com'un. Second, in the jurel fishery the cession flow is an order of magnitude smaller relative to the assigned TAC (−3.5 to +4.3%) and reverses direction across years: 2019–2022 records small artisanal-to-industrial cessions of 11–15~kt per year, which reverse to industrial-to-artisanal cessions of 0.7 and 31.5~kt in 2023 and 2024 respectively.

The implication for the main-text policy interpretation is twofold. First, the structural asymmetry that the paper identifies is estimated on realised landings, which are post-cession; the asymmetry therefore reflects the operational allocation produced by the joint Article₄₇–Article₅₅~T regime rather than the de jure Article~47 fractioning alone. Second, the static counterfactual under Ley~21.752 in Section~5.1 of the main text and in equation~(17) of this appendix should be read as a de jure upper bound on the operational re-allocation: because the industrial sector already cedes the bulk of its assigned small-pelagic TAC to artisanal operators under the pre-reform regime, the 12~percentage-point reduction in the industrial de jure share under items~6 and~7 of Article~1 of Ley~21.752 translates into a proportionally smaller change in realised artisanal landings. The operationally meaningful change in the small pelagics is the re-pricing of the Article₅₅T cession channel under the 1.3~UTM/ton patent of Article~6 rather than the de jure share reassignment.

Table 20: Inter-sectoral cession flow under Article 55 T of the LGPA, Central-South small pelagic fishery, 2013–2024. Source: SERNAPESCA quota control reports.

Year	Species	IND assigned (kt)	IND cession (kt)	Share ceded	Direction
2018	anchoveta	12.8	−10.7	84%	IND→ART
2018	sardina	74.2	−63.2	85%	IND→ART
2018	jurel	326.9	+16.1	5%	ART→IND
2019	anchoveta	27.4	−25.8	94%	IND→ART
2019	sardina	76.0	−69.6	92%	IND→ART
2019	jurel	339.0	+12.0	3%	ART→IND
2020	anchoveta	38.6	−33.3	86%	IND→ART
2020	sardina	67.2	−60.9	91%	IND→ART
2020	jurel	1268.1	+11.8	1%	ART→IND
2021	anchoveta	45.3	−45.1	~100%	IND→ART
2021	sardina	76.9	−76.5	~100%	IND→ART
2021	jurel	448.7	+13.3	3%	ART→IND
2022	anchoveta	52.2	−50.5	97%	IND→ART
2022	sardina	77.7	−71.6	92%	IND→ART
2022	jurel	516.4	+15.2	3%	ART→IND
2023	anchoveta	39.9	−37.1	93%	IND→ART
2023	sardina	63.3	−61.7	98%	IND→ART
2023	jurel	637.0	−0.7	0%	IND→ART
2024	anchoveta	53.8	−45.5	85%	IND→ART
2024	sardina	53.9	−51.0	95%	IND→ART
2024	jurel	728.6	−31.5	4%	IND→ART

Notes. “IND assigned” is the industrial fraction of the Cuota Global de Captura aggregated across sub-macrozones (anchoveta and sardina común: V–Los Lagos; jurel: XV–Los Lagos). “IND cession” sums the *Movimientos* column across industrial regional rows of the SERNAPESCA *RESUMEN* sheet; negative values denote industrial-to-artisanal flows under Article 55 T, positive values denote the reverse direction. “Share ceded” is the absolute ratio $|\text{cession}|/\text{IND assigned}$. The 2018 report was obtained as a per-titular CSV transparency extract from SERNAPESCA and is parsed analogously to the 2019–2024 workbooks. Source workbooks for 2013–2017 use a different layout (per-titular per-period detail without a consolidated summary sheet); cession totals for that subperiod, available on request, are qualitatively consistent with the pattern reported here. The full per-year, per-sub-macrozone CSV is archived in [paper1/portfolio_check/cesiones_intersectoral_2019_2024.csv](#) and [cesiones_consolidated_2013_2024.csv](#).